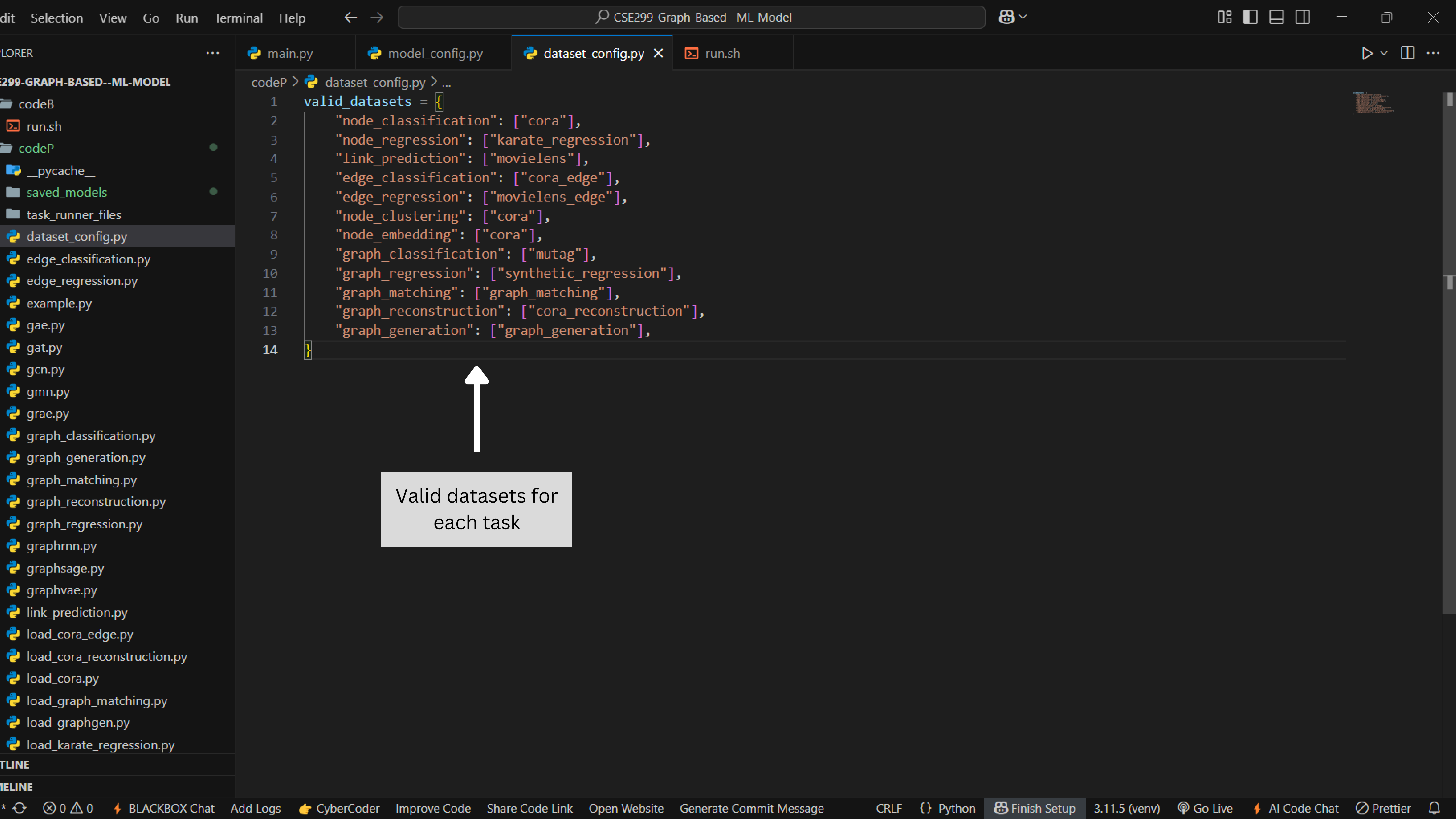
CSE299

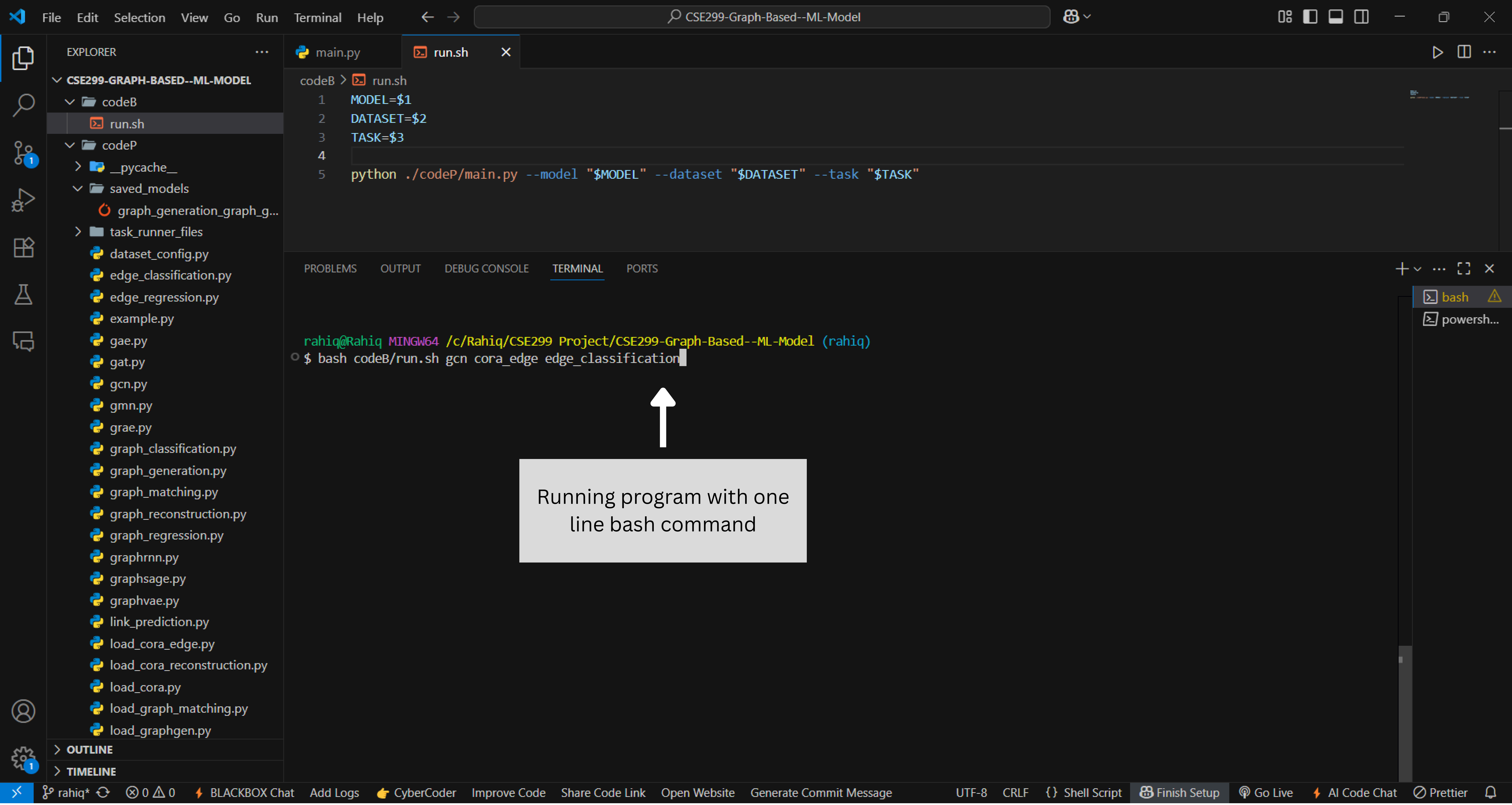
Graph Based ML codebase

(Group - 4)


CSE299-Graph-Based--ML-Model
main.py
model_config.py
dataset_config.py
run.sh
codeP > dataset_config.py > ...
valid_datasets = {
"node_classification": ["cora"],
"node_regression": ["karate_regression"],
"link_prediction": ["movielens"],
"edge_classification": ["cora_edge"],
"edge_regression": ["movielens_edge"],
"node_clustering": ["cora"],
"node_embedding": ["cora"],
"graph_classification": ["mutag"],
"graph_regression": ["synthetic_regression"],
"graph_matching": ["graph_matching"],
"graph_reconstruction": ["cora_reconstruction"],
"graph_generation": ["graph_generation"],
}
Valid datasets for each task
codeB
run.sh
codeP
__pycache__
saved_models
task_runner_files
dataset_config.py
edge_classification.py
edge_regression.py
example.py
gae.py
gat.py
gcn.py
gmn.py
grae.py
graph_classification.py
graph_generation.py
graph_matching.py
graph_reconstruction.py
graph_regression.py
graphrnn.py
graphsage.py
graphvae.py
link_prediction.py
load_cora_edge.py
load_cora_reconstruction.py
load_cora.py
load_graph_matching.py
load_graphgen.py
load_karate_regression.py

All available models for each task

```
# Models
available_models = {
    "gcn": {
        "name": "GCN",
        "module": {
            "link_prediction": link_prediction_gcn,
            "node_classification": node_classification_gcn,
            "node_regression": node_regression_gcn,
            "edge_classification": edge_classification_gcn,
            "edge_regression": edge_regression_gcn,
            "node_clustering": node_clustering_gcn,
            "node_embedding": node_embedding_gcn,
            "graph_classification": graph_classification_gcn,
            "graph_regression": graph_regression_gcn,
        }
    },
    "gat": {
        "name": "GAT",
        "module": {
            "link_prediction": link_prediction_gat,
            "node_classification": node_classification_gat,
            "node_regression": node_regression_gat,
            "edge_classification": edge_classification_gat,
            "edge_regression": edge_regression_gat,
            "node_clustering": node_clustering_gat,
            "node_embedding": node_embedding_gat,
            "graph_classification": graph_classification_gat,
            "graph_regression": graph_regression_gat,
        }
    },
    "graphsage": {
        "name": "GraphSAGE",
        "module": {
            "link_prediction": link_prediction_graphsage,
            "node_classification": node_classification_graphsage,
            "node_regression": node_regression_graphsage,
            "edge_classification": edge_classification_graphsage,
            "edge_regression": edge_regression_graphsage,
            "node_clustering": node_clustering_graphsage,
            "node_embedding": node_embedding_graphsage,
            "graph_classification": graph_classification_graphsage,
            "graph_regression": graph_regression_graphsage,
        }
    },
    "siamese_gcn": {
        "name": "Siamese GCN",
        "module": {
            "graph_matching": graph_matching_siamese_gcn,
        }
    },
    "gae": {
        "name": "GAE",
        "module": {
            "graph_reconstruction": graph_reconstruction_gae,
        }
    },
    "graphvae": {
        "name": "GraphVAE",
        "module": {
            "graph_generation": graph_generation_graphvae,
        }
    },
    "gmn": {
        "module": {
            "graph_matching": graph_matching_gmn,
        }
    },
    "graphrnn": {
        "module": {
            "graph_generation": graph_generation_graphrnn,
        }
    },
     "grae": {
        "module": {
            "graph_reconstruction": graph_reconstruction_grae,
        }
```


CSE299-Graph-Based--ML-Model
EXPLORER
CSE299-GRAPH-BASED--ML-MODEL
codeB
run.sh
codeP
__pycache__
saved_models
graph_generation_graph_g...
task_runner_files
dataset_config.py
edge_classification.py
edge_regression.py
example.py
gae.py
gat.py
gcn.py
gmn.py
grae.py
graph_classification.py
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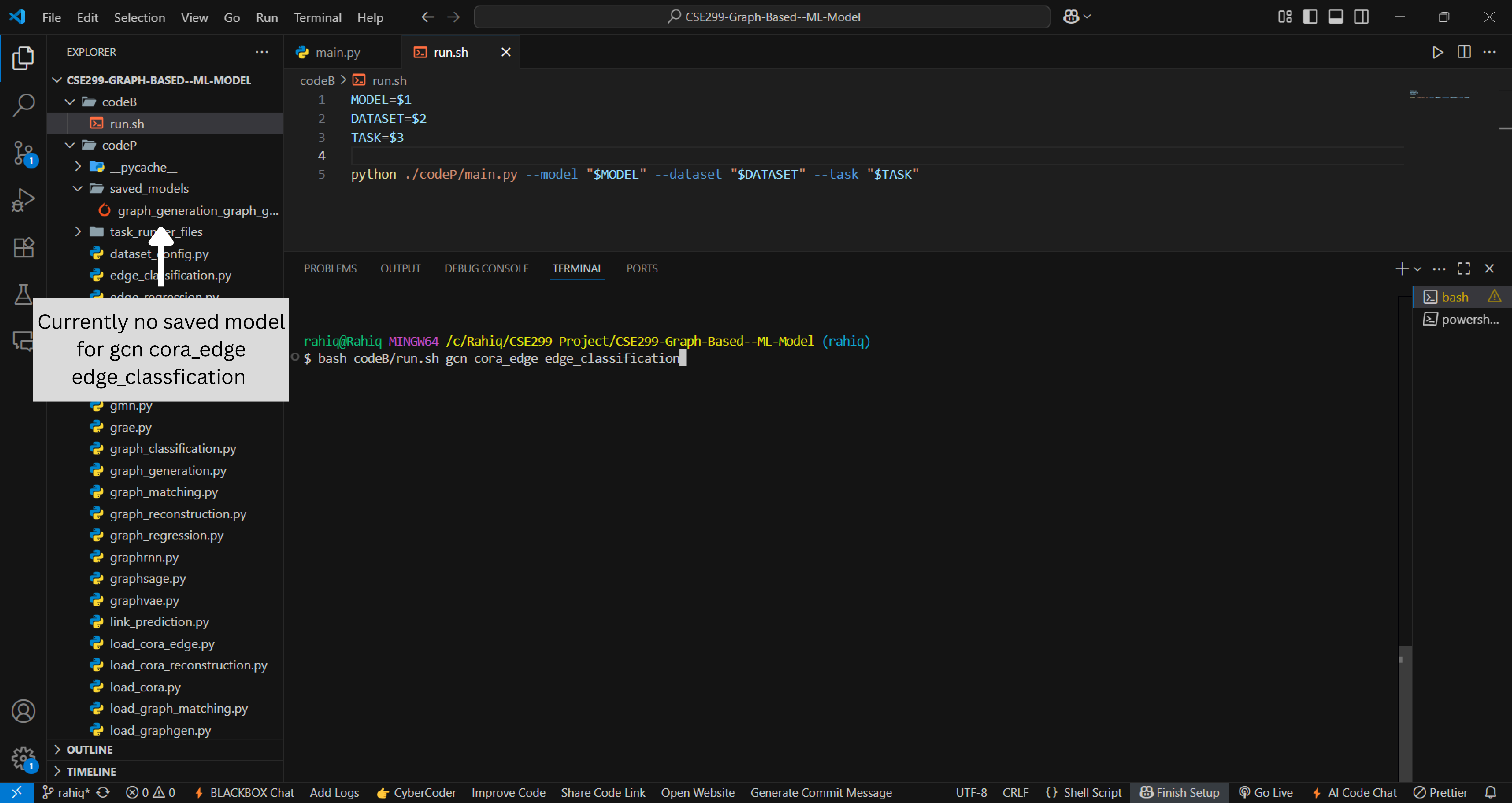
graph_reconstruction.py
graph_regression.py
graphrnn.py
graphsage.py
graphvae.py
link_prediction.py
load_cora_edge.py
load_cora_reconstruction.py
load_cora.py
load_graph_matching.py
load_graphgen.py
OUTLINE
TIMELINE
main.py
run.sh
codeB > run.sh
MODEL=$1
DATASET=$2
TASK=$3
python ./codeP/main.py --model "$MODEL" --dataset "$DATASET" --task "$TASK"
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn cora_edge edge_classification
bash
powersh...
Running program with one line bash command

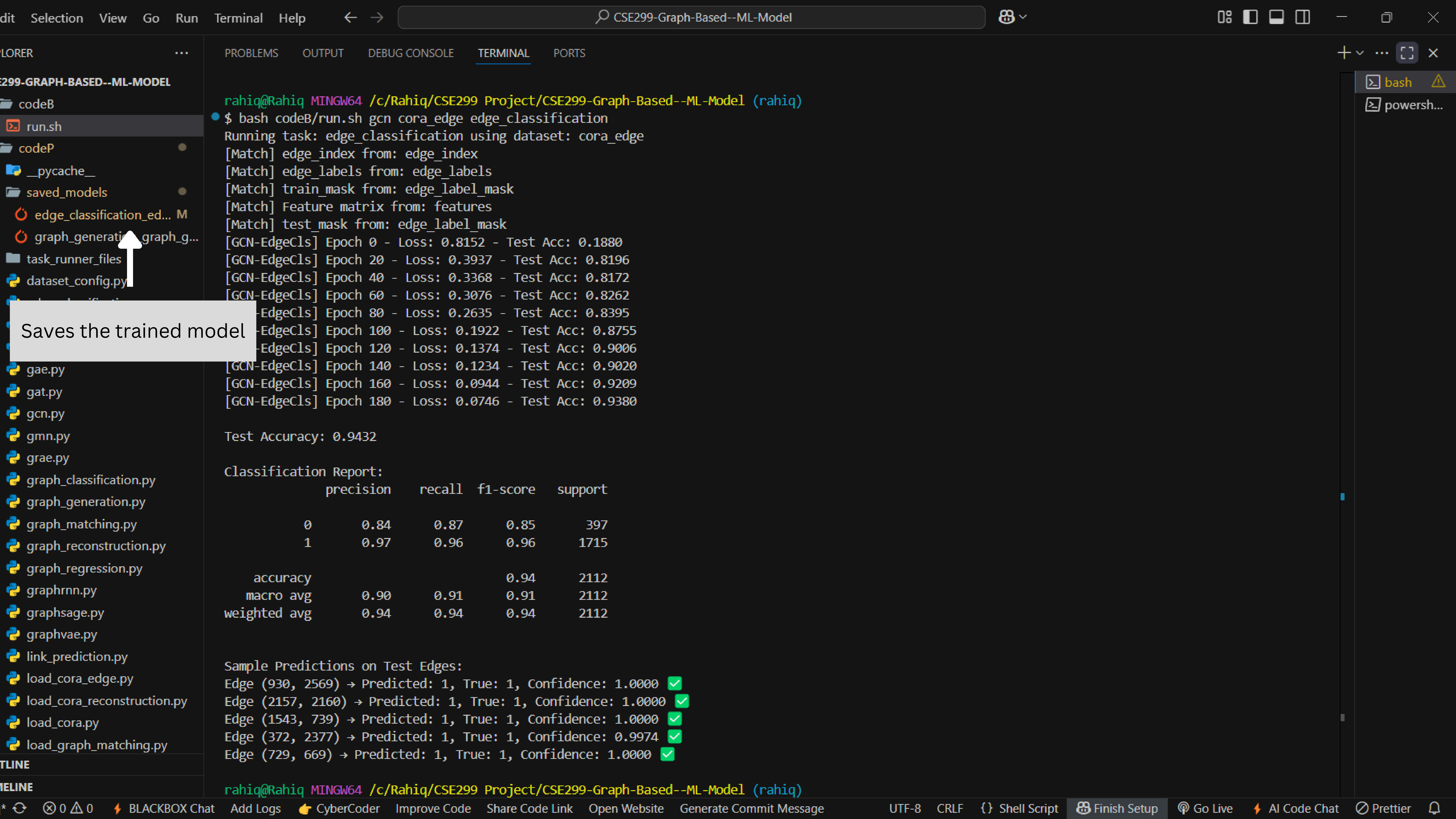
Model training code for edge_classification

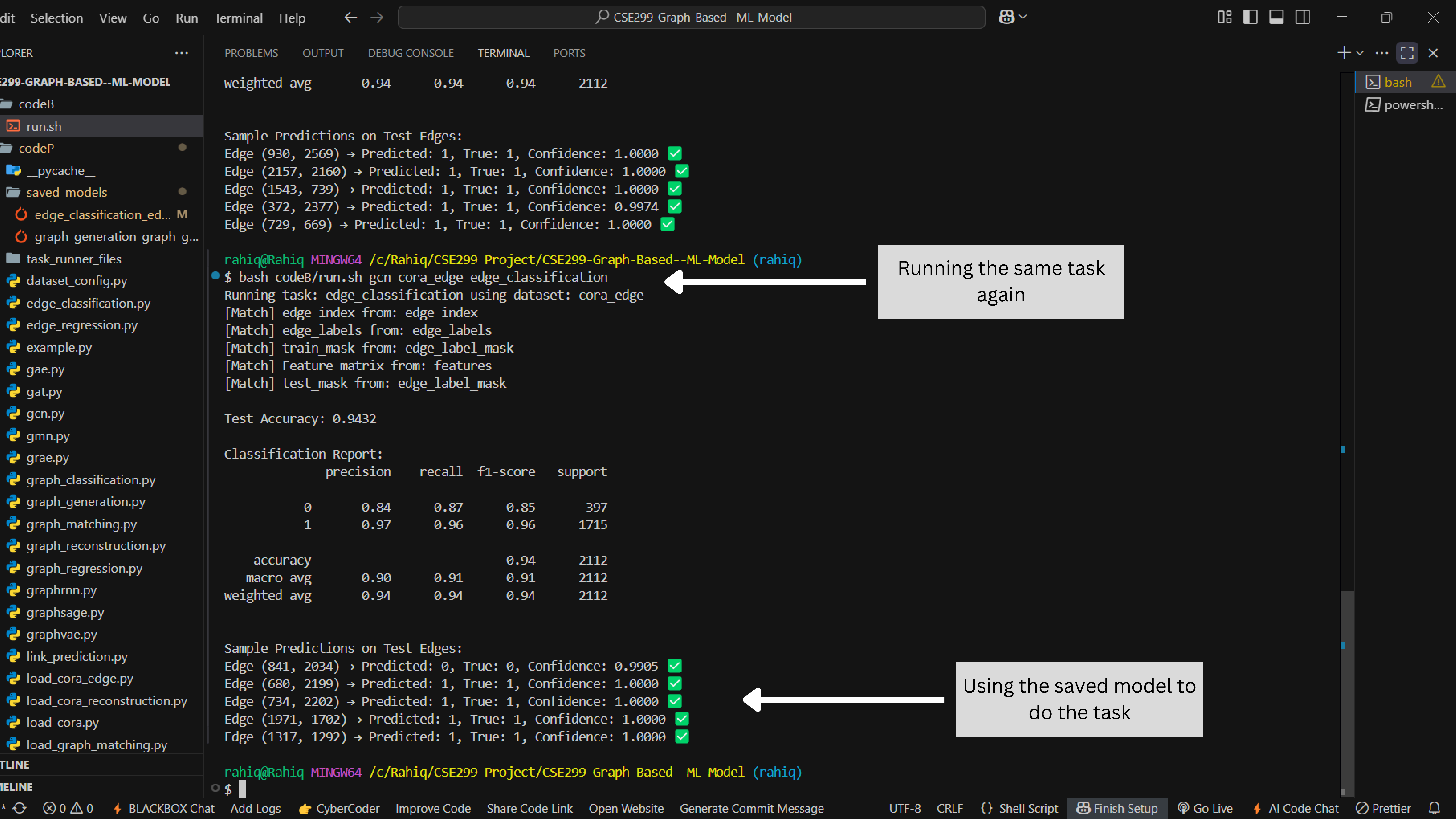
```python


#GCN
def edge_classification_gcn(x, edge_index, edge_labels, train_mask, test_mask, epochs=200):
    model = GCN(input_dim=x.size(1), hidden_dim=16, output_dim=16)
    classifier = torch.nn.Linear(16, 2)
    optimizer = torch.optim.Adam(list(model.parameters()) + list(classifier.parameters()), lr=0.01)

    for epoch in range(epochs):
        model.train()
        classifier.train()
        optimizer.zero_grad()

        node_embeddings = model(x, edge_index)
        src = edge_index[0]
        dst = edge_index[1]
        edge_emb = node_embeddings[src] * node_embeddings[dst]

        edge_logits = classifier(edge_emb)
        loss = F.cross_entropy(edge_logits[train_mask], edge_labels[train_mask])

        loss.backward()
        optimizer.step()

        if epoch % 20 == 0:
            model.eval()
            classifier.eval()
            with torch.no_grad():
                preds = edge_logits[test_mask].argmax(dim=1)
                acc = (preds == edge_labels[test_mask]).float().mean().item()
                print(f"[GCN-EdgeCls] Epoch {epoch} - Loss: {loss.item():.4f} - Test Acc: {acc:.4f}")

    return model, classifier


#GAT
def edge_classification_gat(x, edge_index, edge_labels, train_mask, test_mask, epochs=200):
```


Currently no saved model for gcn cora_edge edge_classfication
MODEL=$1
DATASET=$2
TASK=$3
python ./codeP/main.py --model "$MODEL" --dataset "$DATASET" --task "$TASK"
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn cora_edge edge_classification

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn cora_edge edge_classification
Running task: edge_classification using dataset: cora_edge
[Match] edge_index from: edge_index
[Match] edge_labels from: edge_labels
[Match] train_mask from: edge_label_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_label_mask
[GCN-EdgeCls] Epoch 0 - Loss: 0.8152 - Test Acc: 0.1880
[GCN-EdgeCls] Epoch 20 - Loss: 0.3937 - Test Acc: 0.8196
[GCN-EdgeCls] Epoch 40 - Loss: 0.3368 - Test Acc: 0.8172
[GCN-EdgeCls] Epoch 60 - Loss: 0.3076 - Test Acc: 0.8262
[GCN-EdgeCls] Epoch 80 - Loss: 0.2635 - Test Acc: 0.8395
[GCN-EdgeCls] Epoch 100 - Loss: 0.1922 - Test Acc: 0.8755
[GCN-EdgeCls] Epoch 120 - Loss: 0.1374 - Test Acc: 0.9006
[GCN-EdgeCls] Epoch 140 - Loss: 0.1234 - Test Acc: 0.9020
[GCN-EdgeCls] Epoch 160 - Loss: 0.0944 - Test Acc: 0.9209
[GCN-EdgeCls] Epoch 180 - Loss: 0.0746 - Test Acc: 0.9380

Test Accuracy: 0.9432

Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.87      0.85       397
           1       0.97      0.96      0.96      1715

    accuracy                           0.94      2112
   macro avg       0.90      0.91      0.91      2112
weighted avg       0.94      0.94      0.94      2112


Sample Predictions on Test Edges:
Edge (930, 2569) → Predicted: 1, True: 1, Confidence: 1.0000 ✅
Edge (2157, 2160) → Predicted: 1, True: 1, Confidence: 1.0000 ✅
Edge (1543, 739) → Predicted: 1, True: 1, Confidence: 1.0000 ✅
Edge (372, 2377) → Predicted: 1, True: 1, Confidence: 0.9974 ✅
Edge (729, 669) → Predicted: 1, True: 1, Confidence: 1.0000 ✅

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
```

Training the model

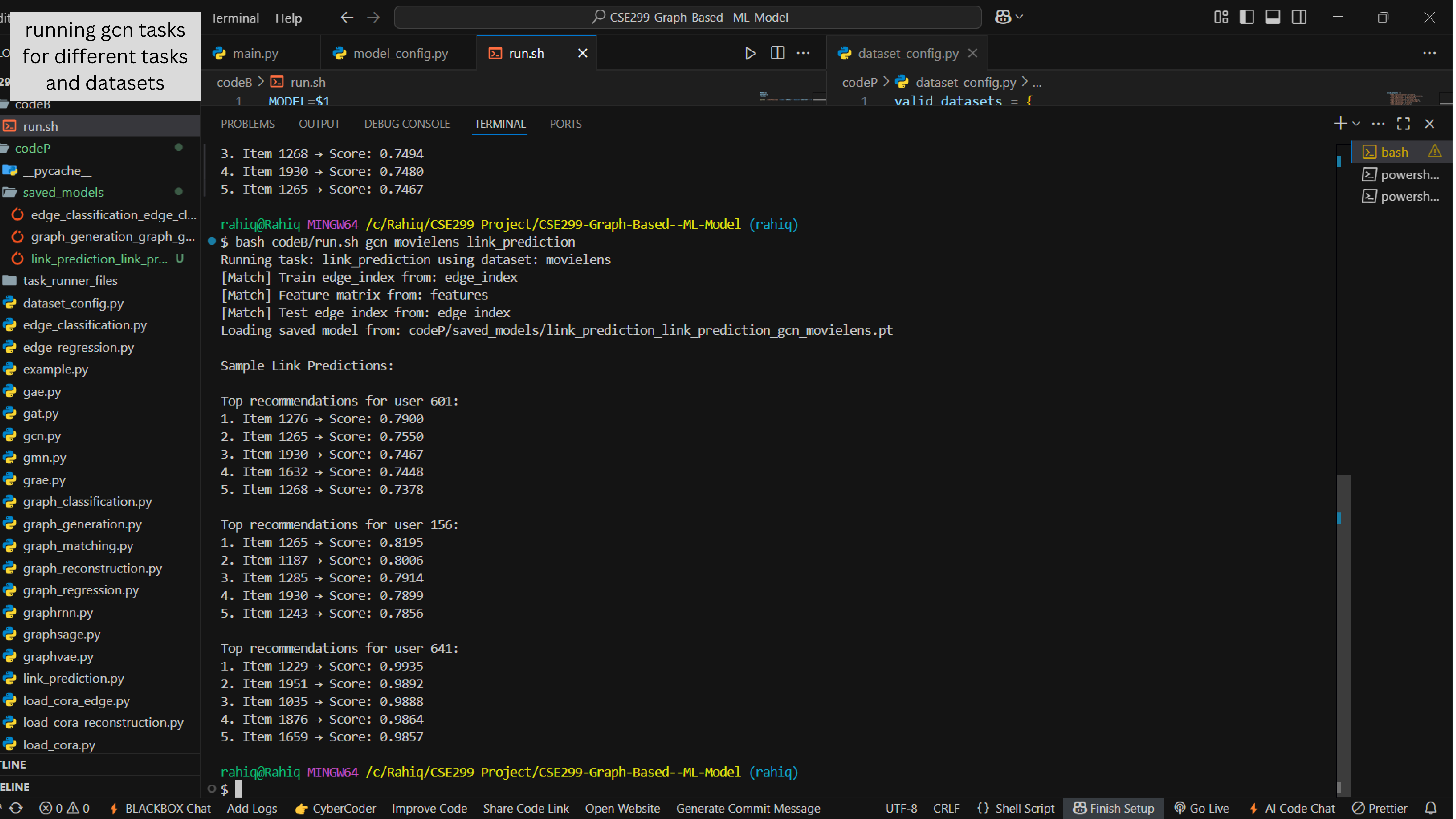

CSE299-Graph-Based--ML-Model
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
codeB
run.sh
codeP
__pycache__
saved_models
edge_classification_ed... M
graph_generati... graph_g...
task_runner_files
dataset_config.py
Saves the trained model
gae.py
gat.py
gcn.py
gmn.py
grae.py
graph_classification.py
graph_generation.py
graph_matching.py
graph_reconstruction.py
graph_regression.py
graphrnn.py
graphsage.py
graphvae.py
link_prediction.py
load_cora_edge.py
load_cora_reconstruction.py
load_cora.py
load_graph_matching.py
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn cora_edge edge_classification
Running task: edge_classification using dataset: cora_edge
[Match] edge_index from: edge_index
[Match] edge_labels from: edge_labels
[Match] train_mask from: edge_label_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_label_mask
[GCN-EdgeCls] Epoch 0 - Loss: 0.8152 - Test Acc: 0.1880
[GCN-EdgeCls] Epoch 20 - Loss: 0.3937 - Test Acc: 0.8196
[GCN-EdgeCls] Epoch 40 - Loss: 0.3368 - Test Acc: 0.8172
[GCN-EdgeCls] Epoch 60 - Loss: 0.3076 - Test Acc: 0.8262
-EdgeCls] Epoch 80 - Loss: 0.2635 - Test Acc: 0.8395
-EdgeCls] Epoch 100 - Loss: 0.1922 - Test Acc: 0.8755
-EdgeCls] Epoch 120 - Loss: 0.1374 - Test Acc: 0.9006
[GCN-EdgeCls] Epoch 140 - Loss: 0.1234 - Test Acc: 0.9020
[GCN-EdgeCls] Epoch 160 - Loss: 0.0944 - Test Acc: 0.9209
[GCN-EdgeCls] Epoch 180 - Loss: 0.0746 - Test Acc: 0.9380
Test Accuracy: 0.9432
Classification Report:
precision recall f1-score support
0 0.84 0.87 0.85 397
1 0.97 0.96 0.96 1715
accuracy 0.94 2112
macro avg 0.90 0.91 0.91 2112
weighted avg 0.94 0.94 0.94 2112
Sample Predictions on Test Edges:
Edge (930, 2569) → Predicted: 1, True: 1, Confidence: 1.0000 ✅
Edge (2157, 2160) → Predicted: 1, True: 1, Confidence: 1.0000 ✅
Edge (1543, 739) → Predicted: 1, True: 1, Confidence: 1.0000 ✅
Edge (372, 2377) → Predicted: 1, True: 1, Confidence: 0.9974 ✅
Edge (729, 669) → Predicted: 1, True: 1, Confidence: 1.0000 ✅
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)

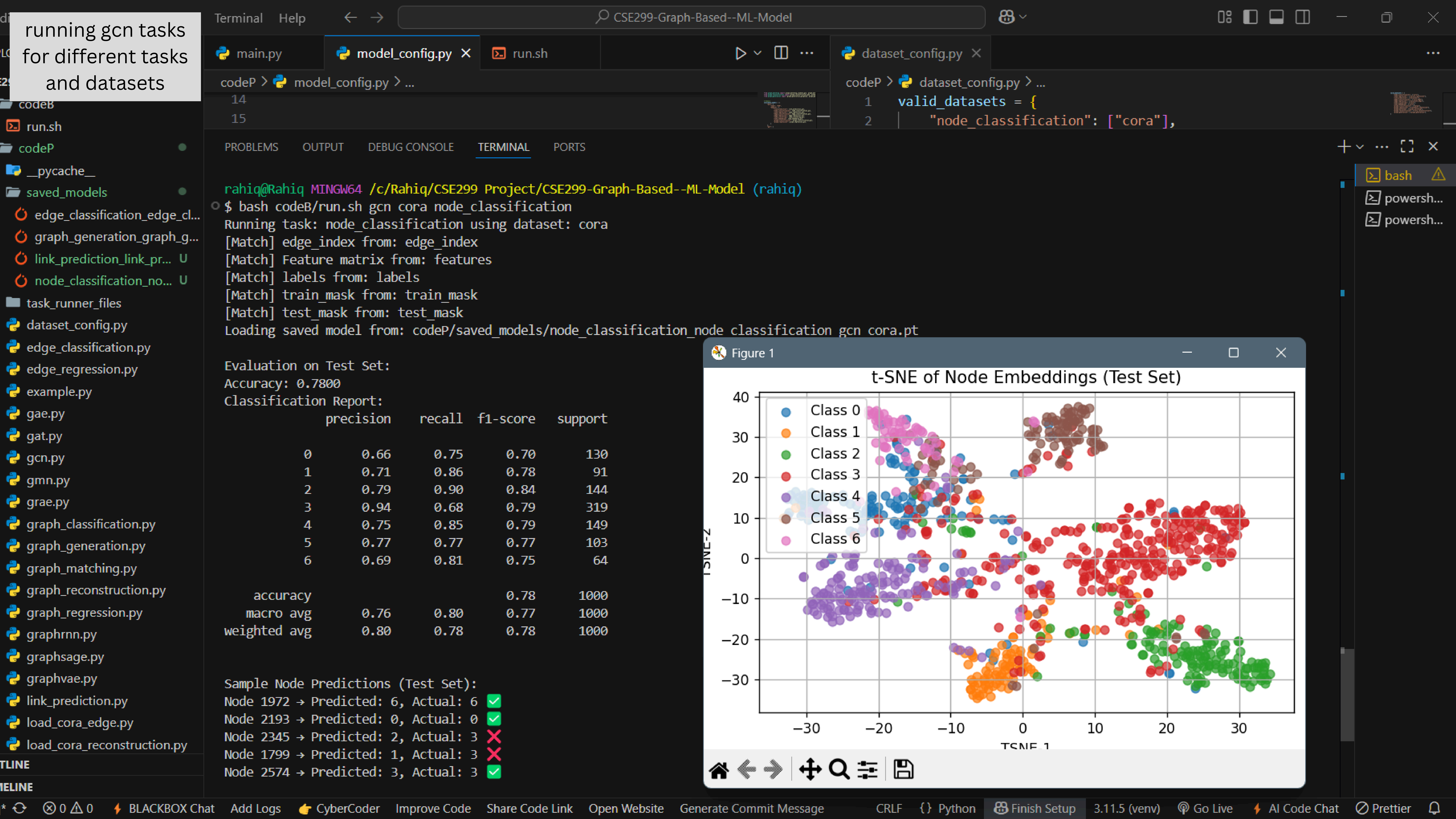

CSE299-Graph-Based--ML-Model
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
weighted avg 0.94 0.94 0.94 2112
Sample Predictions on Test Edges:
Edge (930, 2569) → Predicted: 1, True: 1, Confidence: 1.0000
Edge (2157, 2160) → Predicted: 1, True: 1, Confidence: 1.0000
Edge (1543, 739) → Predicted: 1, True: 1, Confidence: 1.0000
Edge (372, 2377) → Predicted: 1, True: 1, Confidence: 0.9974
Edge (729, 669) → Predicted: 1, True: 1, Confidence: 1.0000
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn cora_edge edge_classification
Running task: edge_classification using dataset: cora_edge
[Match] edge_index from: edge_index
[Match] edge_labels from: edge_labels
[Match] train_mask from: edge_label_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_label_mask
Test Accuracy: 0.9432
Classification Report:
precision recall f1-score support
0 0.84 0.87 0.85 397
1 0.97 0.96 0.96 1715
accuracy 0.94 2112
macro avg 0.90 0.91 0.91 2112
weighted avg 0.94 0.94 0.94 2112
Sample Predictions on Test Edges:
Edge (841, 2034) → Predicted: 0, True: 0, Confidence: 0.9905
Edge (680, 2199) → Predicted: 1, True: 1, Confidence: 1.0000
Edge (734, 2202) → Predicted: 1, True: 1, Confidence: 1.0000
Edge (1971, 1702) → Predicted: 1, True: 1, Confidence: 1.0000
Edge (1317, 1292) → Predicted: 1, True: 1, Confidence: 1.0000
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$
Running the same task again
Using the saved model to do the task

```python
import os
import json
import torch
import pickle
import pandas as pd
import numpy as np
from torch_geometric.data import Data
from torch_geometric.data.data import DataEdgeAttr

torch.serialization.add_safe_globals([Data, DataEdgeAttr])

def try_loaders(path):
    loaders = [
        lambda p: torch.load(p, weights_only=False),
        lambda p: pickle.load(open(p, 'rb')),
        lambda p: np.load(p, allow_pickle=True),
        lambda p: pd.read_csv(p, header=None).values,
        lambda p: json.load(open(p, 'r', encoding='utf-8')),
        lambda p: [json.loads(line) for line in open(p, 'r', encoding='utf-8')],
        lambda p: open(p, 'r', encoding='utf-8').read(),
    ]
    for loader in loaders:
        try:
            return loader(path)
        except Exception:
            continue
    print(f"[Warning] Could not load file: {path}")
    return None


def load_file(path):
    try:
        return try_loaders(path)
    except Exception as e:
        print(f"[Error] Exception while loading {path}: {e}")
        return None
```

Supports different file formats

running gcn tasks for different tasks and datasets

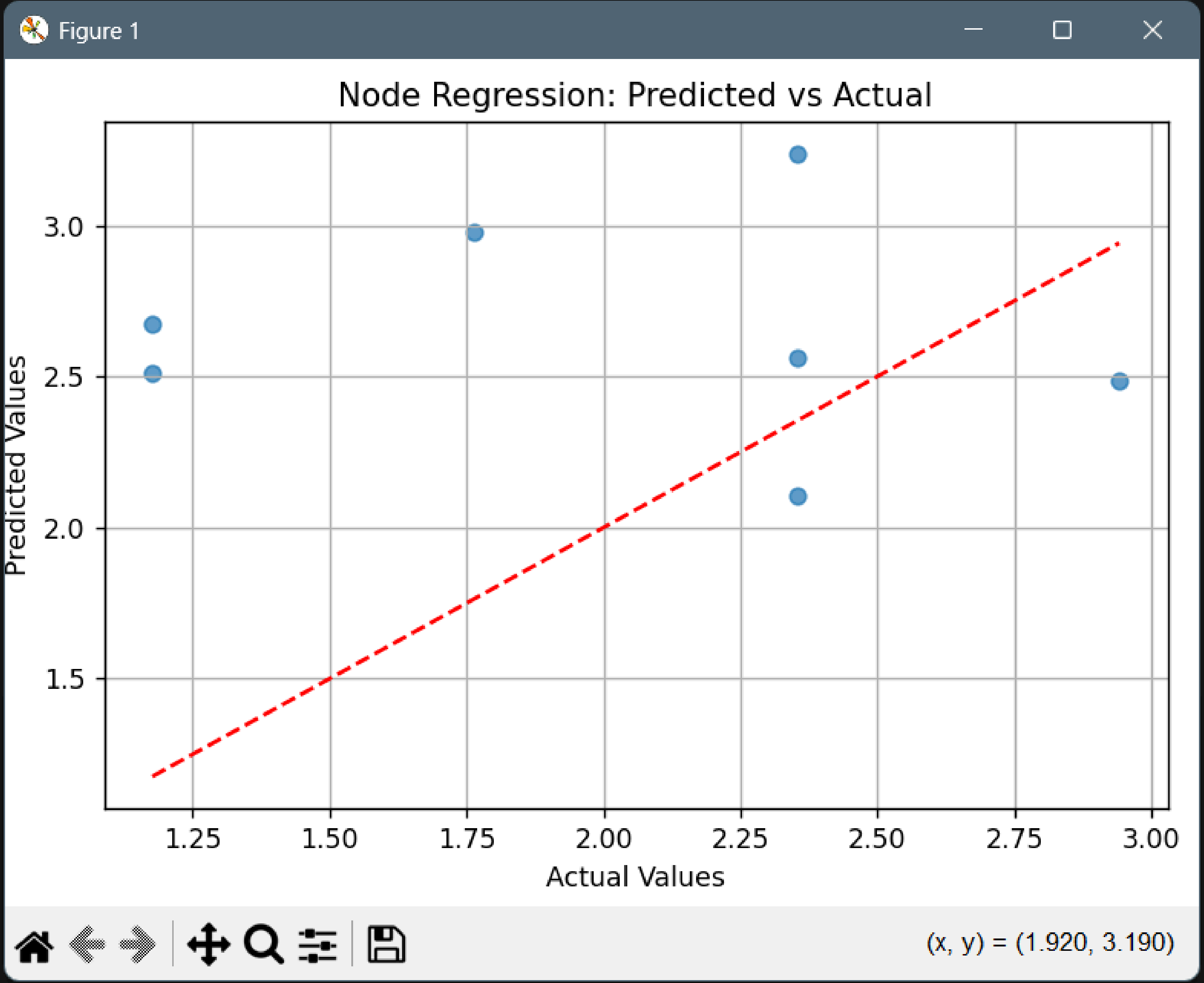




running gcn tasks for different tasks and datasets

running gcn tasks for different tasks and datasets

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn karate_regression node_regression
Running task: node_regression using dataset: karate_regression
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] targets from: targets
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
No saved model found. Training from scratch...
[GCN-NodeReg] Epoch 0 - Loss: 13.7964 - Test MSE: 4.1609
[GCN-NodeReg] Epoch 20 - Loss: 4.6479 - Test MSE: 0.3594
[GCN-NodeReg] Epoch 40 - Loss: 2.5046 - Test MSE: 2.1338
[GCN-NodeReg] Epoch 60 - Loss: 1.9763 - Test MSE: 0.8149
[GCN-NodeReg] Epoch 80 - Loss: 1.7009 - Test MSE: 1.1180
[GCN-NodeReg] Epoch 100 - Loss: 1.4523 - Test MSE: 0.9659
[GCN-NodeReg] Epoch 120 - Loss: 1.2036 - Test MSE: 0.9511
[GCN-NodeReg] Epoch 140 - Loss: 0.9628 - Test MSE: 0.9056
[GCN-NodeReg] Epoch 160 - Loss: 0.7501 - Test MSE: 0.8788
[GCN-NodeReg] Epoch 180 - Loss: 0.5725 - Test MSE: 0.8996

Evaluation on Test Set:
Mean Squared Error (MSE): 0.9407
Mean Absolute Error (MAE): 0.8344
R² Score: -1.4670

Sample Node Regression Results (Test Set):
Node 30 → Predicted: 3.2377 | Actual: 2.3529 | Absolute Error: 0.8848
Node 24 → Predicted: 2.9793 | Actual: 1.7647 | Absolute Error: 1.2146
Node 5 → Predicted: 2.1044 | Actual: 2.3529 | Absolute Error: 0.2486
Node 22 → Predicted: 2.5104 | Actual: 1.1765 | Absolute Error: 1.3339
Node 27 → Predicted: 2.5598 | Actual: 2.3529 | Absolute Error: 0.2069
```



running gcn tasks for different tasks and datasets

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn karate_regression node_regression
Running task: node_regression using dataset: karate_regression
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] targets from: targets
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
No saved model found. Training from scratch...
[GCN-NodeReg] Epoch 0 - Loss: 13.7964 - Test MSE: 4.1609
[GCN-NodeReg] Epoch 20 - Loss: 4.6479 - Test MSE: 0.3594
[GCN-NodeReg] Epoch 40 - Loss: 2.5046 - Test MSE: 2.1338
[GCN-NodeReg] Epoch 60 - Loss: 1.9763 - Test MSE: 0.8149
[GCN-NodeReg] Epoch 80 - Loss: 1.7009 - Test MSE: 1.1180
[GCN-NodeReg] Epoch 100 - Loss: 1.4523 - Test MSE: 0.9659
[GCN-NodeReg] Epoch 120 - Loss: 1.2036 - Test MSE: 0.9511
[GCN-NodeReg] Epoch 140 - Loss: 0.9628 - Test MSE: 0.9056
[GCN-NodeReg] Epoch 160 - Loss: 0.7501 - Test MSE: 0.8788
[GCN-NodeReg] Epoch 180 - Loss: 0.5725 - Test MSE: 0.8996

Evaluation on Test Set:
Mean Squared Error (MSE): 0.9407
Mean Absolute Error (MAE): 0.8344
R² Score: -1.4670

Sample Node Regression Results (Test Set):
Node 30 → Predicted: 3.2377 | Actual: 2.3529 | Absolute Error: 0.8848
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Node 5 → Predicted: 2.1044 | Actual: 2.3529 | Absolute Error: 0.2486
Node 22 → Predicted: 2.5104 | Actual: 1.1765 | Absolute Error: 1.3339
Node 27 → Predicted: 2.5598 | Actual: 2.3529 | Absolute Error: 0.2069
```
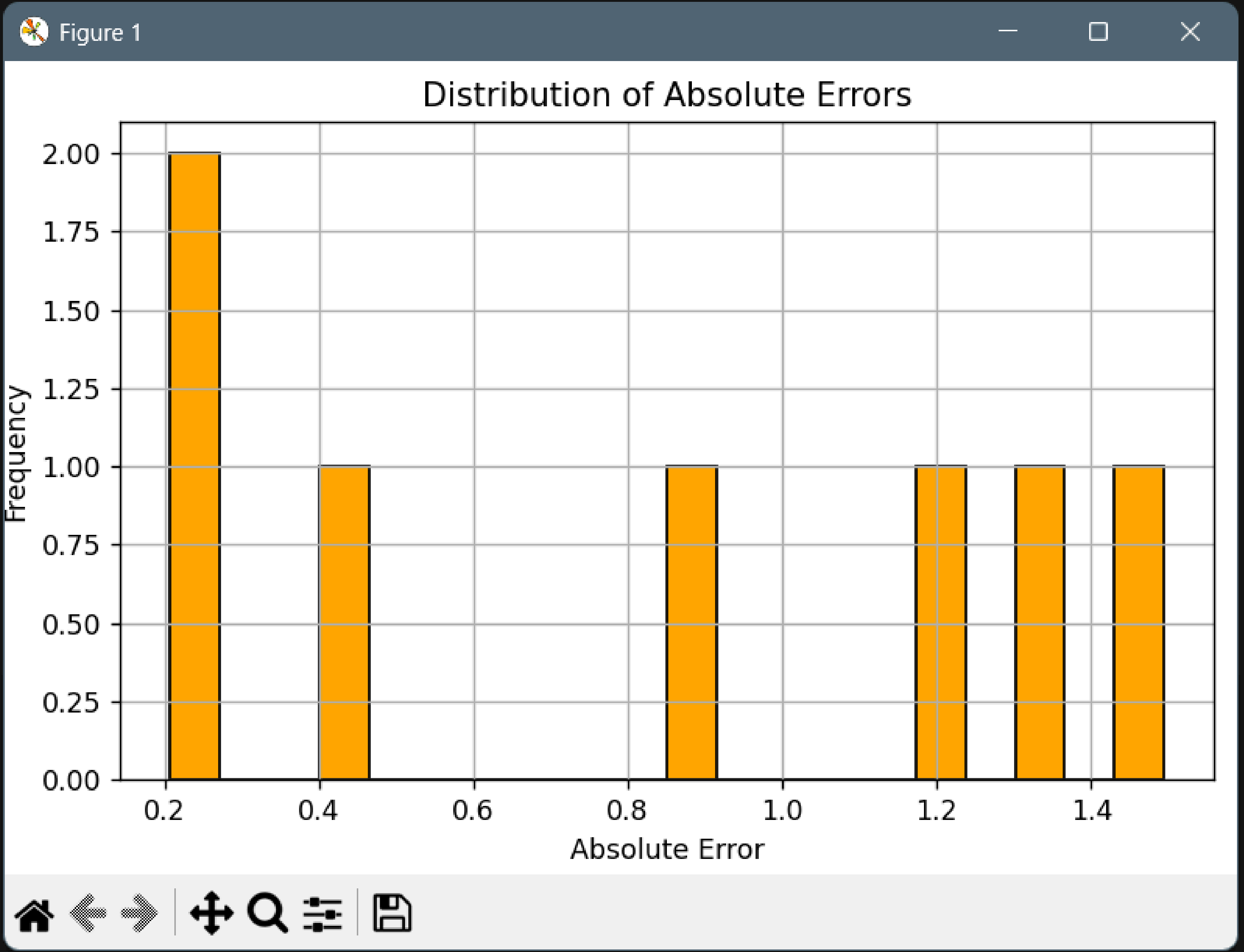
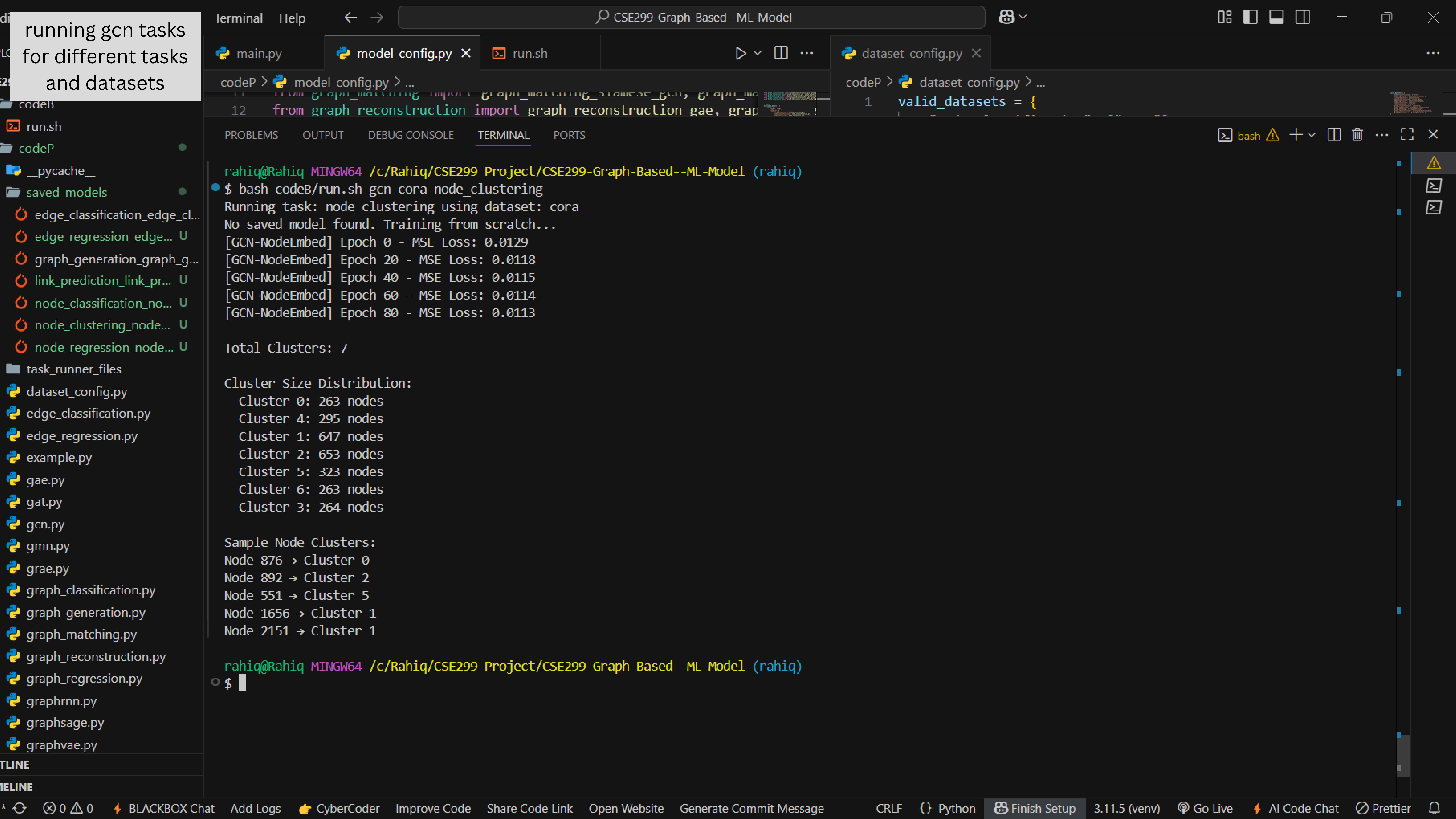



running gcn tasks for different tasks and datasets

```
R² Score: -1.4670

Sample Node Regression Results (Test Set):
Node 30 → Predicted: 3.2377 | Actual: 2.3529 | Absolute Error: 0.8848
Node 24 → Predicted: 2.9793 | Actual: 1.7647 | Absolute Error: 1.2146
Node 5 → Predicted: 2.1044 | Actual: 2.3529 | Absolute Error: 0.2486
Node 22 → Predicted: 2.5104 | Actual: 1.1765 | Absolute Error: 1.3339
Node 27 → Predicted: 2.5598 | Actual: 2.3529 | Absolute Error: 0.2069

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn movielens_edge edge_regression
Running task: edge_regression using dataset: movielens_edge
[Match] edge_index from: edge_index
[Match] edge_ratings from: edge_ratings
[Match] train_mask from: edge_rating_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_rating_mask
[GCN-EdgeReg] Epoch 0 - Loss: 12.9105 - Test MSE: 12.8656
[GCN-EdgeReg] Epoch 20 - Loss: 4.0433 - Test MSE: 4.0406
[GCN-EdgeReg] Epoch 40 - Loss: 2.4905 - Test MSE: 2.4919
[GCN-EdgeReg] Epoch 60 - Loss: 1.9981 - Test MSE: 2.0005
[GCN-EdgeReg] Epoch 80 - Loss: 1.6525 - Test MSE: 1.6552
[GCN-EdgeReg] Epoch 100 - Loss: 1.3328 - Test MSE: 1.3329
[GCN-EdgeReg] Epoch 120 - Loss: 1.2271 - Test MSE: 1.2258
[GCN-EdgeReg] Epoch 140 - Loss: 1.1360 - Test MSE: 1.1368
[GCN-EdgeReg] Epoch 160 - Loss: 1.0577 - Test MSE: 1.0585
[GCN-EdgeReg] Epoch 180 - Loss: 1.0267 - Test MSE: 1.0270

Mean Squared Error (MSE): 1.0074
R² Score: 0.2032

Sample Predictions:
Edge (879, 1024) → Predicted: 3.79, Actual: 3.00, Error: 0.79
Edge (200, 1228) → Predicted: 3.57, Actual: 2.00, Error: 1.57
Edge (1677, 362) → Predicted: 3.60, Actual: 3.00, Error: 0.60
Edge (278, 1572) → Predicted: 3.36, Actual: 4.00, Error: 0.64
Edge (1265, 833) → Predicted: 3.02, Actual: 2.00, Error: 1.02

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$
```

running gcn tasks for different tasks and datasets



running gcn tasks for different tasks and datasets

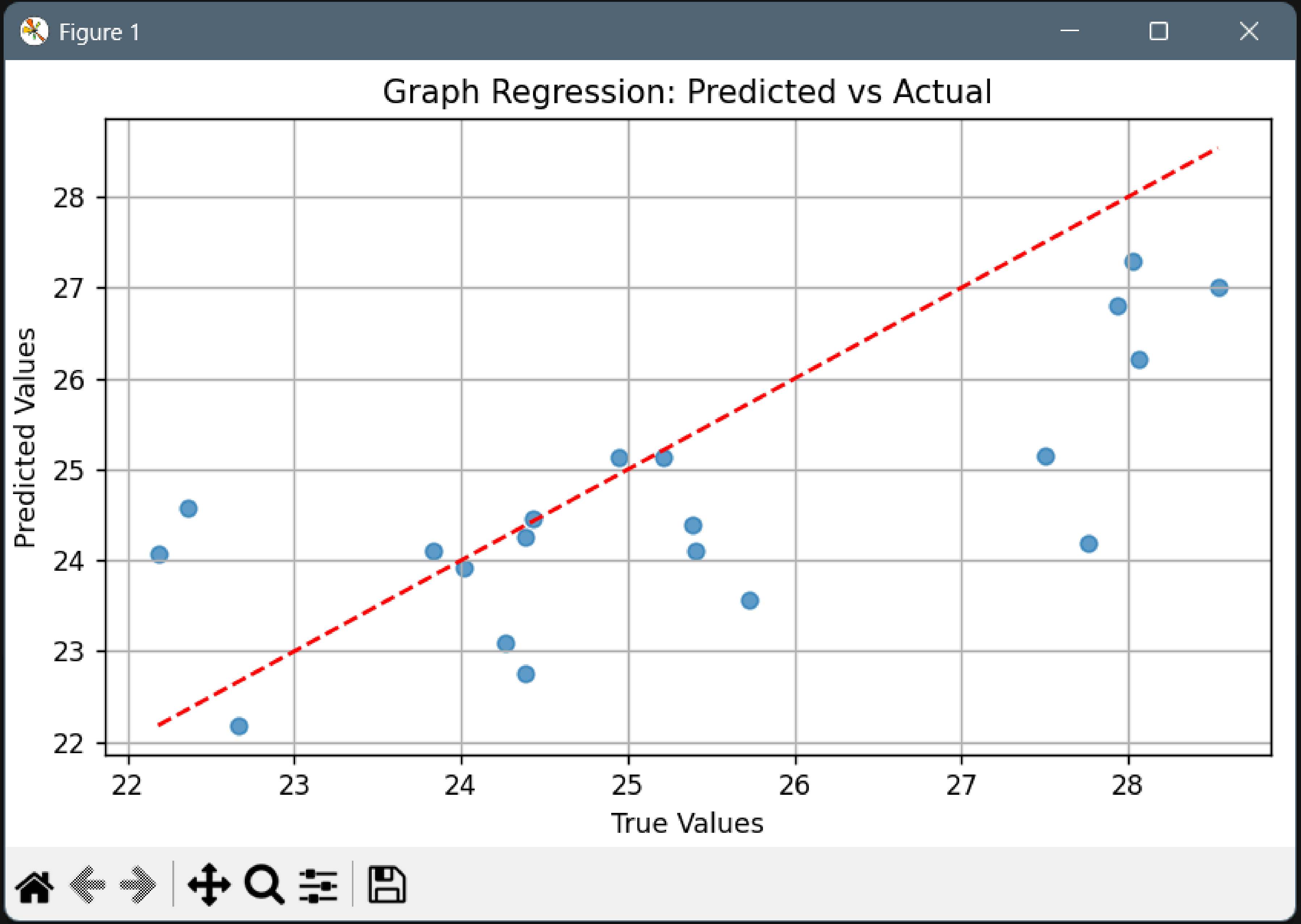
```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn cora node_embedding
Running task: node_embedding using dataset: cora
No saved model found. Training from scratch...
[GCN-NodeEmbed] Epoch 0 - MSE Loss: 0.0130
[GCN-NodeEmbed] Epoch 20 - MSE Loss: 0.0118
[GCN-NodeEmbed] Epoch 40 - MSE Loss: 0.0116
[GCN-NodeEmbed] Epoch 60 - MSE Loss: 0.0114
[GCN-NodeEmbed] Epoch 80 - MSE Loss: 0.0114

Sample Node Embeddings:
Node 0 → Embedding: [0.0024, 0.0217, 0.0256, -0.0004, 0.0297, ...]
Node 1 → Embedding: [-0.0051, -0.0124, -0.0063, 0.0396, -0.0502, ...]
Node 2 → Embedding: [-0.0030, -0.0019, 0.0164, 0.0368, -0.0274, ...]
Node 3 → Embedding: [-0.0009, -0.0048, 0.0119, 0.0371, 0.0227, ...]
Node 4 → Embedding: [0.0073, 0.0125, 0.0227, 0.0262, 0.0915, ...]

Embedding Statistics:
Mean: 0.0127
Std Dev: 0.0340
Min: -0.4595
Max: 2.5026

Cosine Similarity Between Sample Nodes:
Node 0 vs Node 1: 0.5154
Node 0 vs Node 2: 0.6135
Node 0 vs Node 3: 0.2785
Node 0 vs Node 4: 0.4364
Node 1 vs Node 2: 0.9271
Node 1 vs Node 3: 0.7855
Node 1 vs Node 4: 0.6091
Node 2 vs Node 3: 0.8257
Node 2 vs Node 4: 0.5491
Node 3 vs Node 4: 0.5920
```

running gcn tasks for different tasks and datasets

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn mutag graph_classification
Running task: graph_classification using dataset: mutag
No saved model found. Training from scratch...
[GCN-GraphCls] Epoch 0 - Loss: 0.6691 - Train Acc: 0.6467
[GCN-GraphCls] Epoch 10 - Loss: 0.4971 - Train Acc: 0.7467
[GCN-GraphCls] Epoch 20 - Loss: 0.4675 - Train Acc: 0.7800
[GCN-GraphCls] Epoch 30 - Loss: 0.4714 - Train Acc: 0.7867
[GCN-GraphCls] Epoch 40 - Loss: 0.4905 - Train Acc: 0.7667
[GCN-GraphCls] Epoch 50 - Loss: 0.4792 - Train Acc: 0.7933
[GCN-GraphCls] Epoch 60 - Loss: 0.4708 - Train Acc: 0.7933
[GCN-GraphCls] Epoch 70 - Loss: 0.4641 - Train Acc: 0.7800
[GCN-GraphCls] Epoch 80 - Loss: 0.4781 - Train Acc: 0.8000
[GCN-GraphCls] Epoch 90 - Loss: 0.4635 - Train Acc: 0.8067
[GCN-GraphCls] Epoch 99 - Loss: 0.4696 - Train Acc: 0.7933

Evaluation on Test Set:
Accuracy: 0.6316
Classification Report:
              precision    recall  f1-score   support

           0       0.43      0.23      0.30        13
           1       0.68      0.84      0.75        25

    accuracy                           0.63        38
   macro avg       0.55      0.54      0.53        38
weighted avg       0.59      0.63      0.60        38


Sample Graph Classification:
Graph 12 → Predicted: 1 | Actual: 0 ❌
Graph 30 → Predicted: 1 | Actual: 0 ❌
Graph 37 → Predicted: 1 | Actual: 0 ❌
Graph 33 → Predicted: 1 | Actual: 0 ❌
Graph 17 → Predicted: 1 | Actual: 1 ✅
```

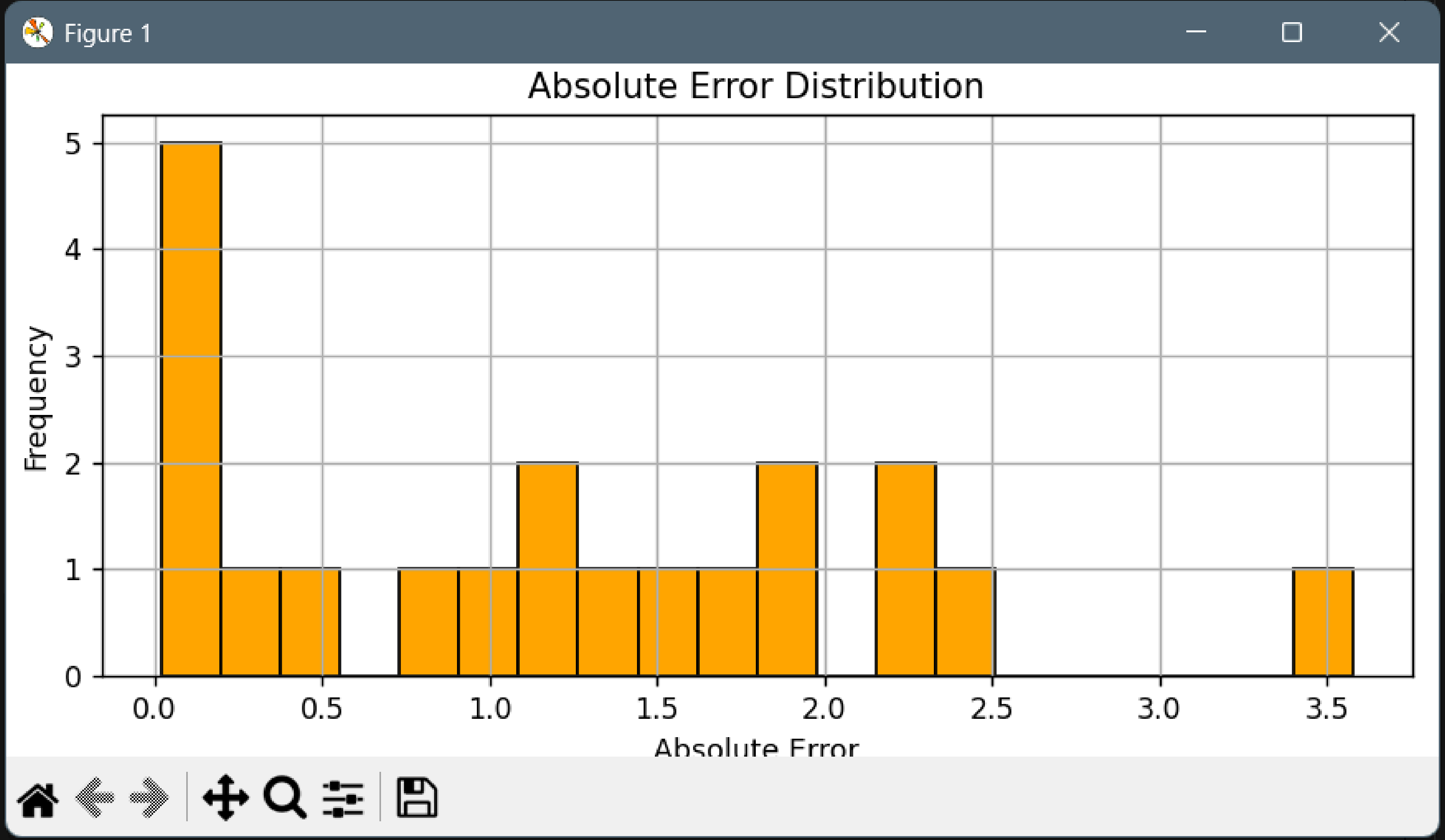
running gcn tasks for different tasks and datasets

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn synthetic_regression graph_regression
Running task: graph_regression using dataset: synthetic_regression
No saved model found. Training from scratch...
[GCN-GraphReg] Epoch 0 - Train MSE: 97.183635
[GCN-GraphReg] Epoch 10 - Train MSE: 5.703851
[GCN-GraphReg] Epoch 20 - Train MSE: 4.431348
[GCN-GraphReg] Epoch 30 - Train MSE: 5.535823
[GCN-GraphReg] Epoch 40 - Train MSE: 3.090857
[GCN-GraphReg] Epoch 50 - Train MSE: 3.097077
[GCN-GraphReg] Epoch 60 - Train MSE: 2.237914
[GCN-GraphReg] Epoch 70 - Train MSE: 1.971153
[GCN-GraphReg] Epoch 80 - Train MSE: 1.883828
[GCN-GraphReg] Epoch 90 - Train MSE: 2.248514
[GCN-GraphReg] Epoch 99 - Train MSE: 1.946905

Evaluation on Test Set:
MSE: 2.3075
MAE: 1.1903
R² Score: 0.3953

Sample Graph Regression:
Graph 17 → Predicted: 24.0982 | True: 23.8365 | Abs Error: 0.2618
Graph 13 → Predicted: 25.1579 | True: 27.5053 | Abs Error: 2.3475
Graph 18 → Predicted: 24.0629 | True: 22.1834 | Abs Error: 1.8795
Graph 2 → Predicted: 23.9181 | True: 24.0147 | Abs Error: 0.0966
Graph 6 → Predicted: 25.1342 | True: 24.9460 | Abs Error: 0.1882
```



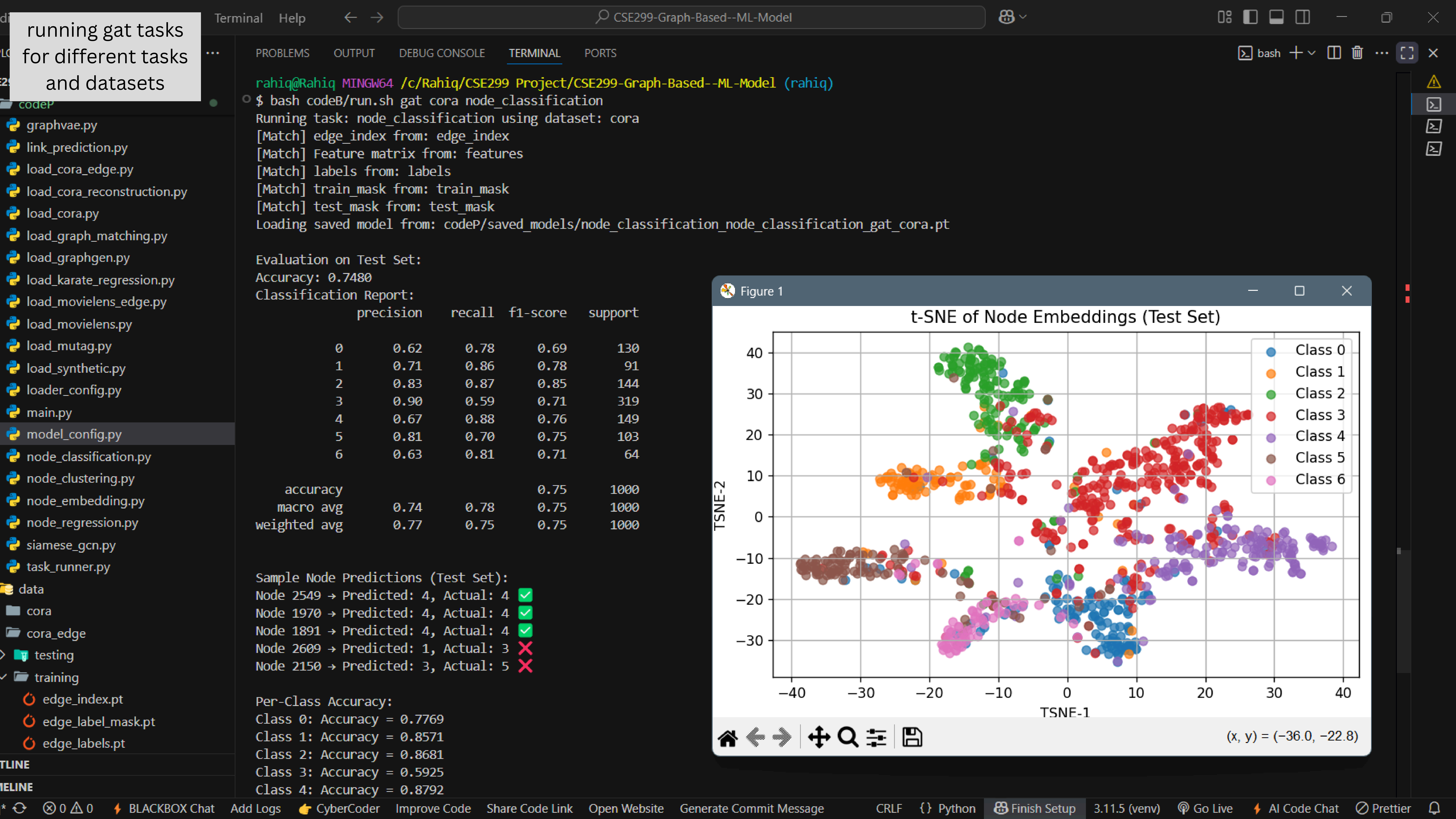
running gcn tasks for different tasks and datasets

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gcn synthetic_regression graph_regression
Running task: graph_regression using dataset: synthetic_regression
No saved model found. Training from scratch...
[GCN-GraphReg] Epoch 0 - Train MSE: 97.183635
[GCN-GraphReg] Epoch 10 - Train MSE: 5.703851
[GCN-GraphReg] Epoch 20 - Train MSE: 4.431348
[GCN-GraphReg] Epoch 30 - Train MSE: 5.535823
[GCN-GraphReg] Epoch 40 - Train MSE: 3.090857
[GCN-GraphReg] Epoch 50 - Train MSE: 3.097077
[GCN-GraphReg] Epoch 60 - Train MSE: 2.237914
[GCN-GraphReg] Epoch 70 - Train MSE: 1.971153
[GCN-GraphReg] Epoch 80 - Train MSE: 1.883828
[GCN-GraphReg] Epoch 90 - Train MSE: 2.248514
[GCN-GraphReg] Epoch 99 - Train MSE: 1.946905

Evaluation on Test Set:
MSE: 2.3075
MAE: 1.1903
R² Score: 0.3953

Sample Graph Regression:
Graph 17 → Predicted: 24.0982 | True: 23.8365 | Abs Error: 0.2618
Graph 13 → Predicted: 25.1579 | True: 27.5053 | Abs Error: 2.3475
Graph 18 → Predicted: 24.0629 | True: 22.1834 | Abs Error: 1.8795
Graph 2 → Predicted: 23.9181 | True: 24.0147 | Abs Error: 0.0966
Graph 6 → Predicted: 25.1342 | True: 24.9460 | Abs Error: 0.1882
```



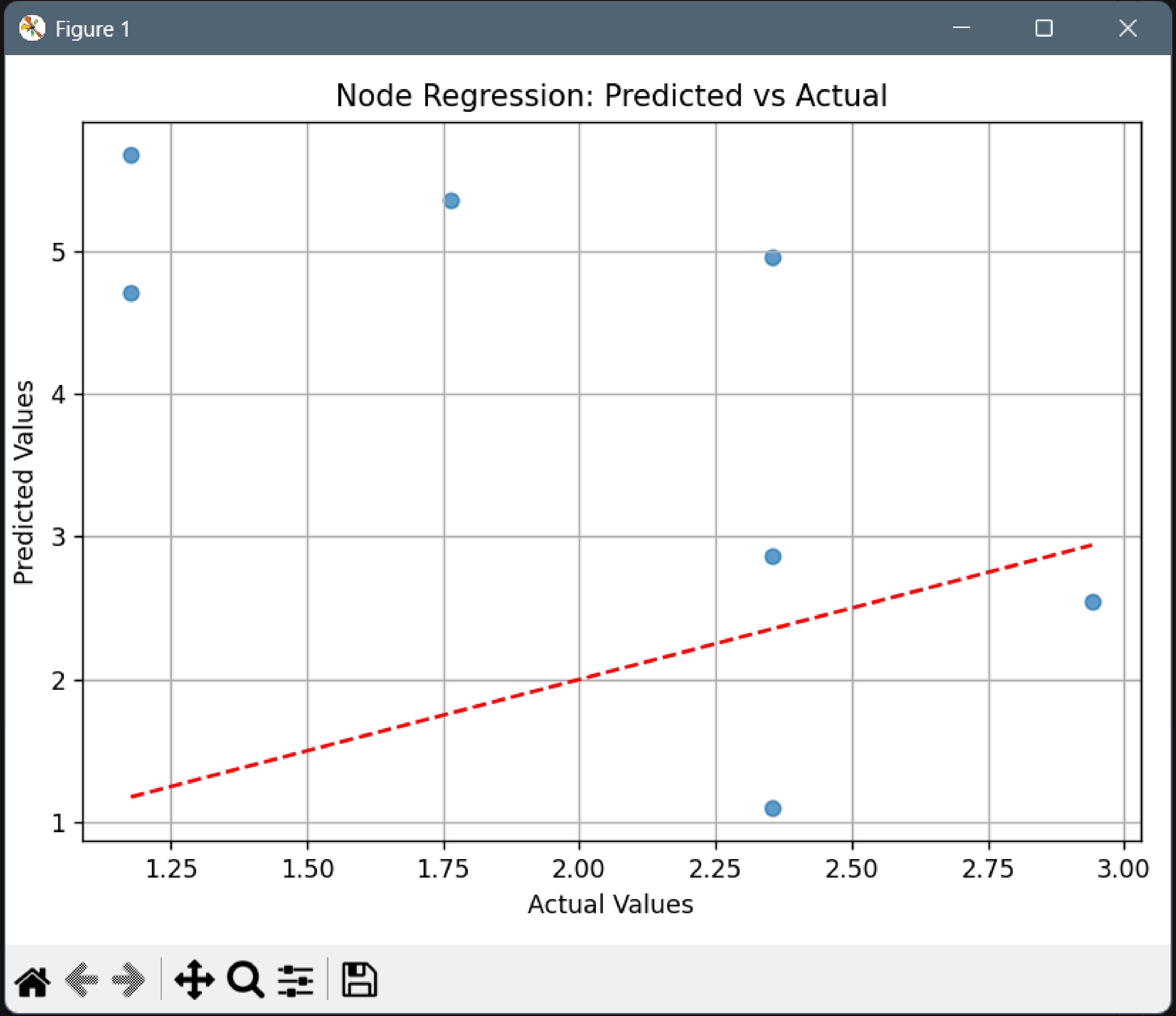
running gat tasks for different tasks and datasets

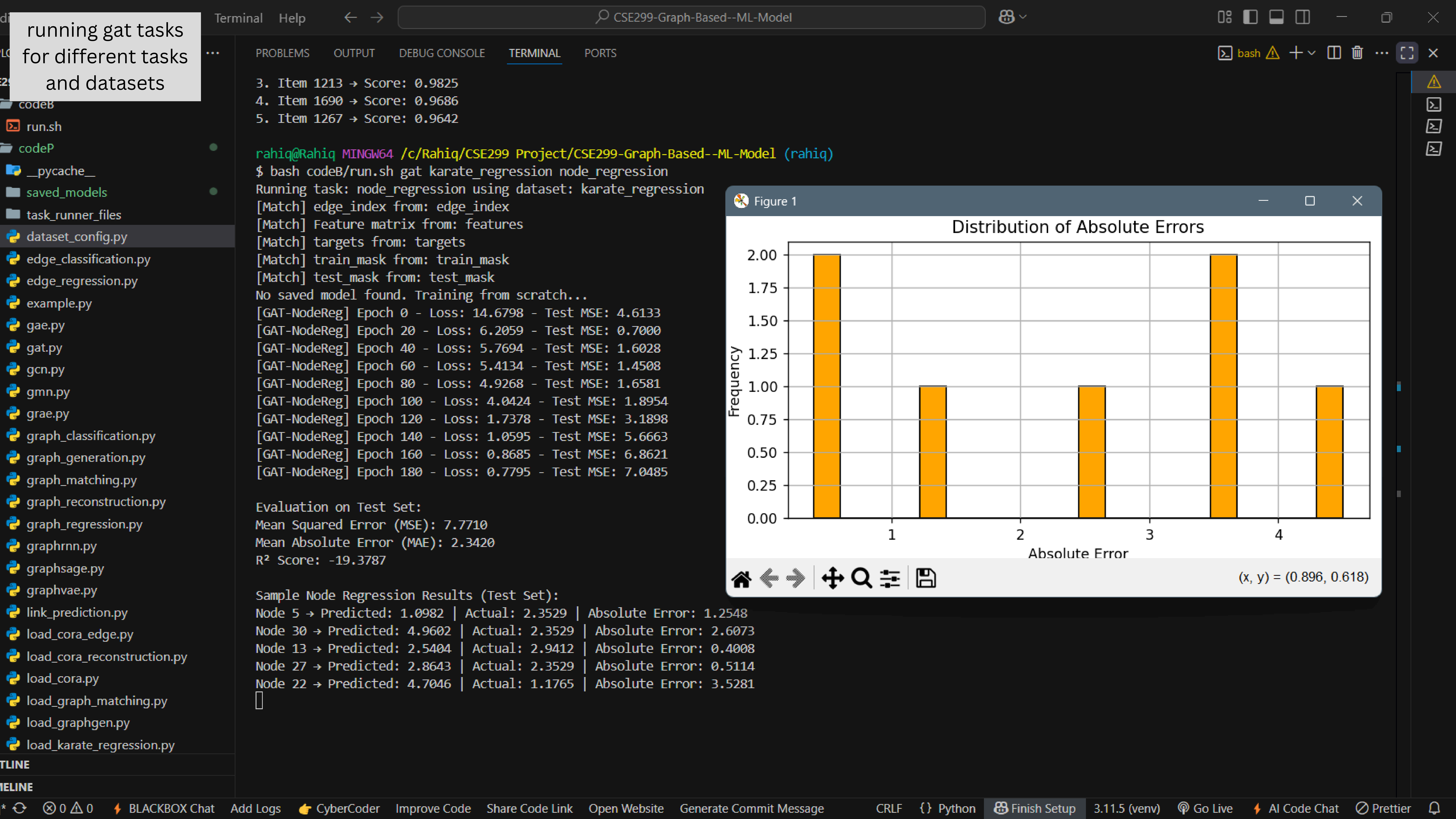
```
5. Item 1213 → Score: 0.9719

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat movielens link_prediction
Running task: link_prediction using dataset: movielens
[Match] Train edge_index from: edge_index
[Match] Feature matrix from: features
[Match] Test edge_index from: edge_index
Loading saved model from: codeP/saved_models/link_prediction_link_prediction_gat_movielens.pt

Sample Link Predictions:

Top recommendations for user 115:
1. Item 1200 → Score: 0.9917
2. Item 1690 → Score: 0.9887
3. Item 1230 → Score: 0.9840
4. Item 1300 → Score: 0.9794
5. Item 1255 → Score: 0.9785

Top recommendations for user 609:
1. Item 1690 → Score: 0.9930
2. Item 1200 → Score: 0.9925
3. Item 1187 → Score: 0.9795
4. Item 1230 → Score: 0.9787
5. Item 1300 → Score: 0.9684

Top recommendations for user 626:
1. Item 1236 → Score: 0.9922
2. Item 1187 → Score: 0.9909
3. Item 1213 → Score: 0.9825
4. Item 1690 → Score: 0.9686
5. Item 1267 → Score: 0.9642

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$
```


running gat tasks for different tasks and datasets
Terminal Help
CSE299-Graph-Based--ML-Model
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash
codeP
graphvae.py
link_prediction.py
load_cora_edge.py
load_cora_reconstruction.py
load_cora.py
load_graph_matching.py
load_graphgen.py
load_karate_regression.py
load_movielens_edge.py
load_movielens.py
load_mutag.py
load_synthetic.py
loader_config.py
main.py
model_config.py
node_classification.py
node_clustering.py
node_embedding.py
node_regression.py
siamese_gcn.py
task_runner.py
data
cora
cora_edge
testing
training
edge_index.pt
edge_label_mask.pt
edge_labels.pt
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat cora node_classification
Running task: node_classification using dataset: cora
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] labels from: labels
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
Loading saved model from: codeP/saved_models/node_classification_node_classification_gat_cora.pt
Evaluation on Test Set:
Accuracy: 0.7480
Classification Report:
precision recall f1-score support
0 0.62 0.78 0.69 130
1 0.71 0.86 0.78 91
2 0.83 0.87 0.85 144
3 0.90 0.59 0.71 319
4 0.67 0.88 0.76 149
5 0.81 0.70 0.75 103
6 0.63 0.81 0.71 64
accuracy 0.75 1000
macro avg 0.74 0.78 0.75 1000
weighted avg 0.77 0.75 0.75 1000
Sample Node Predictions (Test Set):
Node 2549 → Predicted: 4, Actual: 4
Node 1970 → Predicted: 4, Actual: 4
Node 1891 → Predicted: 4, Actual: 4
Node 2609 → Predicted: 1, Actual: 3
Node 2150 → Predicted: 3, Actual: 5
Per-Class Accuracy:
Class 0: Accuracy = 0.7769
Class 1: Accuracy = 0.8571
Class 2: Accuracy = 0.8681
Class 3: Accuracy = 0.5925
Class 4: Accuracy = 0.8792
Figure 1
t-SNE of Node Embeddings (Test Set)
Class 0
Class 1
Class 2
Class 3
Class 4
Class 5
Class 6
TSNE-2
TSNE-1
40 30 20 10 0 −10 −20 −30
−40 −30 −20 −10 0 10 20 30 40
(x, y) = (−36.0, −22.8)
TLINE
ELINE
0 0 BLACKBOX Chat Add Logs CyberCoder Improve Code Share Code Link Open Website Generate Commit Message CRLF {} Python Finish Setup 3.11.5 (venv) Go Live AI Code Chat Prettier

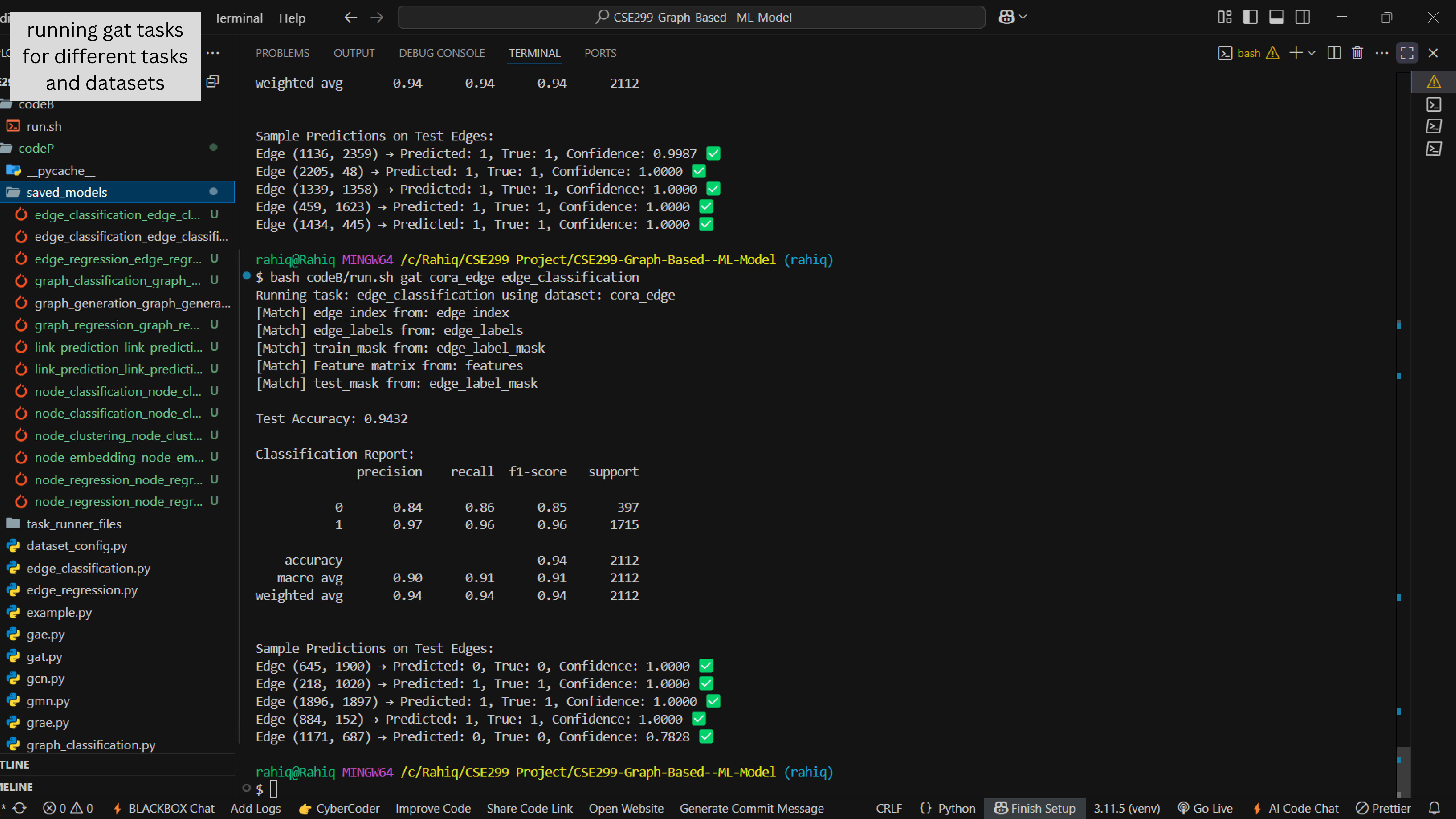
running gat tasks for different tasks and datasets

```
3. Item 1213 → Score: 0.9825
4. Item 1690 → Score: 0.9686
5. Item 1267 → Score: 0.9642

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat karate_regression node_regression
Running task: node_regression using dataset: karate_regression
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] targets from: targets
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
No saved model found. Training from scratch...
[GAT-NodeReg] Epoch 0 - Loss: 14.6798 - Test MSE: 4.6133
[GAT-NodeReg] Epoch 20 - Loss: 6.2059 - Test MSE: 0.7000
[GAT-NodeReg] Epoch 40 - Loss: 5.7694 - Test MSE: 1.6028
[GAT-NodeReg] Epoch 60 - Loss: 5.4134 - Test MSE: 1.4508
[GAT-NodeReg] Epoch 80 - Loss: 4.9268 - Test MSE: 1.6581
[GAT-NodeReg] Epoch 100 - Loss: 4.0424 - Test MSE: 1.8954
[GAT-NodeReg] Epoch 120 - Loss: 1.7378 - Test MSE: 3.1898
[GAT-NodeReg] Epoch 140 - Loss: 1.0595 - Test MSE: 5.6663
[GAT-NodeReg] Epoch 160 - Loss: 0.8685 - Test MSE: 6.8621
[GAT-NodeReg] Epoch 180 - Loss: 0.7795 - Test MSE: 7.0485

Evaluation on Test Set:
Mean Squared Error (MSE): 7.7710
Mean Absolute Error (MAE): 2.3420
R² Score: -19.3787

Sample Node Regression Results (Test Set):
Node 5 → Predicted: 1.0982 | Actual: 2.3529 | Absolute Error: 1.2548
Node 30 → Predicted: 4.9602 | Actual: 2.3529 | Absolute Error: 2.6073
Node 13 → Predicted: 2.5404 | Actual: 2.9412 | Absolute Error: 0.4008
Node 27 → Predicted: 2.8643 | Actual: 2.3529 | Absolute Error: 0.5114
Node 22 → Predicted: 4.7046 | Actual: 1.1765 | Absolute Error: 3.5281
```




running gat tasks for different tasks and datasets
CSE299-Graph-Based--ML-Model
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
3. Item 1213 → Score: 0.9825
4. Item 1690 → Score: 0.9686
5. Item 1267 → Score: 0.9642
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat karate_regression node_regression
Running task: node_regression using dataset: karate_regression
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] targets from: targets
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
No saved model found. Training from scratch...
[GAT-NodeReg] Epoch 0 - Loss: 14.6798 - Test MSE: 4.6133
[GAT-NodeReg] Epoch 20 - Loss: 6.2059 - Test MSE: 0.7000
[GAT-NodeReg] Epoch 40 - Loss: 5.7694 - Test MSE: 1.6028
[GAT-NodeReg] Epoch 60 - Loss: 5.4134 - Test MSE: 1.4508
[GAT-NodeReg] Epoch 80 - Loss: 4.9268 - Test MSE: 1.6581
[GAT-NodeReg] Epoch 100 - Loss: 4.0424 - Test MSE: 1.8954
[GAT-NodeReg] Epoch 120 - Loss: 1.7378 - Test MSE: 3.1898
[GAT-NodeReg] Epoch 140 - Loss: 1.0595 - Test MSE: 5.6663
[GAT-NodeReg] Epoch 160 - Loss: 0.8685 - Test MSE: 6.8621
[GAT-NodeReg] Epoch 180 - Loss: 0.7795 - Test MSE: 7.0485
Evaluation on Test Set:
Mean Squared Error (MSE): 7.7710
Mean Absolute Error (MAE): 2.3420
R² Score: -19.3787
Sample Node Regression Results (Test Set):
Node 5 → Predicted: 1.0982 | Actual: 2.3529 | Absolute Error: 1.2548
Node 30 → Predicted: 4.9602 | Actual: 2.3529 | Absolute Error: 2.6073
Node 13 → Predicted: 2.5404 | Actual: 2.9412 | Absolute Error: 0.4008
Node 27 → Predicted: 2.8643 | Actual: 2.3529 | Absolute Error: 0.5114
Node 22 → Predicted: 4.7046 | Actual: 1.1765 | Absolute Error: 3.5281
Figure 1
Distribution of Absolute Errors
Frequency
Absolute Error
(x, y) = (0.896, 0.618)
codeB
run.sh
codeP
__pycache__
saved_models
task_runner_files
dataset_config.py
edge_classification.py
edge_regression.py
example.py
gae.py
gat.py
gcn.py
gmn.py
grae.py
graph_classification.py
graph_generation.py
graph_matching.py
graph_reconstruction.py
graph_regression.py
graphrnn.py
graphsage.py
graphvae.py
link_prediction.py
load_cora_edge.py
load_cora_reconstruction.py
load_cora.py
load_graph_matching.py
load_graphgen.py
load_karate_regression.py

running gat tasks for different tasks and datasets



running gat tasks for different tasks and datasets

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat movielens_edge  edge_regression
Running task: edge_regression using dataset: movielens_edge
[Match] edge_index from: edge_index
[Match] edge_ratings from: edge_ratings
[Match] train_mask from: edge_rating_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_rating_mask
[GAT-EdgeReg] Epoch 0 - Loss: 14.5650 - Test MSE: 14.5173
[GAT-EdgeReg] Epoch 20 - Loss: 9.1524 - Test MSE: 9.1149
[GAT-EdgeReg] Epoch 40 - Loss: 1.3019 - Test MSE: 1.3010
[GAT-EdgeReg] Epoch 60 - Loss: 1.3112 - Test MSE: 1.3115
[GAT-EdgeReg] Epoch 80 - Loss: 1.2142 - Test MSE: 1.2116
[GAT-EdgeReg] Epoch 100 - Loss: 1.1951 - Test MSE: 1.1924
[GAT-EdgeReg] Epoch 120 - Loss: 1.1753 - Test MSE: 1.1730
[GAT-EdgeReg] Epoch 140 - Loss: 1.1573 - Test MSE: 1.1552
[GAT-EdgeReg] Epoch 160 - Loss: 1.1382 - Test MSE: 1.1364
[GAT-EdgeReg] Epoch 180 - Loss: 1.1098 - Test MSE: 1.1095

Mean Squared Error (MSE): 1.0918
R² Score: 0.1364

Sample Predictions:
Edge (248, 1123) → Predicted: 3.63, Actual: 3.00, Error: 0.63
Edge (1127, 200) → Predicted: 3.67, Actual: 5.00, Error: 1.33
Edge (1984, 654) → Predicted: 2.97, Actual: 2.00, Error: 0.97
Edge (650, 1264) → Predicted: 3.46, Actual: 3.00, Error: 0.46
Edge (503, 1979) → Predicted: 3.18, Actual: 1.00, Error: 2.18

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$
```

running gat tasks for different tasks and datasets

```
        "module": {
            "link_prediction": link_prediction_gat,
            "node_classification": node_classification_gat,
            "node_regression": node_regression_gat,
```

```
valid_datasets = {
    "node_classification": ["cora"],
    "node_regression": ["karate_regression"],
    "link_prediction": ["movielens"],
    "edge_classification": ["cora_edge"],
```

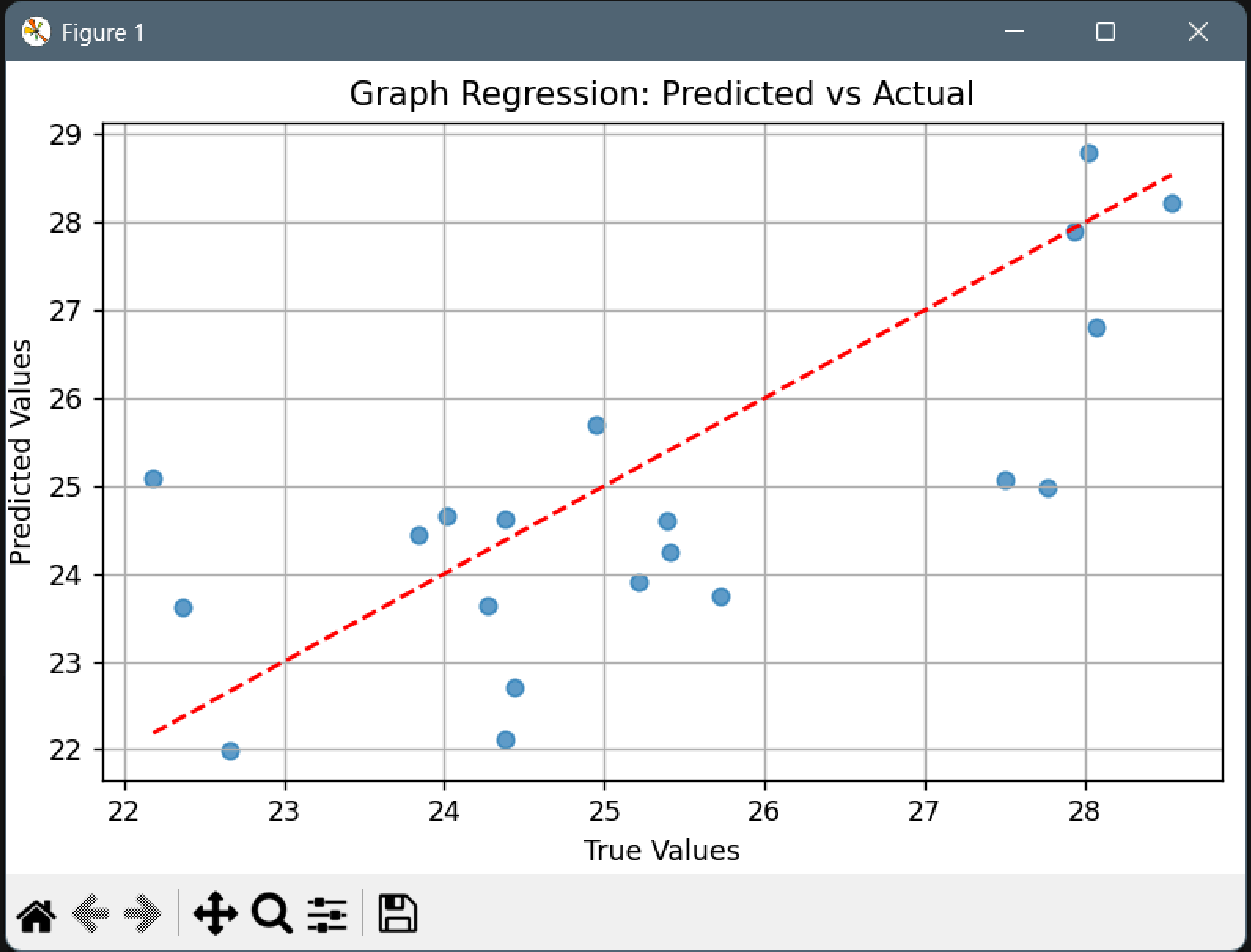
```
Visualizing node embeddings with t-SNE (Test Set)...

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat cora node_clustering
Running task: node_clustering using dataset: cora
No saved model found. Training from scratch...
[GAT-NodeEmbed] Epoch 0 - MSE Loss: 0.0137
[GAT-NodeEmbed] Epoch 20 - MSE Loss: 0.0106
[GAT-NodeEmbed] Epoch 40 - MSE Loss: 0.0095
[GAT-NodeEmbed] Epoch 60 - MSE Loss: 0.0088
[GAT-NodeEmbed] Epoch 80 - MSE Loss: 0.0083

Total Clusters: 7

Cluster Size Distribution:
  Cluster 2: 520 nodes
  Cluster 1: 557 nodes
  Cluster 4: 920 nodes
  Cluster 6: 312 nodes
  Cluster 5: 393 nodes
  Cluster 0: 1 nodes
  Cluster 3: 5 nodes

Sample Node Clusters:
Node 2082 → Cluster 4
Node 1032 → Cluster 4
Node 750 → Cluster 4
Node 1482 → Cluster 4
Node 233 → Cluster 4

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$
```

running gat tasks for different tasks and datasets

```
Node 233 → Cluster 4

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat cora node_embedding
Running task: node_embedding using dataset: cora
No saved model found. Training from scratch...
[GAT-NodeEmbed] Epoch 0 - MSE Loss: 0.0137
[GAT-NodeEmbed] Epoch 20 - MSE Loss: 0.0106
[GAT-NodeEmbed] Epoch 40 - MSE Loss: 0.0096
[GAT-NodeEmbed] Epoch 60 - MSE Loss: 0.0089
[GAT-NodeEmbed] Epoch 80 - MSE Loss: 0.0084

Sample Node Embeddings:
Node 0 → Embedding: [0.0067, 0.0330, 0.0790, -0.0038, -0.0481, ...]
Node 1 → Embedding: [-0.0192, -0.0088, 0.0948, 0.0563, 0.1226, ...]
Node 2 → Embedding: [0.0041, -0.0007, 0.0441, 0.0277, 0.0589, ...]
Node 3 → Embedding: [-0.0036, 0.0033, -0.1214, 0.0114, -0.0461, ...]
Node 4 → Embedding: [-0.0083, 0.1102, -0.0547, 0.0874, 0.0038, ...]

Embedding Statistics:
Mean: 0.0129
Std Dev: 0.0673
Min: -0.3890
Max: 1.5086

Cosine Similarity Between Sample Nodes:
Node 0 vs Node 1: 0.3207
Node 0 vs Node 2: 0.3361
Node 0 vs Node 3: 0.0785
Node 0 vs Node 4: 0.2491
Node 1 vs Node 2: 0.7050
Node 1 vs Node 3: 0.1858
Node 1 vs Node 4: 0.2944
Node 2 vs Node 3: 0.2607
Node 2 vs Node 4: 0.2876
Node 3 vs Node 4: 0.0639
```

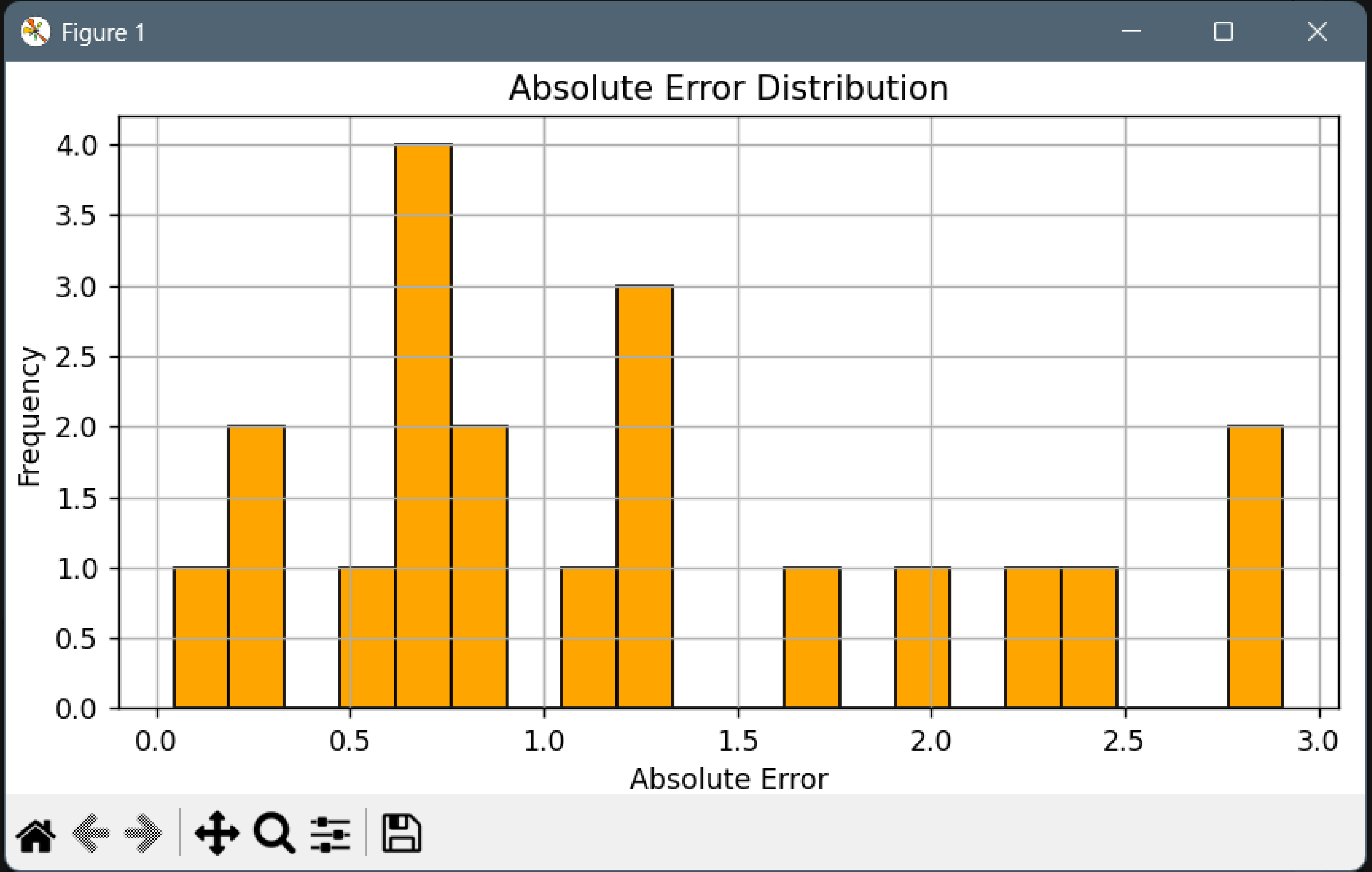
running gat tasks for different tasks and datasets

```
Node 0 vs Node 4: 0.2491
Node 1 vs Node 2: 0.7050
Node 1 vs Node 3: 0.1858
Node 1 vs Node 4: 0.2944
Node 2 vs Node 3: 0.2607
Node 2 vs Node 4: 0.2876
Node 3 vs Node 4: 0.0639

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat synthetic_regression graph_regression
Running task: graph_regression using dataset: synthetic_regression
No saved model found. Training from scratch...
[GAT-GraphReg] Epoch 0 - Train MSE: 43.769165
[GAT-GraphReg] Epoch 10 - Train MSE: 4.442048
[GAT-GraphReg] Epoch 20 - Train MSE: 2.307533
[GAT-GraphReg] Epoch 30 - Train MSE: 2.729299
[GAT-GraphReg] Epoch 40 - Train MSE: 1.827737
[GAT-GraphReg] Epoch 50 - Train MSE: 2.056783
[GAT-GraphReg] Epoch 60 - Train MSE: 1.245393
[GAT-GraphReg] Epoch 70 - Train MSE: 1.990289
[GAT-GraphReg] Epoch 80 - Train MSE: 1.284598
[GAT-GraphReg] Epoch 90 - Train MSE: 0.932240
[GAT-GraphReg] Epoch 99 - Train MSE: 1.143649

Evaluation on Test Set:
MSE: 2.1997
MAE: 1.2271
R² Score: 0.4236

Sample Graph Regression:
Graph 3 → Predicted: 24.6267 | True: 24.3823 | Abs Error: 0.2444
Graph 19 → Predicted: 23.6177 | True: 22.3638 | Abs Error: 1.2539
Graph 16 → Predicted: 22.7005 | True: 24.4355 | Abs Error: 1.7350
Graph 18 → Predicted: 25.0921 | True: 22.1834 | Abs Error: 2.9087
Graph 10 → Predicted: 23.9009 | True: 25.2133 | Abs Error: 1.3124
```

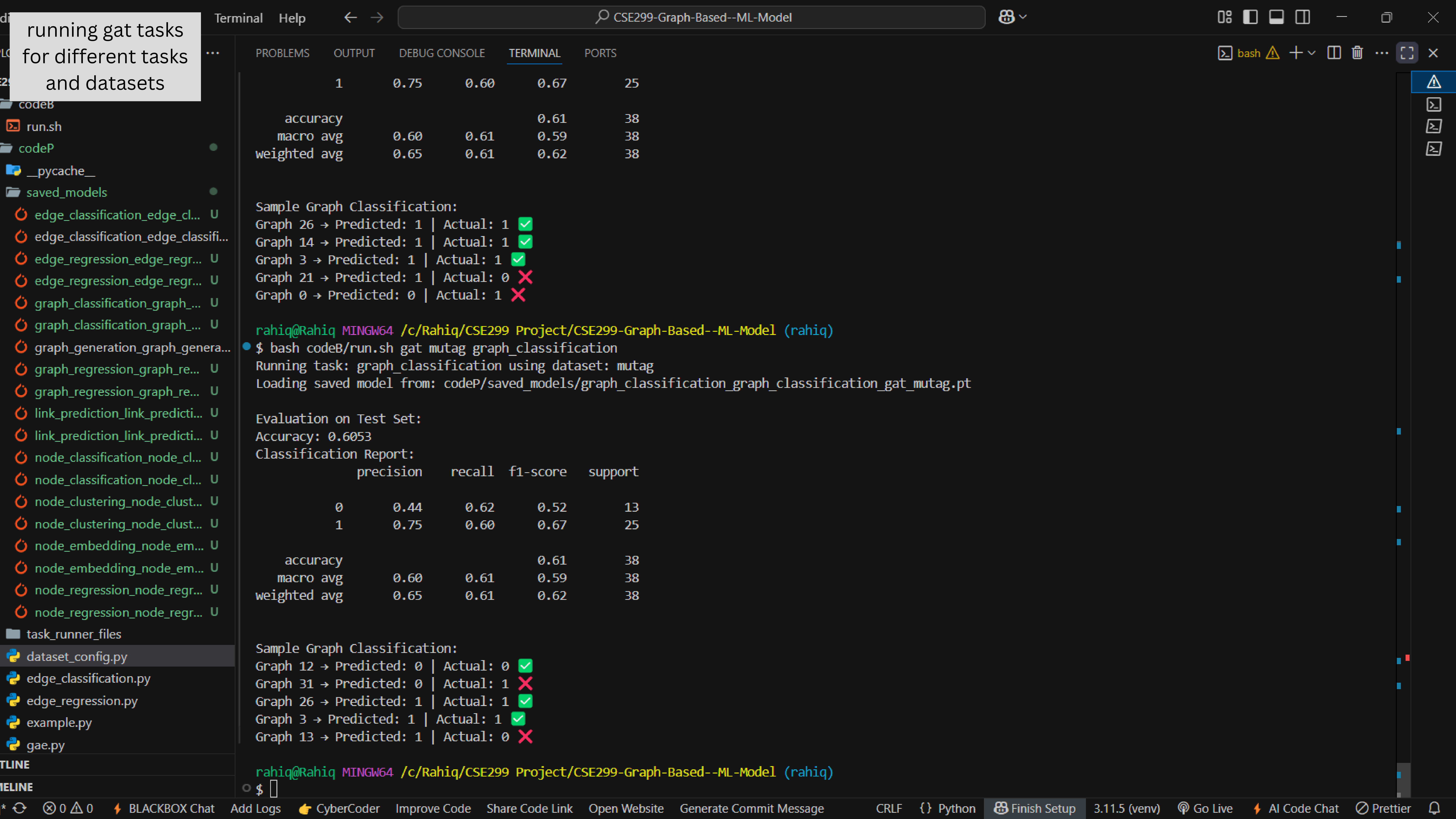


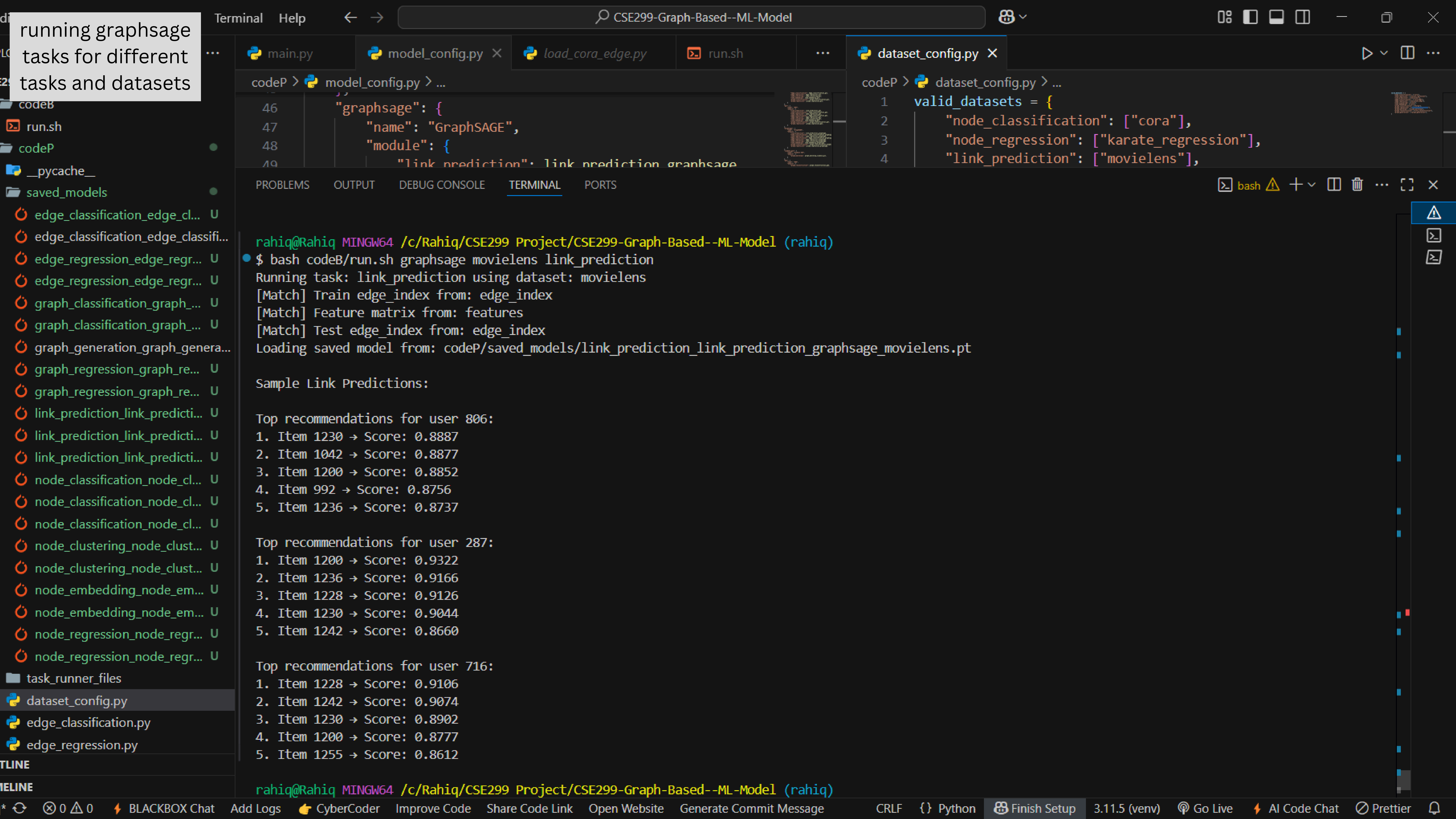
running gat tasks for different tasks and datasets

```
Node 0 vs Node 4: 0.2491
Node 1 vs Node 2: 0.7050
Node 1 vs Node 3: 0.1858
Node 1 vs Node 4: 0.2944
Node 2 vs Node 3: 0.2607
Node 2 vs Node 4: 0.2876
Node 3 vs Node 4: 0.0639

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat synthetic_regression graph_regression
Running task: graph_regression using dataset: synthetic_regression
No saved model found. Training from scratch...
[GAT-GraphReg] Epoch 0 - Train MSE: 43.769165
[GAT-GraphReg] Epoch 10 - Train MSE: 4.442048
[GAT-GraphReg] Epoch 20 - Train MSE: 2.307533
[GAT-GraphReg] Epoch 30 - Train MSE: 2.729299
[GAT-GraphReg] Epoch 40 - Train MSE: 1.827737
[GAT-GraphReg] Epoch 50 - Train MSE: 2.056783
[GAT-GraphReg] Epoch 60 - Train MSE: 1.245393
[GAT-GraphReg] Epoch 70 - Train MSE: 1.990289
[GAT-GraphReg] Epoch 80 - Train MSE: 1.284598
[GAT-GraphReg] Epoch 90 - Train MSE: 0.932240
[GAT-GraphReg] Epoch 99 - Train MSE: 1.143649

Evaluation on Test Set:
MSE: 2.1997
MAE: 1.2271
R² Score: 0.4236

Sample Graph Regression:
Graph 3 → Predicted: 24.6267 | True: 24.3823 | Abs Error: 0.2444
Graph 19 → Predicted: 23.6177 | True: 22.3638 | Abs Error: 1.2539
Graph 16 → Predicted: 22.7005 | True: 24.4355 | Abs Error: 1.7350
Graph 18 → Predicted: 25.0921 | True: 22.1834 | Abs Error: 2.9087
Graph 10 → Predicted: 23.9009 | True: 25.2133 | Abs Error: 1.3124
```



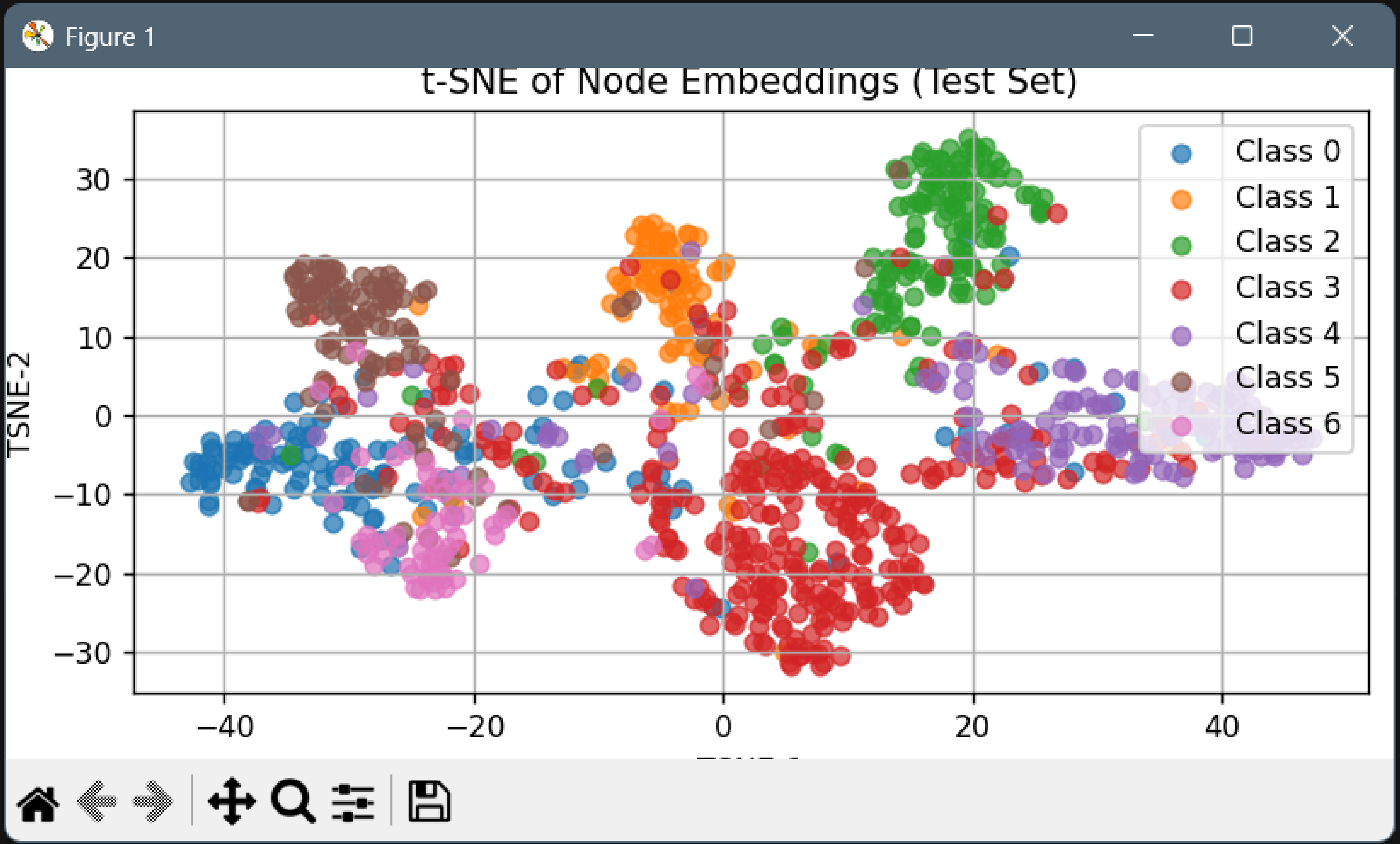
running gat tasks for different tasks and datasets




running graphsage tasks for different tasks and datasets
CSE299-Graph-Based--ML-Model
main.py
model_config.py
load_cora_edge.py
run.sh
dataset_config.py
"graphsage": {
"name": "GraphSAGE",
"module": {
valid_datasets = {
"node_classification": ["cora"],
"node_regression": ["karate_regression"],
"link_prediction": ["movielens"],
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh graphsage movielens link_prediction
Running task: link_prediction using dataset: movielens
[Match] Train edge_index from: edge_index
[Match] Feature matrix from: features
[Match] Test edge_index from: edge_index
Loading saved model from: codeP/saved_models/link_prediction_link_prediction_graphsage_movielens.pt
Sample Link Predictions:
Top recommendations for user 806:
1. Item 1230 → Score: 0.8887
2. Item 1042 → Score: 0.8877
3. Item 1200 → Score: 0.8852
4. Item 992 → Score: 0.8756
5. Item 1236 → Score: 0.8737
Top recommendations for user 287:
1. Item 1200 → Score: 0.9322
2. Item 1236 → Score: 0.9166
3. Item 1228 → Score: 0.9126
4. Item 1230 → Score: 0.9044
5. Item 1242 → Score: 0.8660
Top recommendations for user 716:
1. Item 1228 → Score: 0.9106
2. Item 1242 → Score: 0.9074
3. Item 1230 → Score: 0.8902
4. Item 1200 → Score: 0.8777
5. Item 1255 → Score: 0.8612
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)

running graphsage tasks for different tasks and datasets

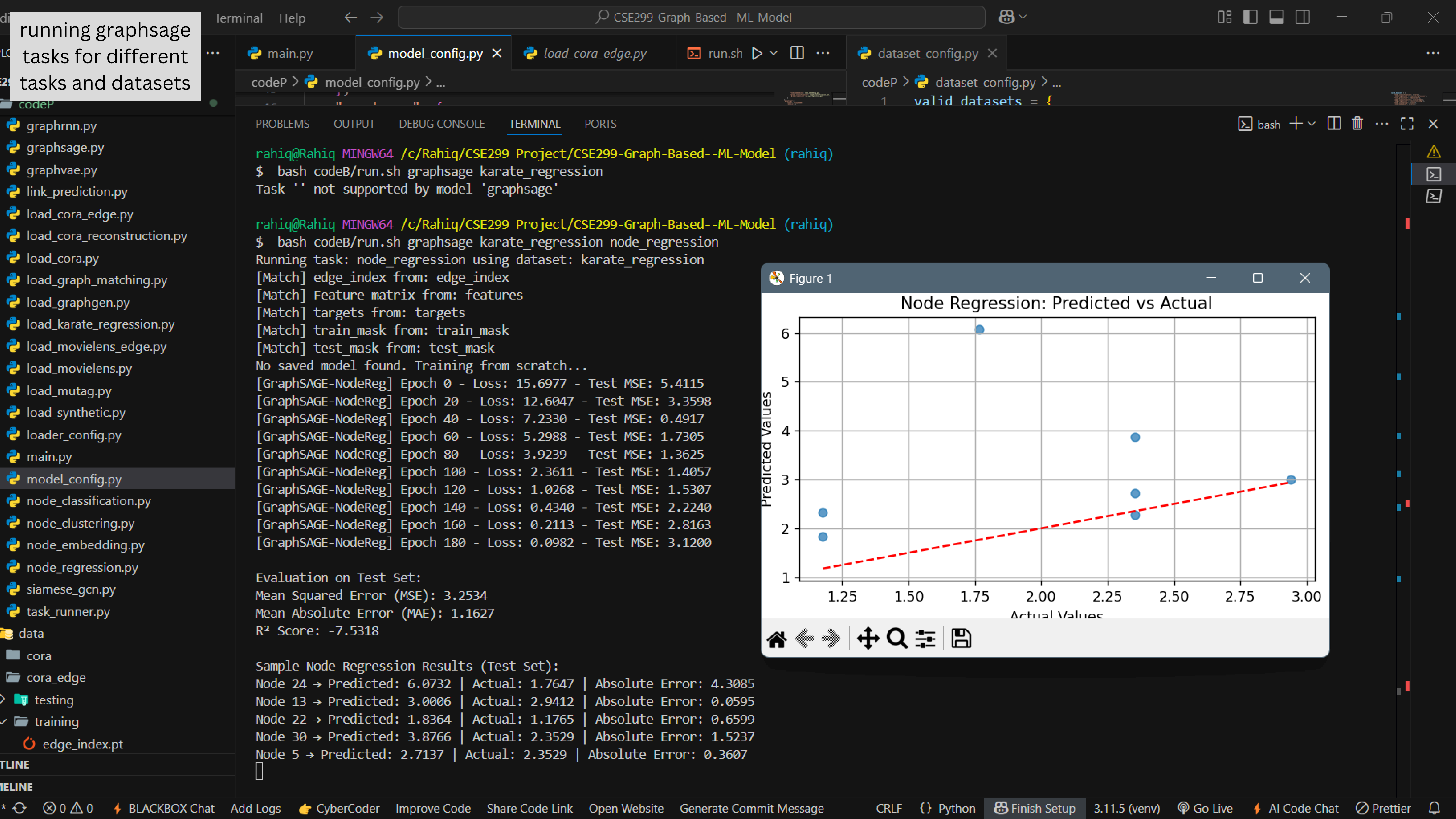
```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh graphsage cora node_classification
Running task: node_classification using dataset: cora
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] labels from: labels
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
Loading saved model from: codeP/saved_models/node_classification_node_classification_graphsage_cora.pt

Evaluation on Test Set:
Accuracy: 0.7510
Classification Report:
              precision    recall  f1-score   support

           0       0.71      0.61      0.65       130
           1       0.66      0.85      0.74        91
           2       0.76      0.89      0.82       144
           3       0.90      0.66      0.76       319
           4       0.75      0.87      0.81       149
           5       0.80      0.74      0.77       103
           6       0.50      0.81      0.62        64

    accuracy                           0.75      1000
   macro avg       0.73      0.77      0.74      1000
weighted avg       0.78      0.75      0.75      1000


Sample Node Predictions (Test Set):
Node 2207 → Predicted: 3, Actual: 3 ✅
Node 1830 → Predicted: 2, Actual: 2 ✅
Node 1756 → Predicted: 2, Actual: 2 ✅
Node 1933 → Predicted: 3, Actual: 3 ✅
Node 2027 → Predicted: 0, Actual: 3 ❌

Per-Class Accuracy:
Class 0: Accuracy = 0.6077
Class 1: Accuracy = 0.8462
Class 2: Accuracy = 0.8889
Class 3: Accuracy = 0.6552
Class 4: Accuracy = 0.8725
```



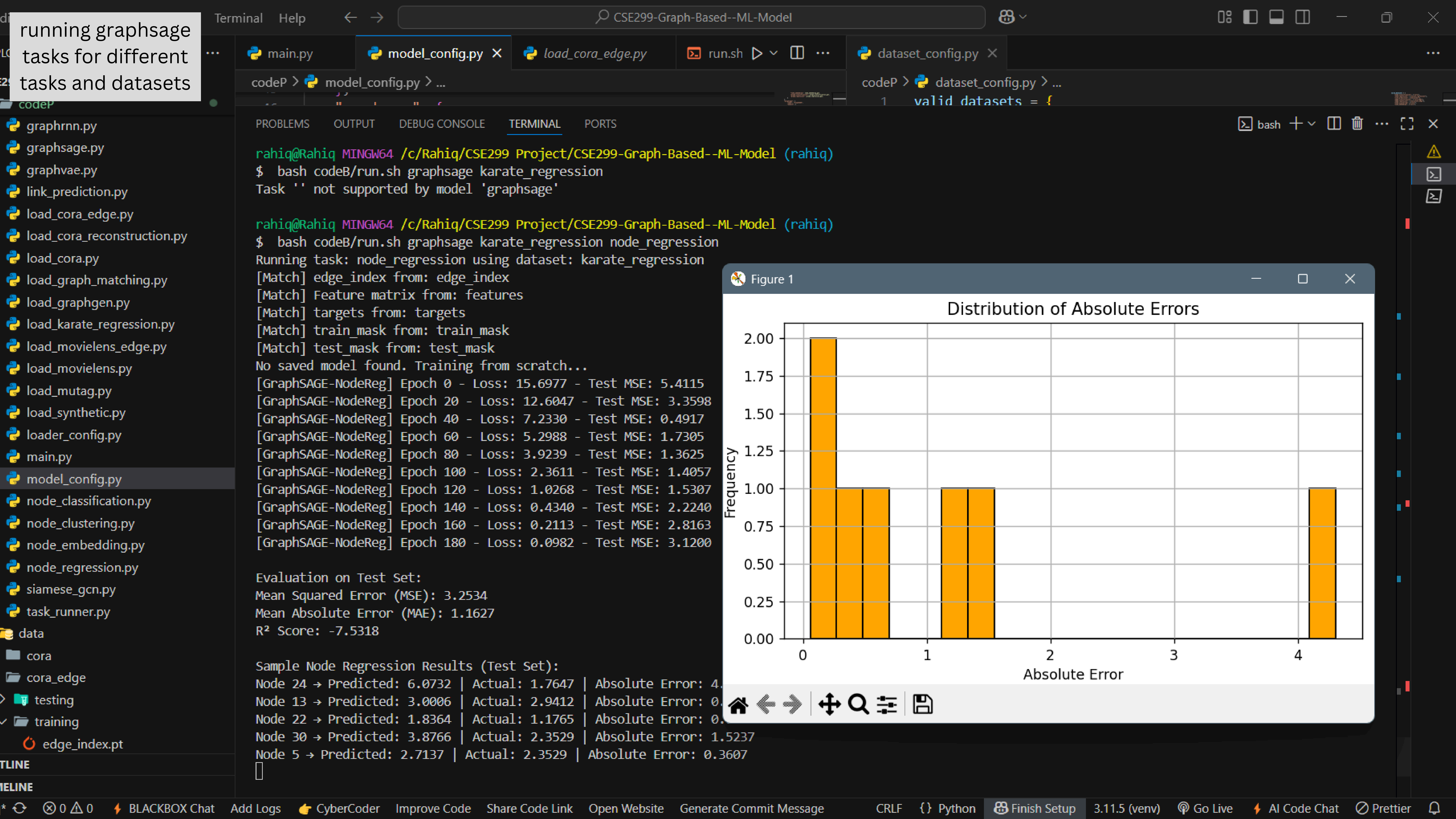
running graphsage tasks for different tasks and datasets

```
Edge (1976, 502) → Predicted: 1, True: 0, Confidence: 0.9996
Edge (487, 1773) → Predicted: 1, True: 1, Confidence: 1.0000

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh graphsage cora_edge edge_classification
Running task: edge_classification using dataset: cora_edge
[Match] edge_index from: edge_index
[Match] edge_labels from: edge_labels
[Match] train_mask from: edge_label_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_label_mask

Test Accuracy: 0.9238

Classification Report:
              precision    recall  f1-score   support

           0       0.78      0.83      0.80       397
           1       0.96      0.95      0.95      1715

    accuracy                           0.92      2112
   macro avg       0.87      0.89      0.88      2112
weighted avg       0.93      0.92      0.92      2112


Sample Predictions on Test Edges:
Edge (108, 1647) → Predicted: 1, True: 1, Confidence: 1.0000
Edge (306, 294) → Predicted: 1, True: 1, Confidence: 0.9778
Edge (306, 236) → Predicted: 1, True: 1, Confidence: 0.9740
Edge (1331, 1810) → Predicted: 1, True: 1, Confidence: 1.0000
Edge (2639, 917) → Predicted: 1, True: 1, Confidence: 1.0000

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$
```

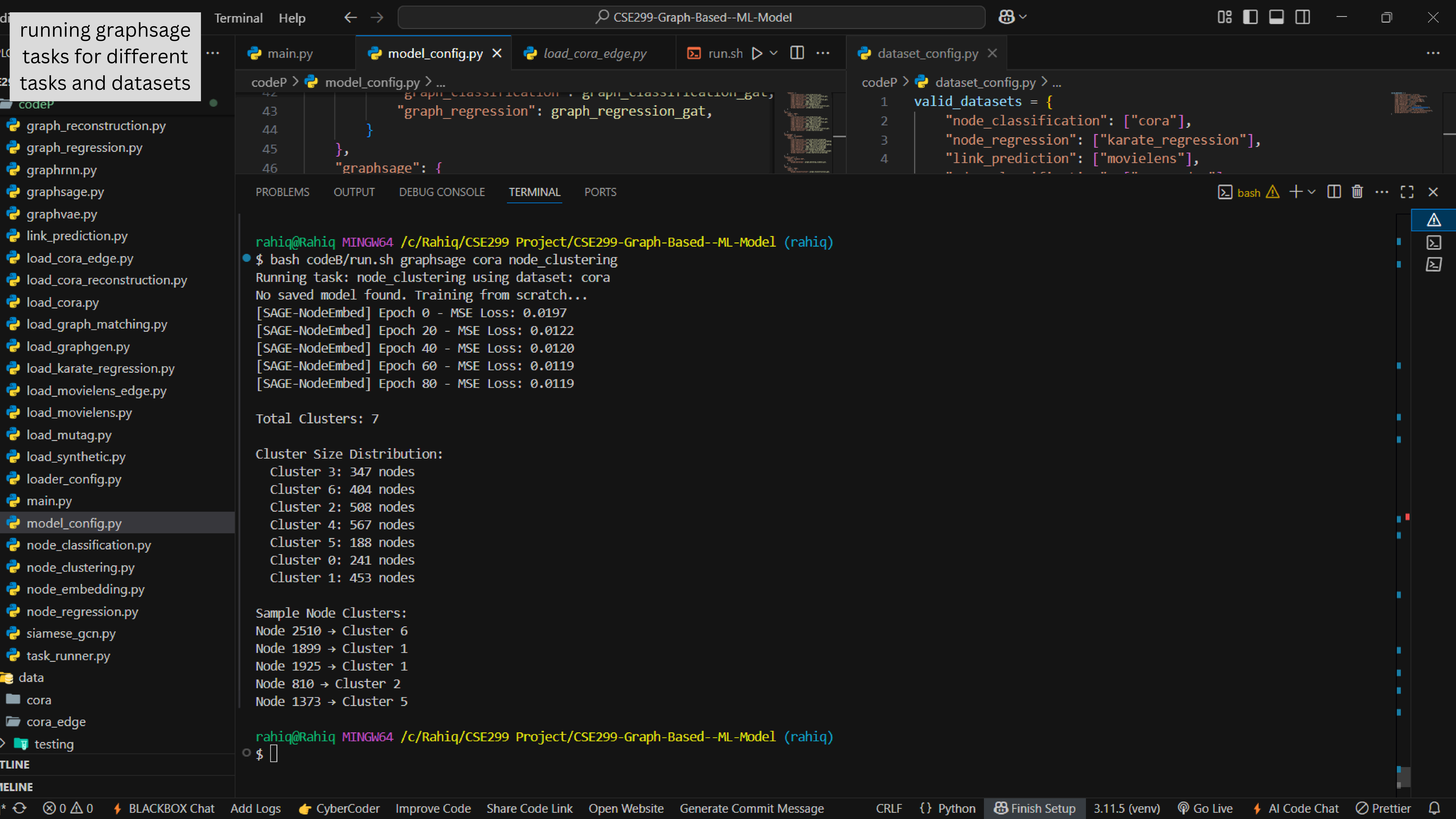
running graphsage tasks for different tasks and datasets



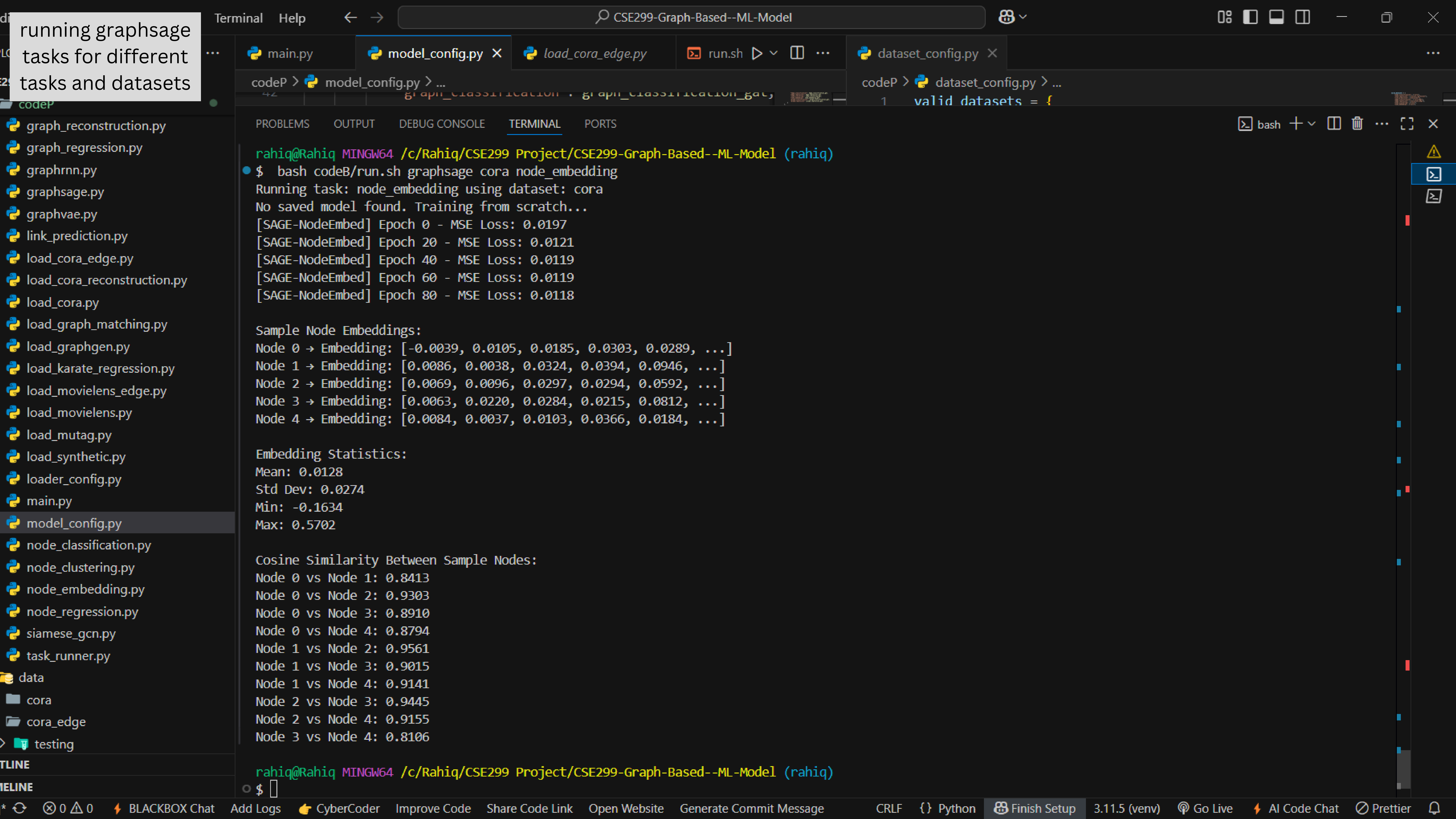
running graphsage tasks for different tasks and datasets

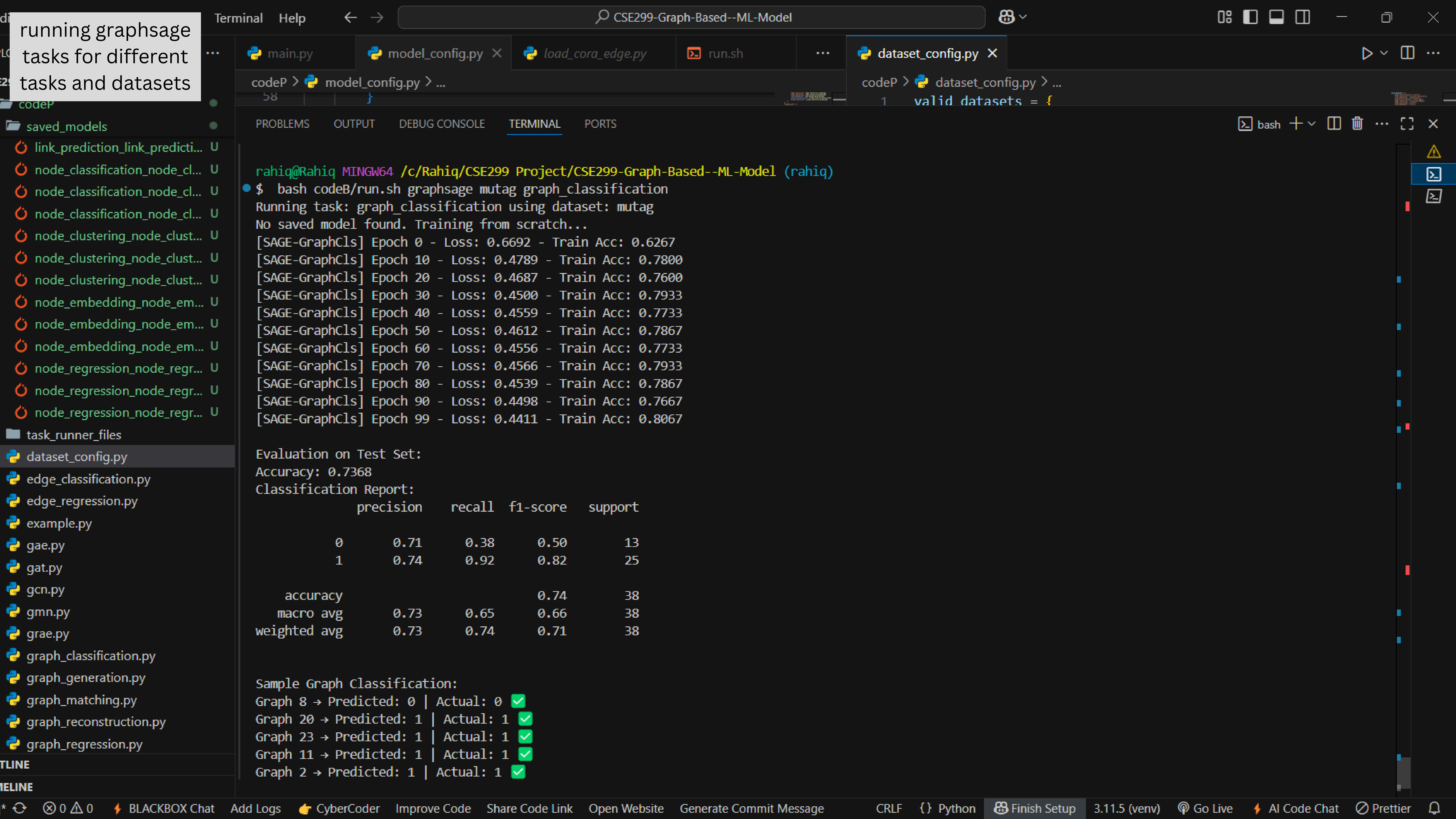


running graphsage tasks for different tasks and datasets



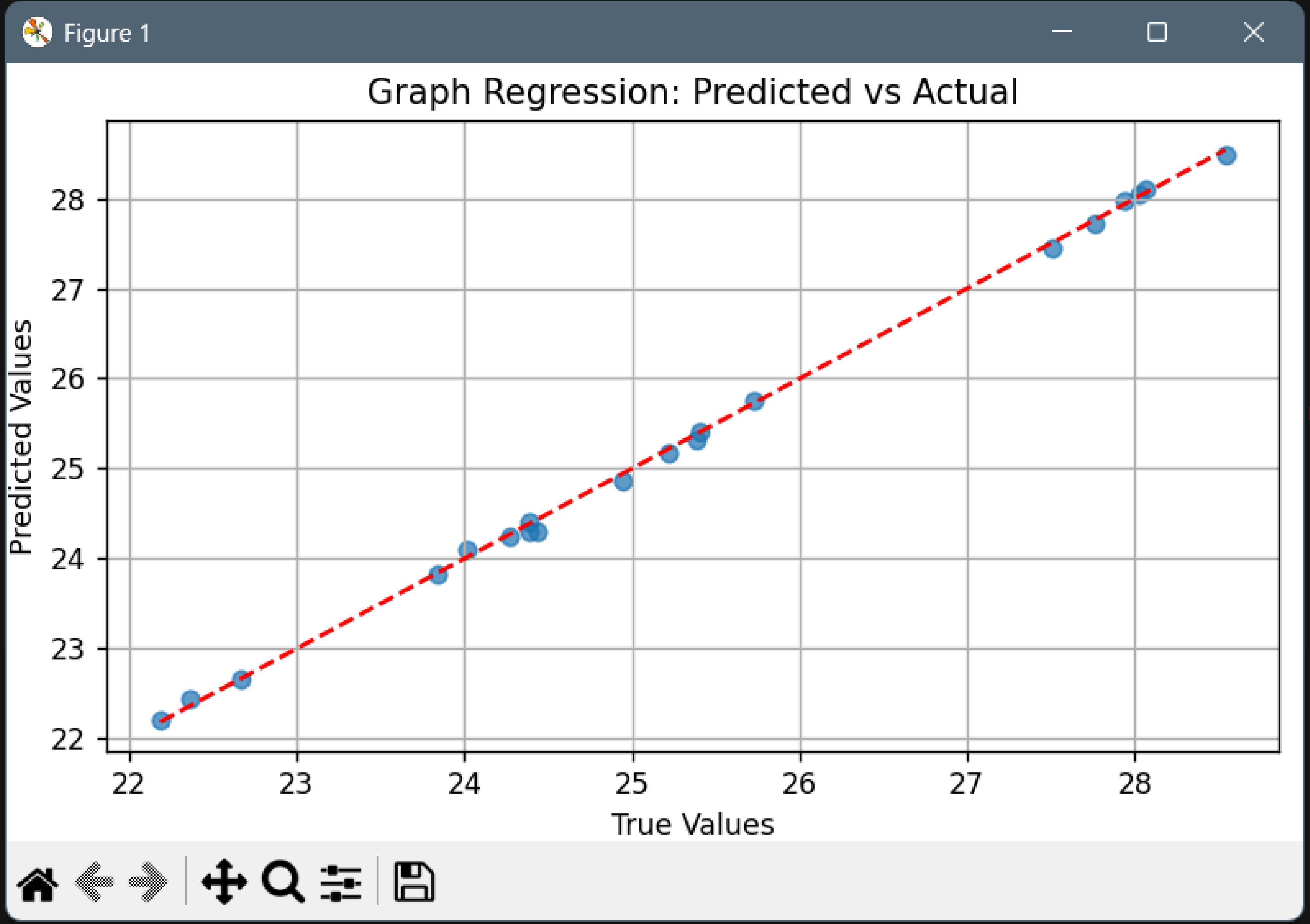
running graphsage tasks for different tasks and datasets



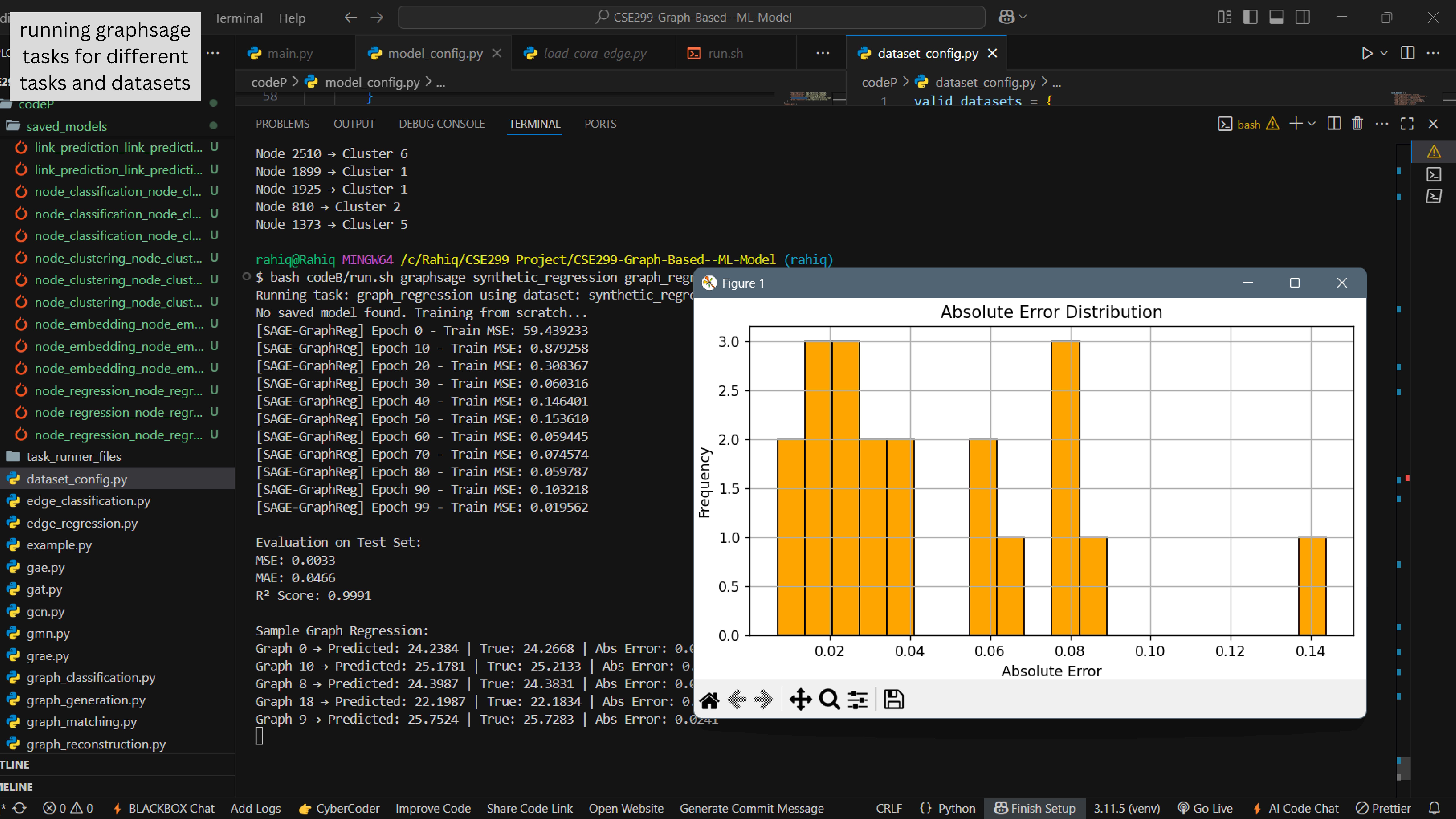

running graphsage tasks for different tasks and datasets
Terminal
Help
CSE299-Graph-Based--ML-Model
main.py
model_config.py
load_cora_edge.py
run.sh
dataset_config.py
codeP > model_config.py > ...
codeP > dataset_config.py > ...
valid_datasets = {
codeP
saved_models
link_prediction_link_predicti...
node_classification_node_cl...
node_classification_node_cl...
node_classification_node_cl...
node_clustering_node_clust...
node_clustering_node_clust...
node_clustering_node_clust...
node_embedding_node_em...
node_embedding_node_em...
node_embedding_node_em...
node_regression_node_regr...
node_regression_node_regr...
node_regression_node_regr...
task_runner_files
dataset_config.py
edge_classification.py
edge_regression.py
example.py
gae.py
gat.py
gcn.py
gmn.py
grae.py
graph_classification.py
graph_generation.py
graph_matching.py
graph_reconstruction.py
graph_regression.py
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
bash
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh graphsage mutag graph_classification
Running task: graph_classification using dataset: mutag
No saved model found. Training from scratch...
[SAGE-GraphCls] Epoch 0 - Loss: 0.6692 - Train Acc: 0.6267
[SAGE-GraphCls] Epoch 10 - Loss: 0.4789 - Train Acc: 0.7800
[SAGE-GraphCls] Epoch 20 - Loss: 0.4687 - Train Acc: 0.7600
[SAGE-GraphCls] Epoch 30 - Loss: 0.4500 - Train Acc: 0.7933
[SAGE-GraphCls] Epoch 40 - Loss: 0.4559 - Train Acc: 0.7733
[SAGE-GraphCls] Epoch 50 - Loss: 0.4612 - Train Acc: 0.7867
[SAGE-GraphCls] Epoch 60 - Loss: 0.4556 - Train Acc: 0.7733
[SAGE-GraphCls] Epoch 70 - Loss: 0.4566 - Train Acc: 0.7933
[SAGE-GraphCls] Epoch 80 - Loss: 0.4539 - Train Acc: 0.7867
[SAGE-GraphCls] Epoch 90 - Loss: 0.4498 - Train Acc: 0.7667
[SAGE-GraphCls] Epoch 99 - Loss: 0.4411 - Train Acc: 0.8067
Evaluation on Test Set:
Accuracy: 0.7368
Classification Report:
precision recall f1-score support
0 0.71 0.38 0.50 13
1 0.74 0.92 0.82 25
accuracy 0.74 38
macro avg 0.73 0.65 0.66 38
weighted avg 0.73 0.74 0.71 38
Sample Graph Classification:
Graph 8 → Predicted: 0 | Actual: 0 ✅
Graph 20 → Predicted: 1 | Actual: 1 ✅
Graph 23 → Predicted: 1 | Actual: 1 ✅
Graph 11 → Predicted: 1 | Actual: 1 ✅
Graph 2 → Predicted: 1 | Actual: 1 ✅
BLACKBOX Chat
Add Logs
CyberCoder
Improve Code
Share Code Link
Open Website
Generate Commit Message
CRLF
Python
Finish Setup
3.11.5 (venv)
Go Live
AI Code Chat
Prettier

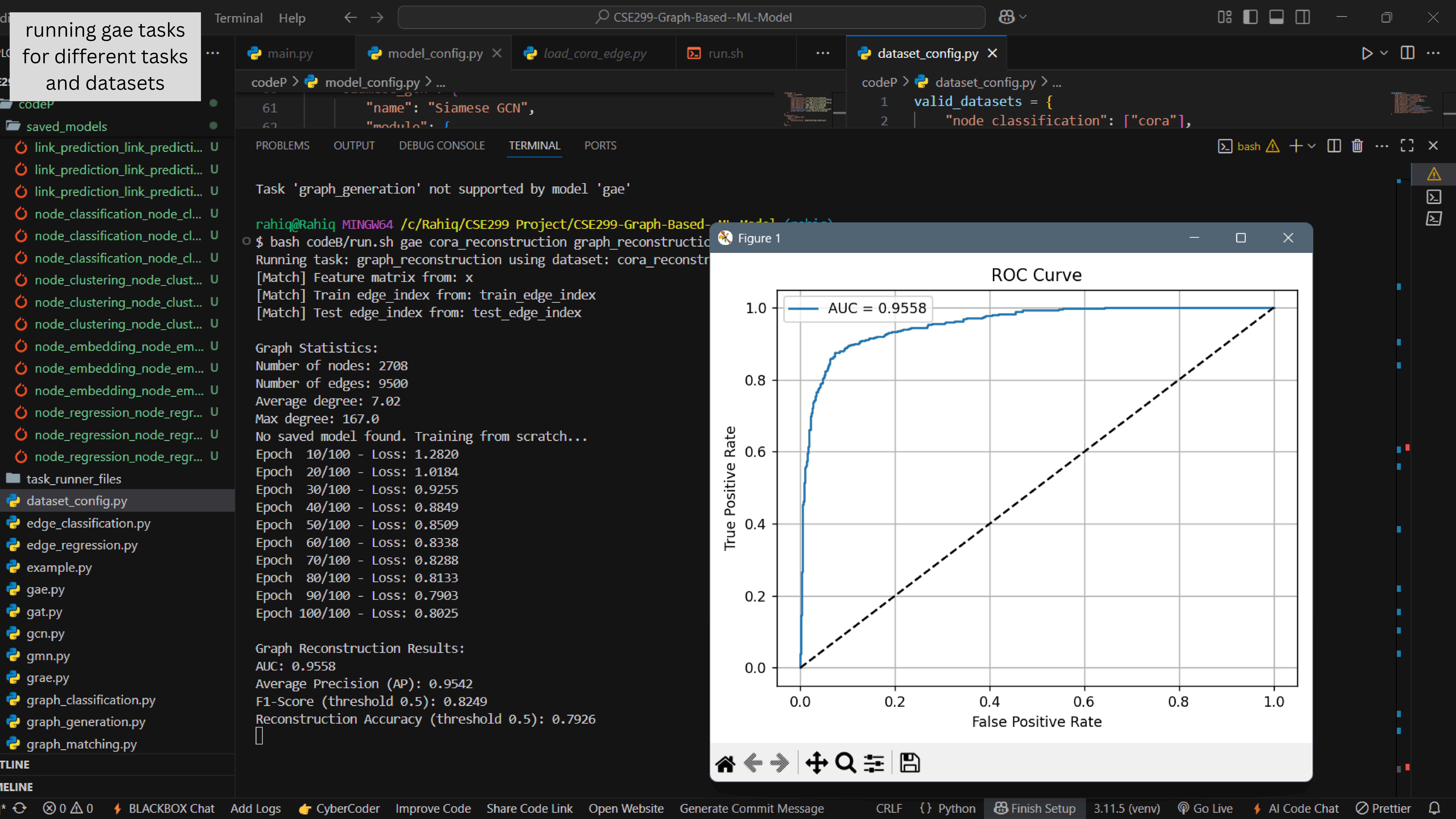
running graphsage tasks for different tasks and datasets

```
Node 2510 → Cluster 6
Node 1899 → Cluster 1
Node 1925 → Cluster 1
Node 810 → Cluster 2
Node 1373 → Cluster 5

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh graphsage synthetic_regression graph_regression
Running task: graph_regression using dataset: synthetic_regression
No saved model found. Training from scratch...
[SAGE-GraphReg] Epoch 0 - Train MSE: 59.439233
[SAGE-GraphReg] Epoch 10 - Train MSE: 0.879258
[SAGE-GraphReg] Epoch 20 - Train MSE: 0.308367
[SAGE-GraphReg] Epoch 30 - Train MSE: 0.060316
[SAGE-GraphReg] Epoch 40 - Train MSE: 0.146401
[SAGE-GraphReg] Epoch 50 - Train MSE: 0.153610
[SAGE-GraphReg] Epoch 60 - Train MSE: 0.059445
[SAGE-GraphReg] Epoch 70 - Train MSE: 0.074574
[SAGE-GraphReg] Epoch 80 - Train MSE: 0.059787
[SAGE-GraphReg] Epoch 90 - Train MSE: 0.103218
[SAGE-GraphReg] Epoch 99 - Train MSE: 0.019562

Evaluation on Test Set:
MSE: 0.0033
MAE: 0.0466
R² Score: 0.9991

Sample Graph Regression:
Graph 0 → Predicted: 24.2384 | True: 24.2668 | Abs Error: 0.0285
Graph 10 → Predicted: 25.1781 | True: 25.2133 | Abs Error: 0.0352
Graph 8 → Predicted: 24.3987 | True: 24.3831 | Abs Error: 0.0156
Graph 18 → Predicted: 22.1987 | True: 22.1834 | Abs Error: 0.0153
Graph 9 → Predicted: 25.7524 | True: 25.7283 | Abs Error: 0.0241
```



running graphsage tasks for different tasks and datasets





running gae tasks for different tasks and datasets

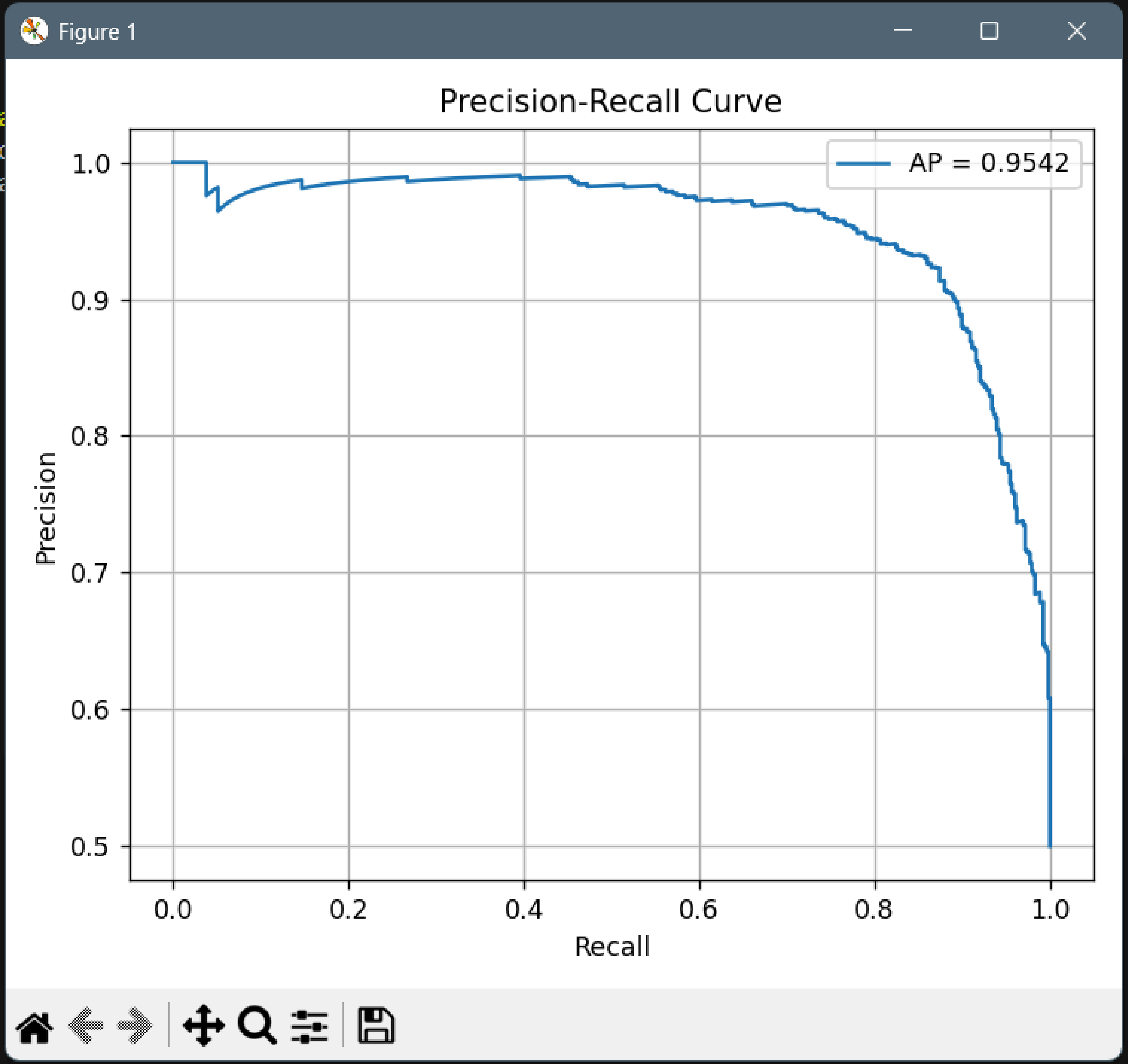
running gae tasks for different tasks and datasets

- saved_models
  - graph_regression_graph_re...
  - link_prediction_link_predicti...
  - link_prediction_link_predicti...
  - link_prediction_link_predicti...
  - node_classification_node_cl...
  - node_classification_node_cl...
  - node_classification_node_cl...
  - node_clustering_node_clust...
  - node_clustering_node_clust...
  - node_clustering_node_clust...
  - node_embedding_node_em...
  - node_embedding_node_em...
  - node_embedding_node_em...
  - node_regression_node_regr...
  - node_regression_node_regr...
  - node_regression_node_regr...
- task_runner_files
- dataset_config.py
- edge_classification.py
- edge_regression.py
- example.py
- gae.py
- gat.py
- gcn.py
- gmn.py
- grae.py
- graph_classification.py
- graph_generation.py

```
"name": "Siamese GCN",
```

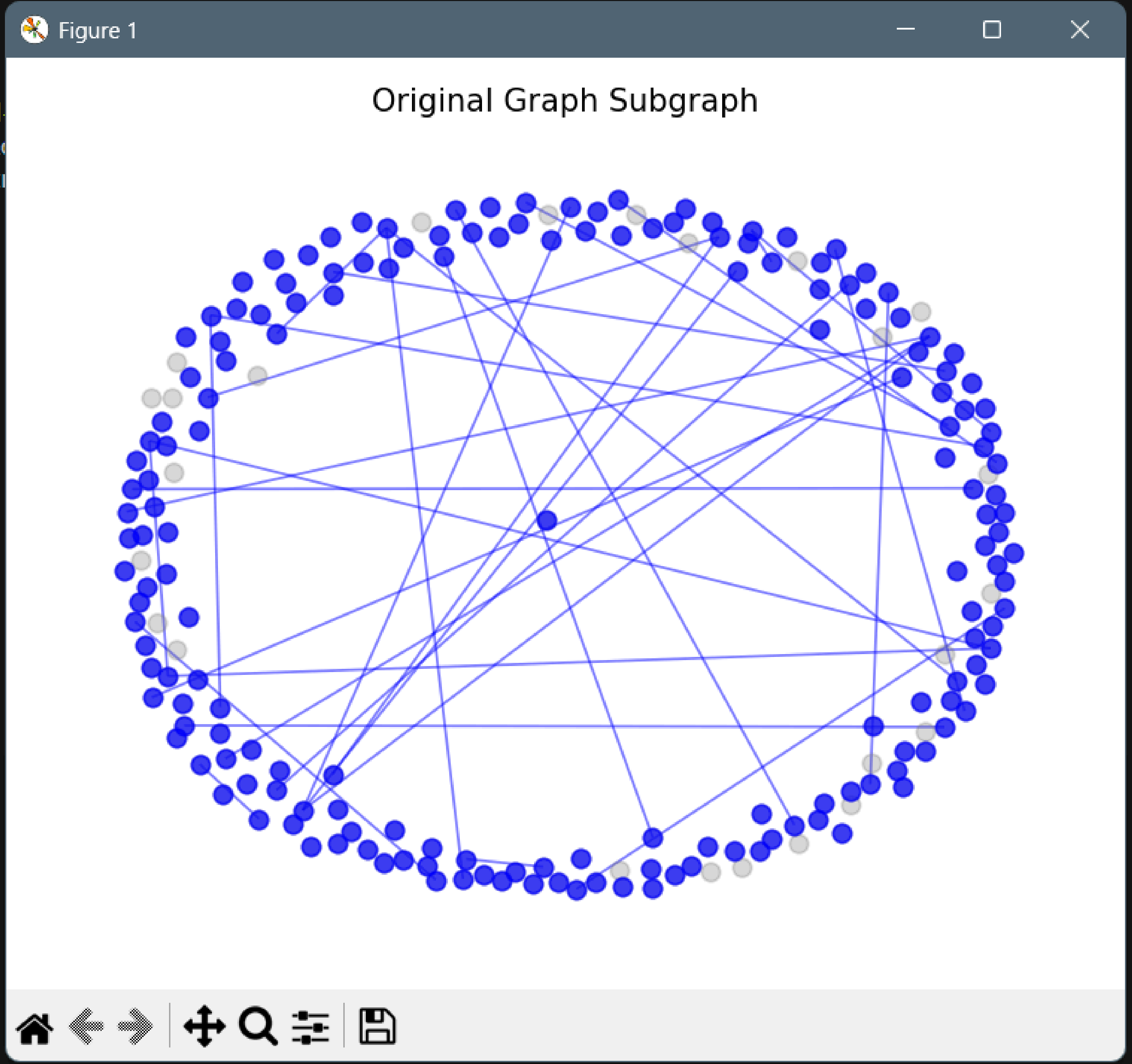
```
valid_datasets = {
    "node classification": ["cora"],
```

```
Task 'graph_generation' not supported by model 'gae'

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Gra
$ bash codeB/run.sh gae cora_reconstruction graph_rec
Running task: graph_reconstruction using dataset: cora
[Match] Feature matrix from: x
[Match] Train edge_index from: train_edge_index
[Match] Test edge_index from: test_edge_index

Graph Statistics:
Number of nodes: 2708
Number of edges: 9500
Average degree: 7.02
Max degree: 167.0
No saved model found. Training from scratch...
Epoch  10/100 - Loss: 1.2820
Epoch  20/100 - Loss: 1.0184
Epoch  30/100 - Loss: 0.9255
Epoch  40/100 - Loss: 0.8849
Epoch  50/100 - Loss: 0.8509
Epoch  60/100 - Loss: 0.8338
Epoch  70/100 - Loss: 0.8288
Epoch  80/100 - Loss: 0.8133
Epoch  90/100 - Loss: 0.7903
Epoch 100/100 - Loss: 0.8025

Graph Reconstruction Results:
AUC: 0.9558
Average Precision (AP): 0.9542
F1-Score (threshold 0.5): 0.8249
Reconstruction Accuracy (threshold 0.5): 0.7926
```

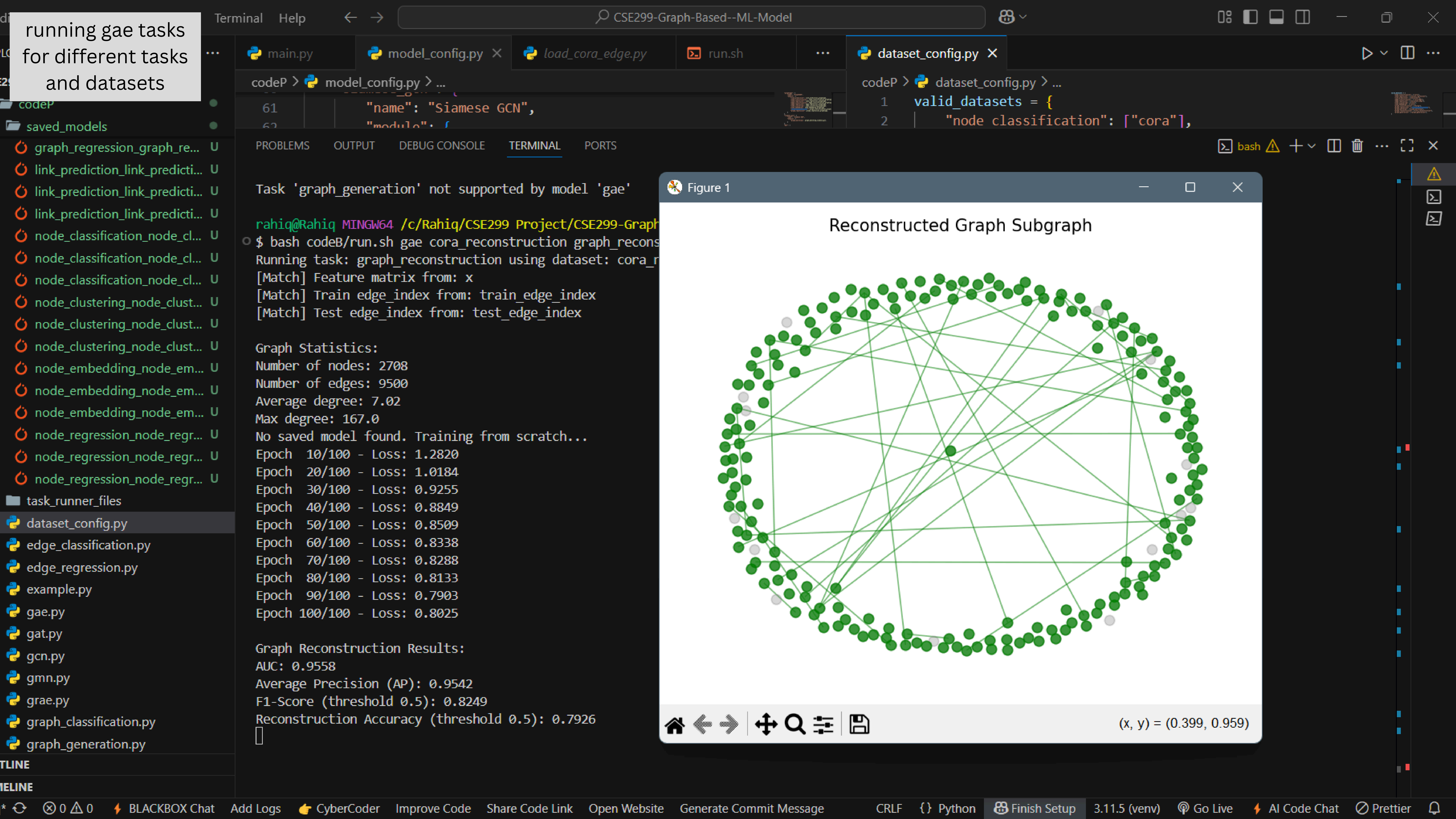


running gae tasks for different tasks and datasets

```
Task 'graph_generation' not supported by model 'gae'

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based
$ bash codeB/run.sh gae cora_reconstruction graph_reconstructi
Running task: graph_reconstruction using dataset: cora_reconst
[Match] Feature matrix from: x
[Match] Train edge_index from: train_edge_index
[Match] Test edge_index from: test_edge_index

Graph Statistics:
Number of nodes: 2708
Number of edges: 9500
Average degree: 7.02
Max degree: 167.0
No saved model found. Training from scratch...
Epoch  10/100 - Loss: 1.2820
Epoch  20/100 - Loss: 1.0184
Epoch  30/100 - Loss: 0.9255
Epoch  40/100 - Loss: 0.8849
Epoch  50/100 - Loss: 0.8509
Epoch  60/100 - Loss: 0.8338
Epoch  70/100 - Loss: 0.8288
Epoch  80/100 - Loss: 0.8133
Epoch  90/100 - Loss: 0.7903
Epoch 100/100 - Loss: 0.8025

Graph Reconstruction Results:
AUC: 0.9558
Average Precision (AP): 0.9542
F1-Score (threshold 0.5): 0.8249
Reconstruction Accuracy (threshold 0.5): 0.7926
```

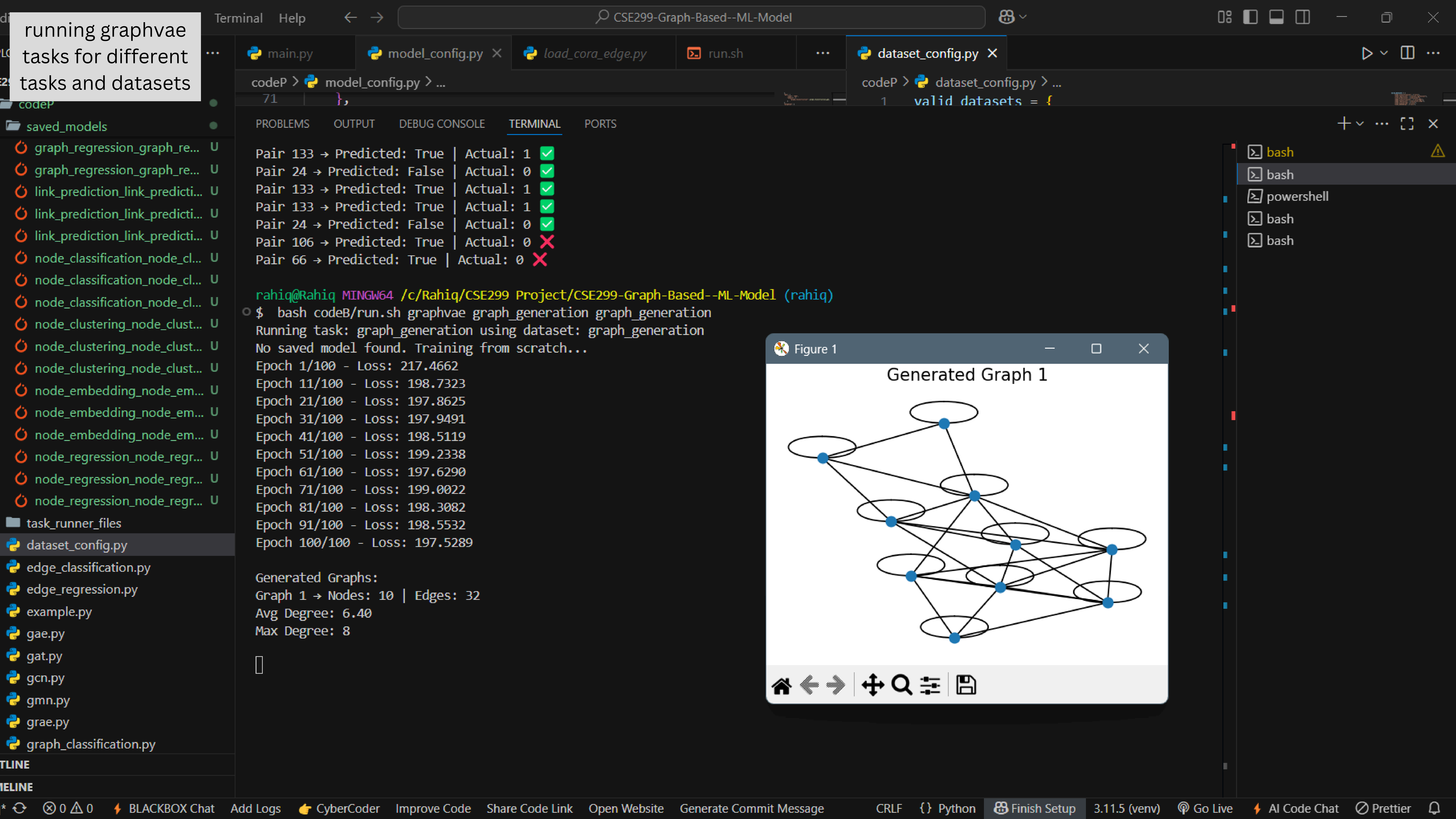


running gae tasks for different tasks and datasets



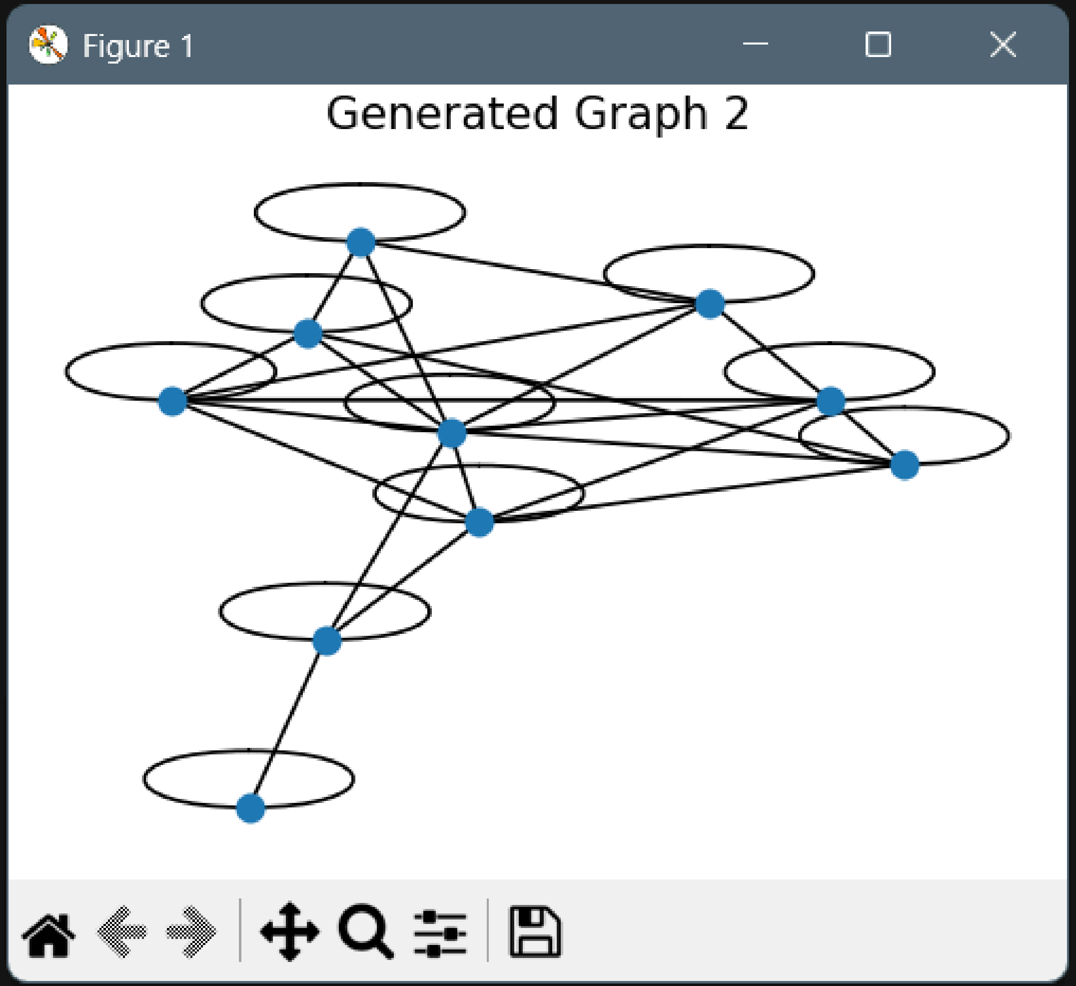
running siamese_gcn tasks for different tasks and datasets

```
Pair 67 → Predicted: True | Actual: 1 ✅
Pair 125 → Predicted: True | Actual: 1 ✅

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh siamese_gcn graph_matching graph_matching
Running task: graph_matching using dataset: graph_matching
Loading saved model from: codeP/saved_models/graph_matching_graph_matching_siamese_gcn_graph_matching.pt

Evaluation on Test Set:
Accuracy: 0.6750
Classification Report:
              precision    recall  f1-score   support

           0       0.68      0.47      0.55        86
           1       0.67      0.83      0.75       114

    accuracy                           0.68       200
   macro avg       0.68      0.65      0.65       200
weighted avg       0.68      0.68      0.66       200


Sample Graph Matching Results:
Pair 75 → Predicted: False | Actual: 1 ❌
Pair 133 → Predicted: True | Actual: 1 ✅
Pair 24 → Predicted: False | Actual: 0 ✅
Pair 106 → Predicted: True | Actual: 0 ❌
Pair 66 → Predicted: True | Actual: 0 ❌

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$
```


running graphvae tasks for different tasks and datasets
Terminal
Help
CSE299-Graph-Based--ML-Model
main.py
model_config.py
load_cora_edge.py
run.sh
dataset_config.py
codeP > model_config.py > ...
codeP > dataset_config.py > ...
valid_datasets = {
codeP
saved_models
graph_regression_graph_re...
graph_regression_graph_re...
link_prediction_link_predicti...
link_prediction_link_predicti...
link_prediction_link_predicti...
node_classification_node_cl...
node_classification_node_cl...
node_classification_node_cl...
node_clustering_node_clust...
node_clustering_node_clust...
node_clustering_node_clust...
node_embedding_node_em...
node_embedding_node_em...
node_embedding_node_em...
node_regression_node_regr...
node_regression_node_regr...
node_regression_node_regr...
task_runner_files
dataset_config.py
edge_classification.py
edge_regression.py
example.py
gae.py
gat.py
gcn.py
gmn.py
grae.py
graph_classification.py
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Pair 133 → Predicted: True | Actual: 1 ✅
Pair 24 → Predicted: False | Actual: 0 ✅
Pair 133 → Predicted: True | Actual: 1 ✅
Pair 133 → Predicted: True | Actual: 1 ✅
Pair 24 → Predicted: False | Actual: 0 ✅
Pair 106 → Predicted: True | Actual: 0 ❌
Pair 66 → Predicted: True | Actual: 0 ❌
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh graphvae graph_generation graph_generation
Running task: graph_generation using dataset: graph_generation
No saved model found. Training from scratch...
Epoch 1/100 - Loss: 217.4662
Epoch 11/100 - Loss: 198.7323
Epoch 21/100 - Loss: 197.8625
Epoch 31/100 - Loss: 197.9491
Epoch 41/100 - Loss: 198.5119
Epoch 51/100 - Loss: 199.2338
Epoch 61/100 - Loss: 197.6290
Epoch 71/100 - Loss: 199.0022
Epoch 81/100 - Loss: 198.3082
Epoch 91/100 - Loss: 198.5532
Epoch 100/100 - Loss: 197.5289
Generated Graphs:
Graph 1 → Nodes: 10 | Edges: 32
Avg Degree: 6.40
Max Degree: 8
Figure 1
Generated Graph 1
bash
bash
powershell
bash
bash
TLINE
IELINE
0 0
BLACKBOX Chat
Add Logs
CyberCoder
Improve Code
Share Code Link
Open Website
Generate Commit Message
CRLF
{} Python
Finish Setup
3.11.5 (venv)
Go Live
AI Code Chat
Prettier

running graphvae tasks for different tasks and datasets

```
Pair 133 → Predicted: True | Actual: 1 ✅
Pair 24 → Predicted: False | Actual: 0 ✅
Pair 133 → Predicted: True | Actual: 1 ✅
Pair 133 → Predicted: True | Actual: 1 ✅
Pair 24 → Predicted: False | Actual: 0 ✅
Pair 106 → Predicted: True | Actual: 0 ❌
Pair 66 → Predicted: True | Actual: 0 ❌

rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$  bash codeB/run.sh graphvae graph_generation graph_generation
Running task: graph_generation using dataset: graph_generation
No saved model found. Training from scratch...
Epoch 1/100 - Loss: 217.4662
Epoch 11/100 - Loss: 198.7323
Epoch 21/100 - Loss: 197.8625
Epoch 31/100 - Loss: 197.9491
Epoch 41/100 - Loss: 198.5119
Epoch 51/100 - Loss: 199.2338
Epoch 61/100 - Loss: 197.6290
Epoch 71/100 - Loss: 199.0022
Epoch 81/100 - Loss: 198.3082
Epoch 91/100 - Loss: 198.5532
Epoch 100/100 - Loss: 197.5289

Generated Graphs:
Graph 1 → Nodes: 10 | Edges: 32
Avg Degree: 6.40
Max Degree: 8

Graph 2 → Nodes: 10 | Edges: 31
Avg Degree: 6.20
Max Degree: 10
```



running graphvae tasks for different tasks and datasets

```
Pair 133 → Predicted: True | Actual: 1 ✅
Pair 24 → Predicted: False | Actual: 0 ✅
Pair 106 → Predicted: True | Actual: 0 ❌
Pair 66 → Predicted: True | Actual: 0 ❌

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Epoch 11/100 - Loss: 198.7323
Epoch 21/100 - Loss: 197.8625
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Epoch 51/100 - Loss: 199.2338
Epoch 61/100 - Loss: 197.6290
Epoch 71/100 - Loss: 199.0022
Epoch 81/100 - Loss: 198.3082
Epoch 91/100 - Loss: 198.5532
Epoch 100/100 - Loss: 197.5289

Generated Graphs:
Graph 1 → Nodes: 10 | Edges: 32
Avg Degree: 6.40
Max Degree: 8

Graph 2 → Nodes: 10 | Edges: 31
Avg Degree: 6.20
Max Degree: 10

Graph 3 → Nodes: 10 | Edges: 35
Avg Degree: 7.00
Max Degree: 9
```
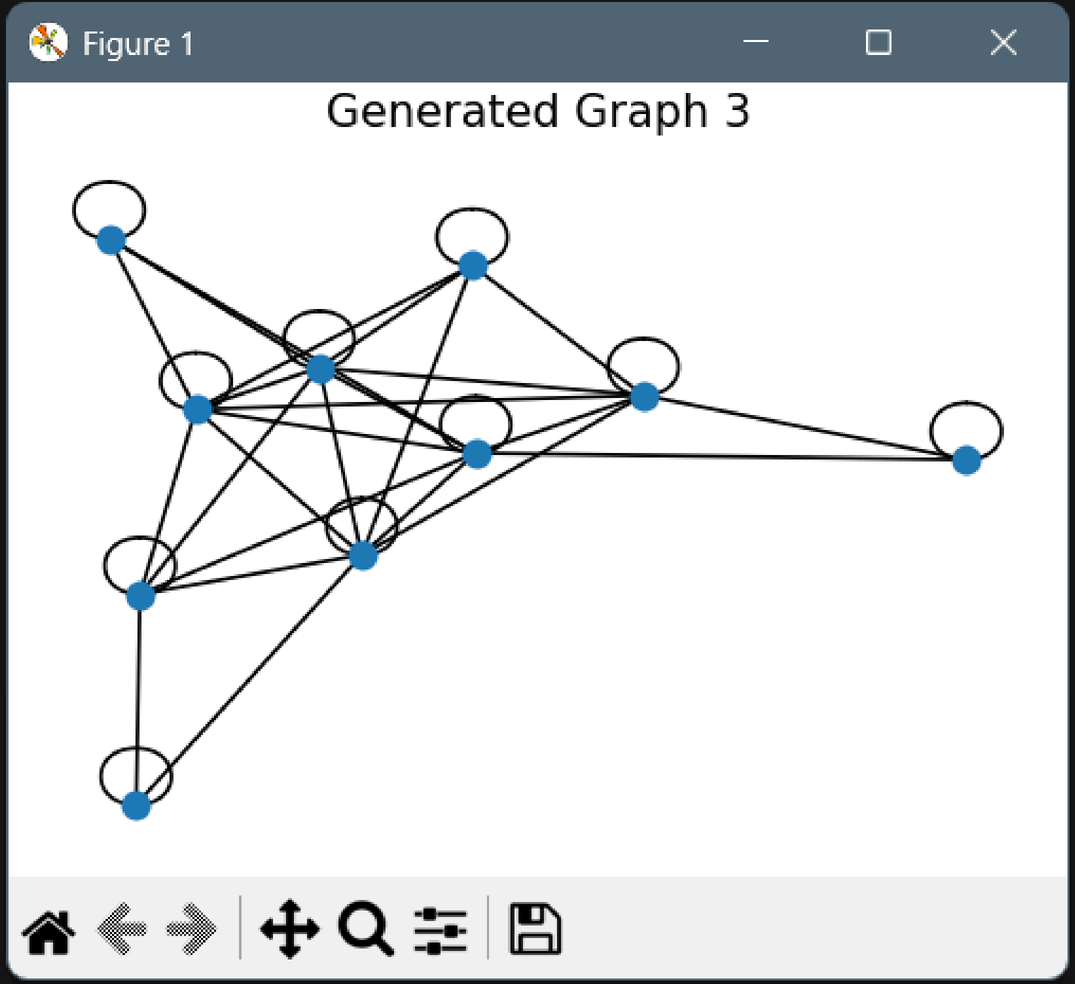
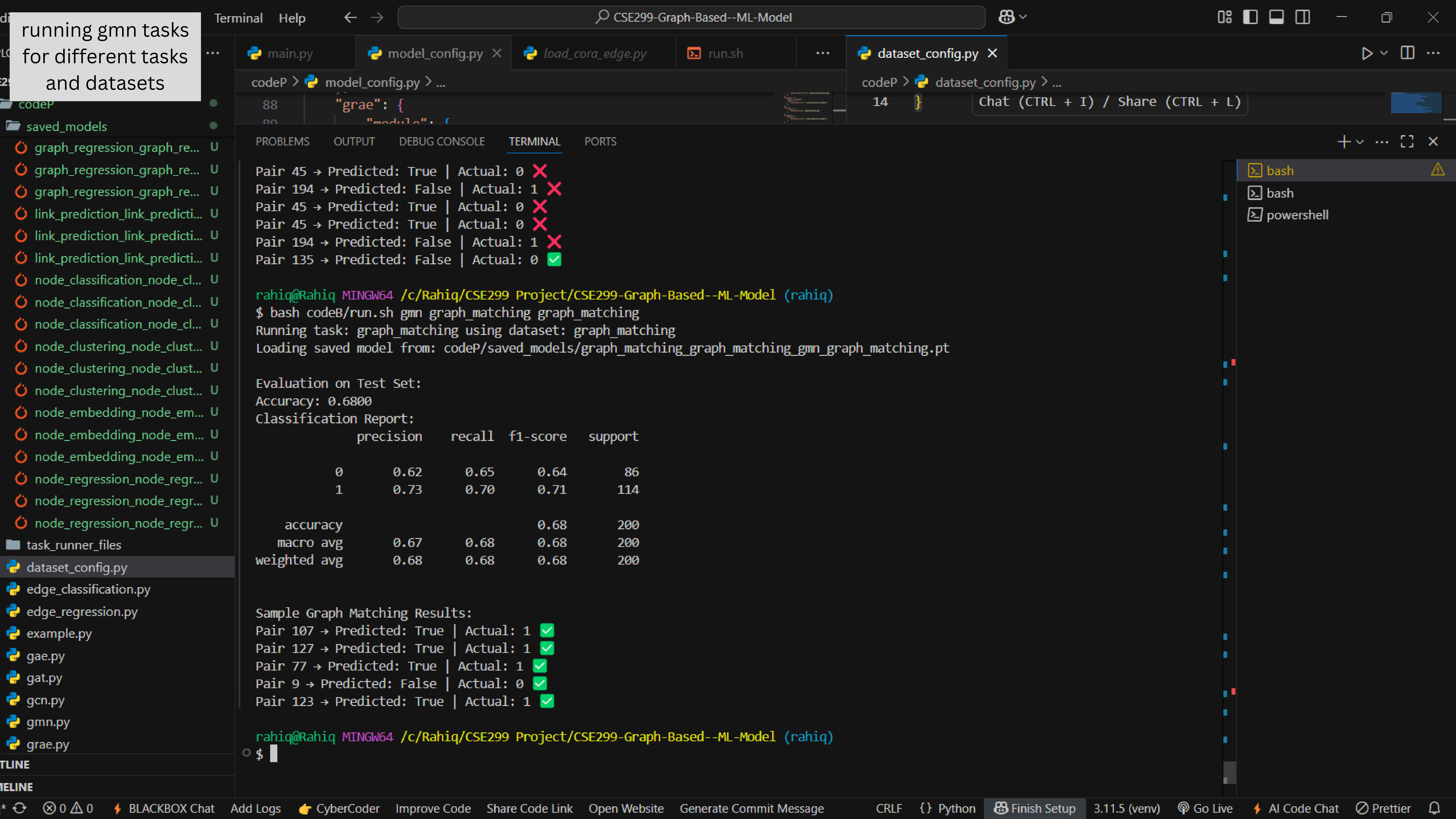
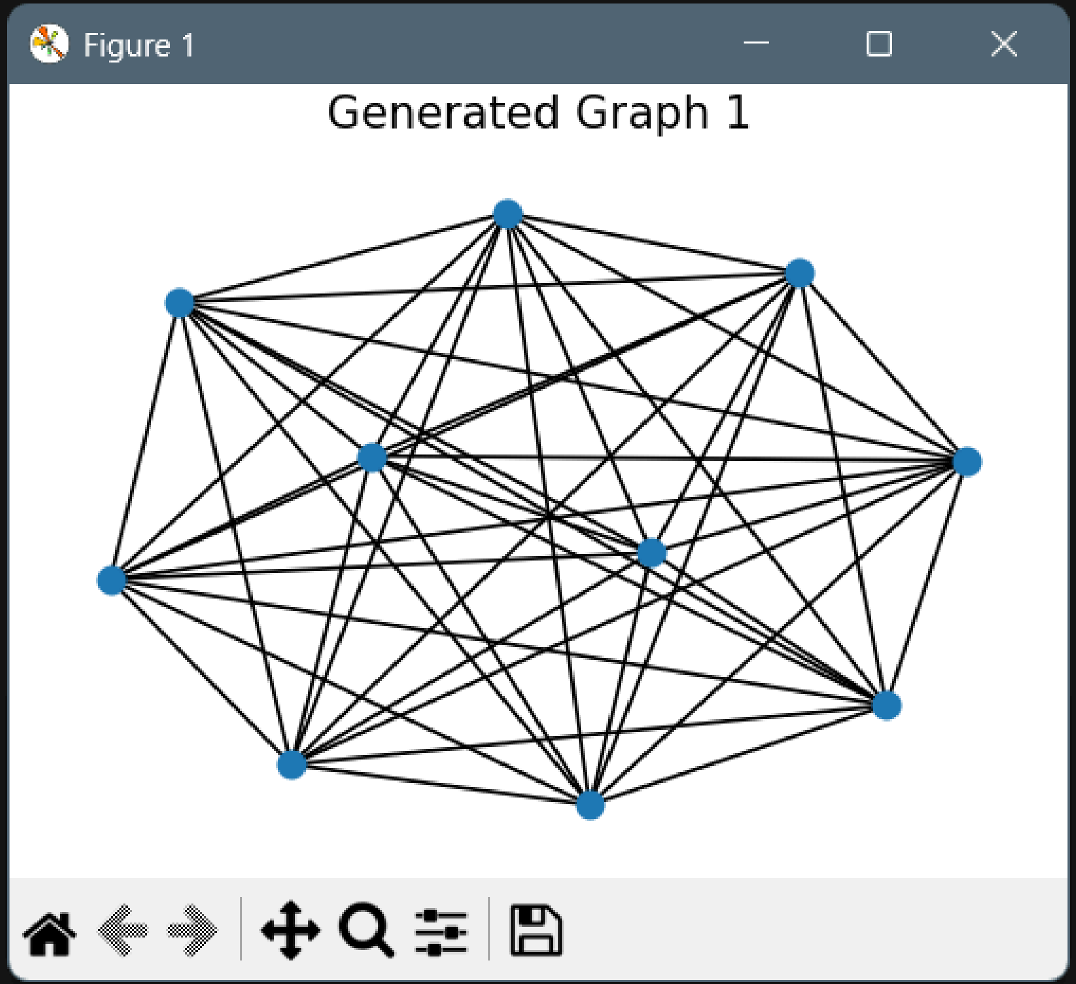



running gmn tasks for different tasks and datasets

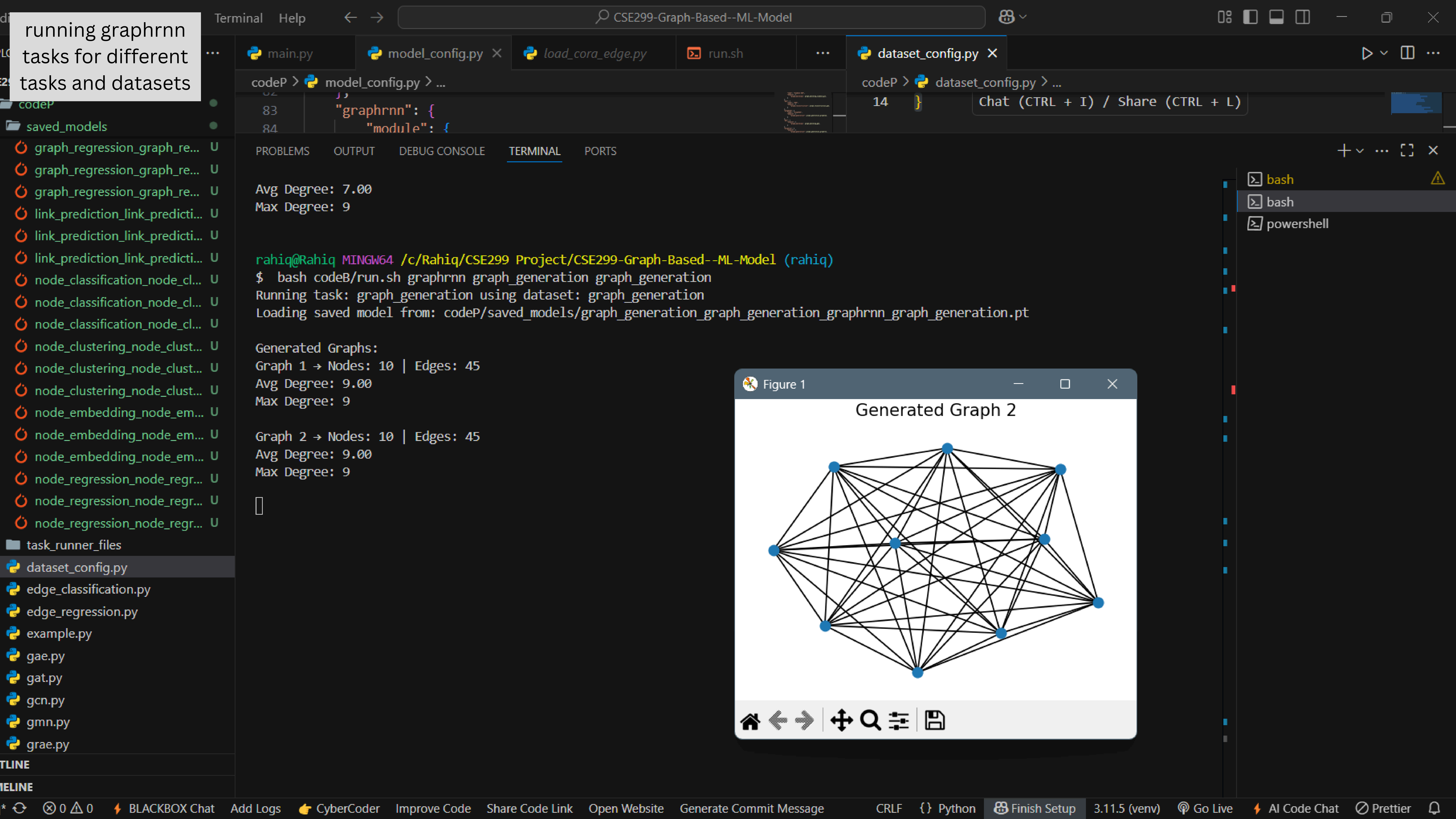


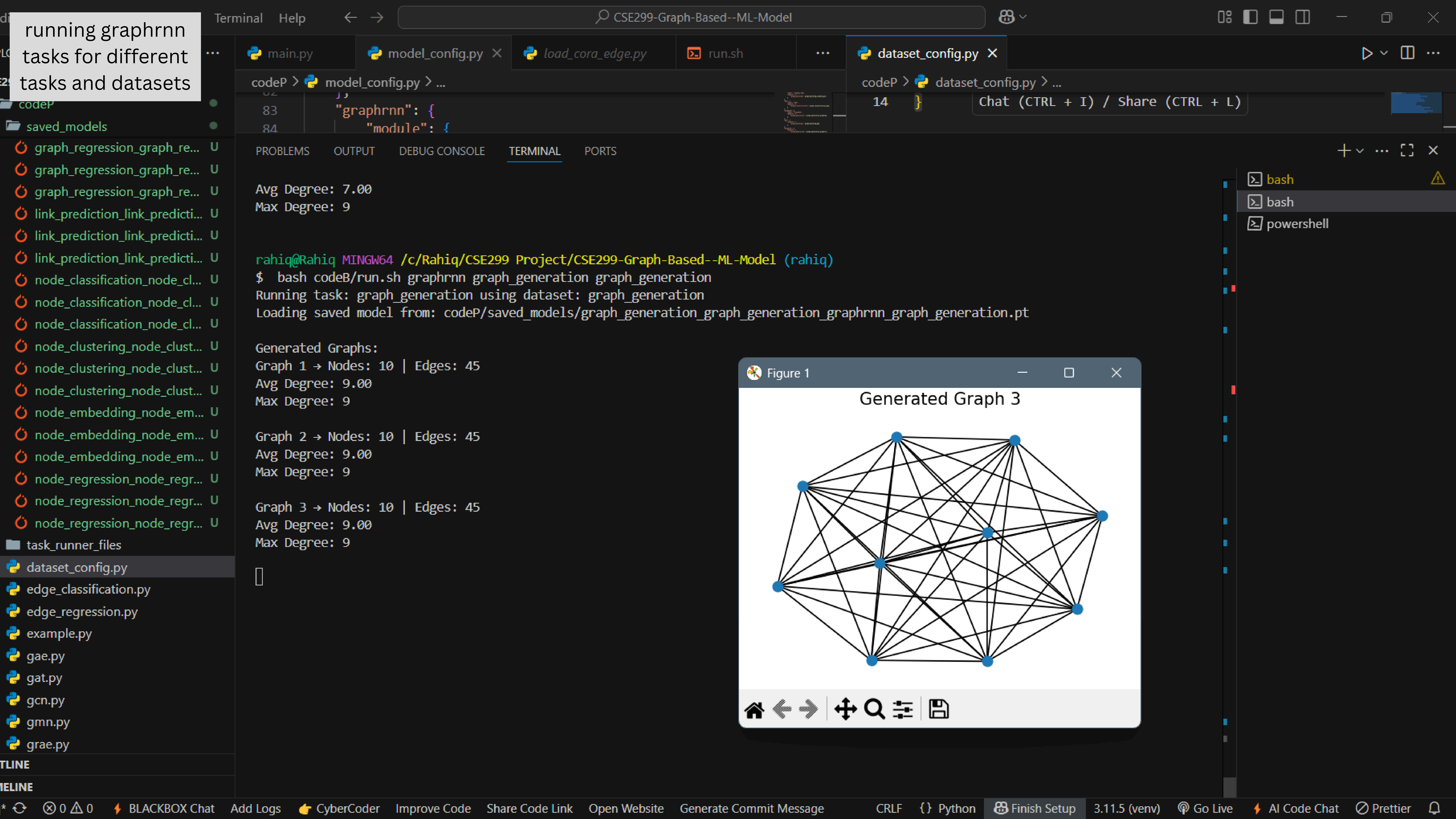
running graphrnn tasks for different tasks and datasets

```
Avg Degree: 7.00
Max Degree: 9


rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$  bash codeB/run.sh graphrnn graph_generation graph_generation
Running task: graph_generation using dataset: graph_generation
Loading saved model from: codeP/saved_models/graph_generation_graph_generation_graphrnn_graph_generation.pt

Generated Graphs:
Graph 1 → Nodes: 10 | Edges: 45
Avg Degree: 9.00
Max Degree: 9
```



running graphrnn tasks for different tasks and datasets



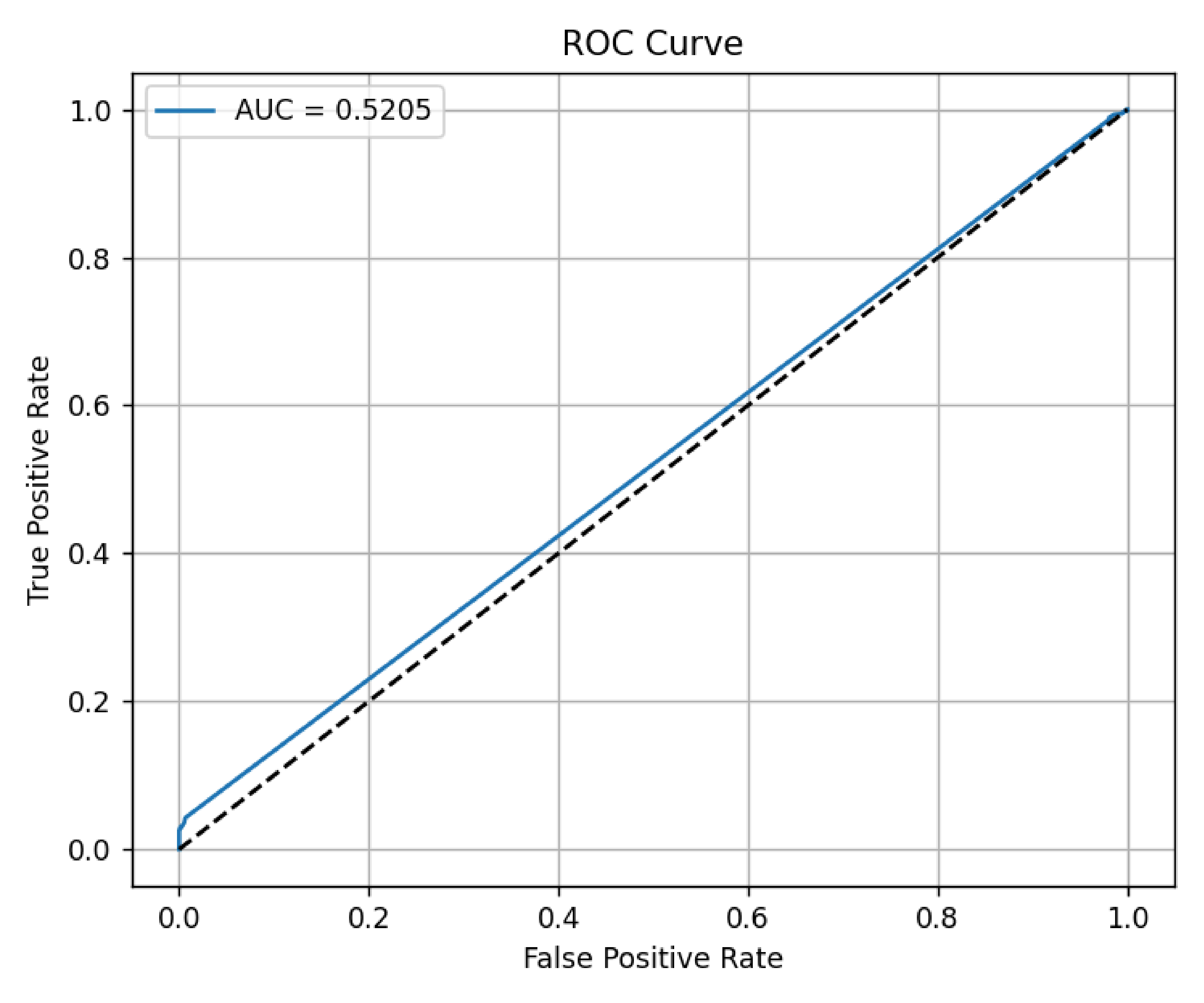


running graphrnn tasks for different tasks and datasets

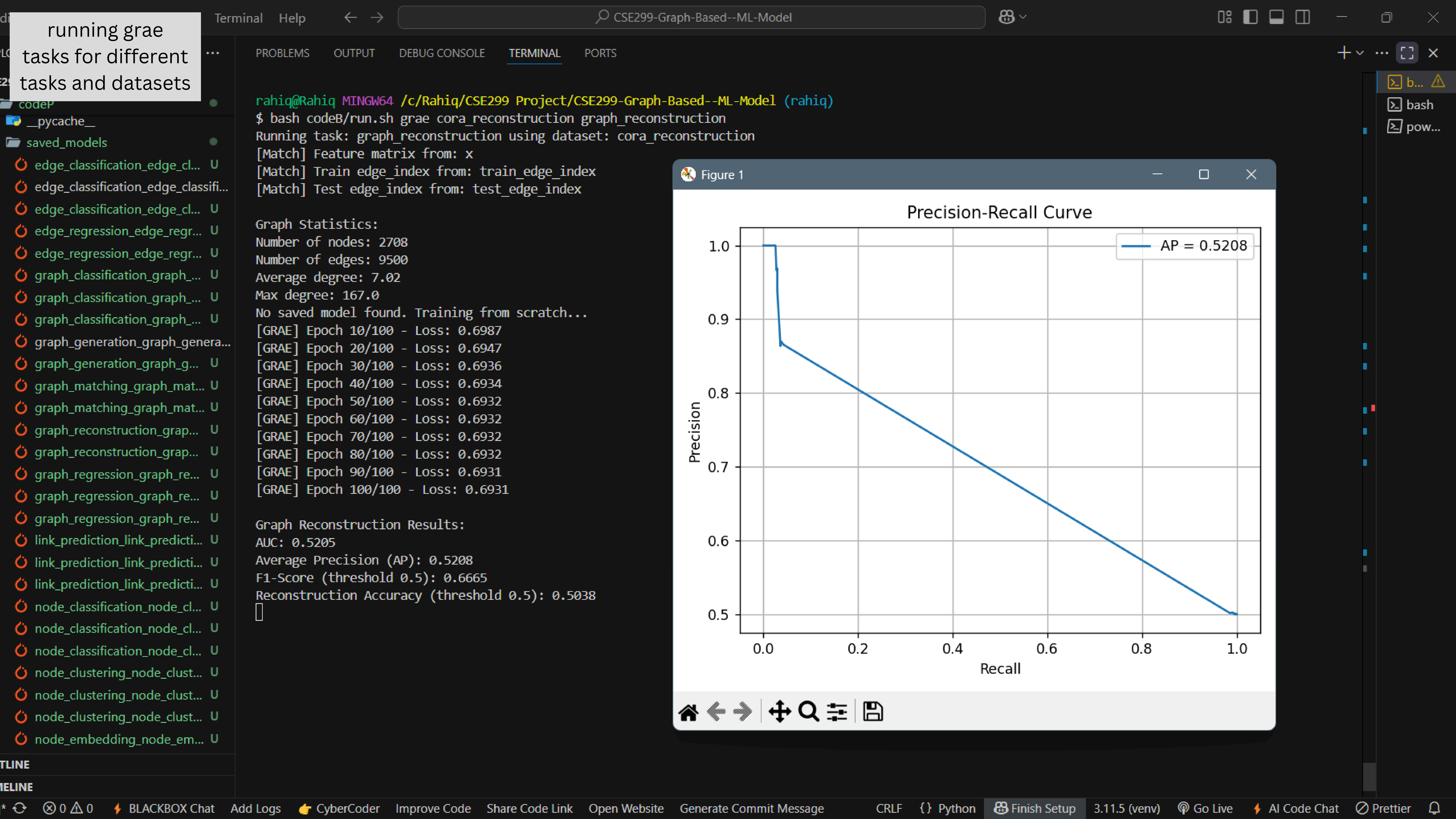
running grae tasks for different tasks and datasets

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh grae cora_reconstruction graph_reconstruction
Running task: graph_reconstruction using dataset: cora_reconstruction
[Match] Feature matrix from: x
[Match] Train edge_index from: train_edge_index
[Match] Test edge_index from: test_edge_index

Graph Statistics:
Number of nodes: 2708
Number of edges: 9500
Average degree: 7.02
Max degree: 167.0
No saved model found. Training from scratch...
[GRAE] Epoch 10/100 - Loss: 0.6987
[GRAE] Epoch 20/100 - Loss: 0.6947
[GRAE] Epoch 30/100 - Loss: 0.6936
[GRAE] Epoch 40/100 - Loss: 0.6934
[GRAE] Epoch 50/100 - Loss: 0.6932
[GRAE] Epoch 60/100 - Loss: 0.6932
[GRAE] Epoch 70/100 - Loss: 0.6932
[GRAE] Epoch 80/100 - Loss: 0.6932
[GRAE] Epoch 90/100 - Loss: 0.6931
[GRAE] Epoch 100/100 - Loss: 0.6931

Graph Reconstruction Results:
AUC: 0.5205
Average Precision (AP): 0.5208
F1-Score (threshold 0.5): 0.6665
Reconstruction Accuracy (threshold 0.5): 0.5038
```

Figure 1



running grae tasks for different tasks and datasets



running grae tasks for different tasks and datasets

```
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Running task: graph_reconstruction using dataset: cora_reconstruction
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[GRAE] Epoch 40/100 - Loss: 0.6934
[GRAE] Epoch 50/100 - Loss: 0.6932
[GRAE] Epoch 60/100 - Loss: 0.6932
[GRAE] Epoch 70/100 - Loss: 0.6932
[GRAE] Epoch 80/100 - Loss: 0.6932
[GRAE] Epoch 90/100 - Loss: 0.6931
[GRAE] Epoch 100/100 - Loss: 0.6931

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```

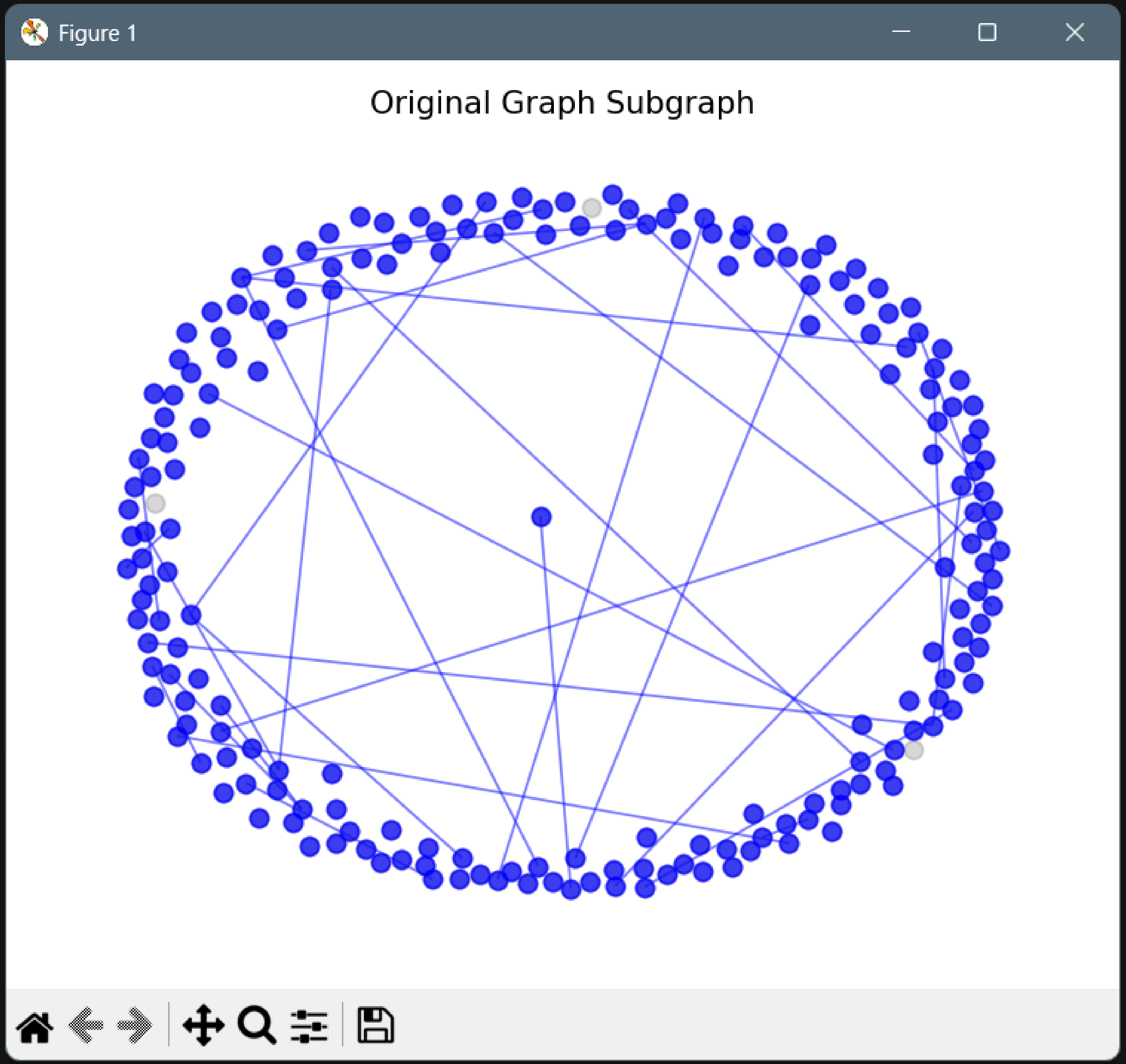

running grae tasks for different tasks and datasets

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh grae cora_reconstruction graph_reconstruction
Running task: graph_reconstruction using dataset: cora_reconstruction
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```